CSE 429 · COMPUTER VISION AND PATTERN RECOGNITION · MIDTERM REFERENCE

Seven lectures, six sheets. Here's the short road through them.

For the midterm on **Tuesday 10 November 2026, 13:30** · Scope: Lectures 1–7 and Sheets 1–6 · Gonzalez & Woods, *Digital Image Processing* (3rd ed.), Chapters 1–3 · written for someone starting from zero.

Is it doable by tomorrow?

Yes, if you follow the core path and don't chase every slide. Most of the marks are methods: count a histogram and equalize it, find the shortest m-path, stretch a 3 × 3 image, run a filter over a small matrix, take a derivative of an array. Each one is taught here on a small problem first, then the real sheet problem is boxed for you to solve, with the full answer at the end of the lecture.

All seven lectures add up to about **5 hours**. That is deliberately lean: slides that are pictures or history are reduced to a few lines, and every sheet question is kept. Keep about **5½ hours of sleep**: a rested brain does arithmetic faster and with fewer slips, and this exam is arithmetic.

A suggested plan (exam Tuesday 13:30)

When	What	⏱
22:45	Lecture 1: what vision is	10
22:55	Lecture 6: spatial filtering + Sheet 5 Q3–Q5, Sheet 6 Q1–Q6	55
23:50	Lecture 2: digital images + Sheets 1 and 3	30
00:20	Lecture 3, part one: pixels, paths, distances + Sheet 2	35
00:55	Lecture 3, part two: interpolation, operations + Sheet 4 Q1–Q3	25
01:25	*Sleep*	
07:00	Lecture 4: stretching, slicing, equalization	35
07:35	Lecture 5: histogram matching + Sheet 4 Q4–Q6, Sheet 5 Q1–Q2	45
08:20	Lecture 7, part 1: derivatives and the Laplacian (Sheet 6 Q7–Q10)	30
12:30	Lecture 7, part 2 (Q11–Q13) + the formula sheet from memory	40

Lecture 6 comes early on purpose: it stands on its own, and it carries the most sheet questions.

How to read this document

📖 Read & understand
No pen. Read it once slowly, then say the idea back to yourself in one sentence.

✏️ Pen & paper
The method is taught on a smaller problem first. Then the real sheet problem is boxed for you to solve on paper. Its full answer is at the end of the lecture.

🔁 Recall drill
Quick questions. Answer before you look.

Tier A must do · Tier B should do · Tier C only if you're ahead. Minutes (⏱) are for a tired student, not a fresh one. Each lecture opens with an "Only 20 minutes?" path.

Links. ↗ Understand more links open the interactive study guide (`vision-study.html`, same folder) or a short outside explanation. Use them when a card didn't click.

🧭 **Analogies** tie each idea to something familiar and say where the comparison stops working.

Mistakes in the course material are flagged in each lecture's ⚠️ Traps box, with the correct version.

Contents

1. **Lecture 1** · What computer vision is, and why it's hard · 4 cards · ⏱ 10 min
2. **Lecture 2** · Digital image fundamentals: from light to numbers · 6 cards · ⏱ 30 min
3. **Lecture 3 (pp. 1–19)** · Pixel relationships: neighbours, adjacency, paths, regions, distances · 5 cards · ⏱ 35 min
4. **Lecture 3 (pp. 20–42)** · Interpolation and operations on whole images · 5 cards · ⏱ 25 min
5. **Lecture 4** · Image enhancement: point transforms and histograms · 5 cards · ⏱ 35 min
6. **Lecture 5** · Histogram processing: equalization and matching · 7 cards · ⏱ 45 min
7. **Lecture 6** · Spatial filtering: correlation, convolution, smoothing · 7 cards · ⏱ 55 min
8. **Lecture 7** · Sharpening: derivatives, the Laplacian, unsharp masking, Sobel · 10 cards · ⏱ 53 min

LECTURE 1

What computer vision is, and why it's hard

Tier B · should ⏱ 10 min Slides: Lec 1 - Overview of Computer Vision and Pattern Recognition.pdf (72 p, mostly pictures)

📘 Guide Tutorial 1

Lecture 1 is the map of the course: what the field tries to do, how its problems are sorted into levels, and why a task a toddler does without effort is still research. There are no calculations here, only definitions and examples, so it is quick marks if you read it once.

What the exam can ask from this lecture

- Computer vision vs image processing vs computer graphics, in one line each.
- The three levels of vision and an example task at each level.
- Why vision is ill-posed, and the named challenges (viewpoint, illumination, occlusion…).
- An example application for a named area (biometrics, automotive safety, medicine…).

⏳ **Only 5 minutes?** Read card 2, then do the drill.

📖 READ & UNDERSTAND · ⏱ 3 min · Tier B

1 · Three fields that all touch images

Take one photo of a street with a car, a horse and a pole. **Image processing** makes the photo better for a person to look at: sharper, brighter, less noisy. The output is still an image.

Computer vision has to say what is in it: "a car, a horse, a person; the car is in front of the pole". The output is a description that a computer acts on. **Computer graphics** runs the other way: it starts from a model and creates the picture.

Between them sits **image analysis**, which has two parts: **feature extraction** (turning pixels into higher-level information, such as edges or shapes) and **pattern classification** (using that information to name objects). The course title, *Computer Vision and Pattern Recognition*, is those two halves.

🧭 **Like a phone camera app.** The beauty filter that smooths your skin is image processing; the face-unlock that decides "this is the owner" is computer vision; the AR sticker drawn on your face is graphics. *Where it breaks:* real apps chain all three, so one product rarely belongs to only one field.

↗ Understand more: guide Tutorial 1, step 1

📖 READ & UNDERSTAND · ⏱ 3 min · Tier B

2 · Low, mid and high: sorting vision tasks by how much they understand

Level	Tasks (Lec 1)	On the street photo
Low	measurements, enhancement, region segmentation, features	the white patch, the shadow, the edges
Mid	reconstruction, depth, motion estimation	the car is about 1 m in front of the pole
High	category detection, activity recognition, pose, deep understanding	a car, a horse, a man riding past

Lecture 2 uses a second, similar list from the textbook, about **processes**: low-level (noise reduction, contrast, sharpening; image in, image out), mid-level (segmentation, classifying objects; image in, attributes out), high-level (making sense of the scene). Notice that segmentation sits at **low** level in Lecture 1's vision list and at **mid** level in Lecture 2's processing list. Answer with whichever list the question names.

🧭 **Like reading a page.** Low level is seeing the ink strokes, mid level is grouping them into words and noticing the page is tilted, high level is understanding the story. *Where it breaks:* in vision the levels feed back into each other; knowing "that's a car" helps find its edges.

↗ Understand more: guide Tutorial 1, step 2

📖 READ & UNDERSTAND · ⏱ 2 min · Tier C

3 · Why a summer project became a whole field

In 1966 Marvin Minsky asked an undergraduate to connect a camera to a computer over one summer and make it "describe what it sees". It is still research. The core reason: a camera flattens a 3-D world into a 2-D picture and throws depth away, so **many different scenes give exactly the same image**. Recovering the scene from the image has no unique answer without extra assumptions; that is what **ill-posed** means.

Lecture 1 names nine practical challenges. The pictures make them easy to remember:

viewpoint	one statue from three angles
illumination	one face, lit two ways
occlusion	a rider half-hidden by trees
scale	a tall and a short man holding laptops
deformation	galloping horses
background clutter	a figure dissolving into a patterned painting
intra-class variation	six very different chairs, all "chair"
local ambiguity	a blurry patch that only makes sense in context
the world behind the image	knowing a train is long and heavy without seeing it

🧭 **Like a shadow puppet.** A hand, a cardboard cut-out and a real dog can all cast the same shadow; from the shadow alone you can't be sure which it was. *Where it breaks:* a camera keeps colour and texture too, which is why vision is hard but not hopeless.

↗ Understand more: guide Tutorial 1, steps 3–4

🔁 RECALL DRILL · ⏱ 2 min · Tier B

4 · Quick recall

1. A system estimates how fast a car is moving between two video frames. Which level of vision?
2. A dog recogniser works in daylight and fails at dusk. Which challenge?
3. Six photos of different chairs are all labelled "chair". Which challenge?
4. In one line: why is vision called ill-posed?
5. Creating images from a 3-D model is which field?

✍️ Answers: S1.1 (end of this lecture)

⚠️ Traps

- Segmentation is "low level" in Lecture 1's vision list but a "mid-level process" in Lecture 2's list. Use the list the question asks about.
- Image processing outputs an image for people; computer vision outputs a description for a machine. Don't swap them.

Solutions and answers — Lecture 1

S1.1 · 4 · Quick recall — Answers

1. Mid level (motion estimation).
2. Illumination.
3. Intra-class variation.
4. Many 3-D scenes project to the same 2-D image, so the answer isn't unique.
5. Computer graphics.

↩ back to the card

LECTURE 2

Digital image fundamentals: from light to numbers

Tier A · must ⏱ 30 min Slides: Lec 2-Digital Image Fundamentals.pdf (63 p) · Sheet 1 (16 MCQs), Sheet 3 Q1–Q6

↗ Szeliski Ch 1–2

Before any filter or histogram can touch a picture, the picture has to become a grid of numbers. This lecture is that conversion: how a camera samples and quantizes a scene, how many bits the result needs, and how long it takes to send. Everything after it assumes you can picture an image as $f(x, y)$, a table of integers.

What the exam can ask from this lecture

- Image size in bits, bytes and MB for a given resolution, number of levels and grey/RGB (Lec 2 p37–40, Sheet 3 Q1–Q2).
- Transmission time over a modem with start/stop bits (Sheet 3 Q3–Q4); HDTV movie storage (Q5–Q6).
- Definitions: computer vision vs image processing vs graphics, low/mid/high level, sampling vs quantization, spatial vs intensity resolution, dynamic range vs contrast.
- MCQs in the style of Sheet 1, mostly on the key stages.

⏳ **Only 20 minutes?** Do cards 3, 6 and 2, in that order.

📖 READ & UNDERSTAND · ⏱ 4 min · Tier A

1 · Who is the picture for: a person or a computer?

The slides split the field by one question: **who consumes the output** (L2 p4). In **image processing** the output is an image for a human to look at (a brighter X-ray, a sharper photo). In **computer vision** the output is used by a computer ("a car at (120, 40)", "this is digit 7"). **Computer graphics** runs the other way: it *creates* images from a model (L2 p6).

Work is also sorted into three **levels** by what goes in and comes out (L2 p5):

level	examples	input → output
low	noise reduction, contrast enhancement, sharpening	image → image
mid	segmentation, classifying individual objects	image → attributes (regions, labels)
high	image analysis, the "understanding" of human vision	attributes → meaning

Quick test: sharpening returns an image, so it is **low-level**; cutting a photo into "road" and "sky" returns regions, so it is **mid-level**. Within computer vision, **image analysis** = feature extraction (pull edges, shapes out of pixels) + pattern classification (name the objects from those features, L2 p8).

🧭 **Like a phone camera app.** The beauty filter that smooths your skin is image processing (you look at it); the face detector that draws the focus box is computer vision (the phone uses it); the cartoon sticker is graphics. *Where it breaks:* real apps chain all three; the exam wants the textbook split.

More: Szeliski Ch 1 (szeliski.org/Book); the guide's Chapter 2 (vision-study.html).

📖 READ & UNDERSTAND · ⏱ 5 min · Tier A

2 · Two roundings turn a scene into a table

A sensor's output is a **continuous waveform**: any position, any value (L2 p30). The slides model an image as a **2-D function** $f(x, y)$: (x, y) is a position and f the **intensity** (grey level) there (L2 p14). It is a **digital image** once x, y *and* f are all discrete; each cell is a **pixel**, and an $M \times N$ image has MN of them (L2 p15–17).

Two separate steps get you there (L2 p30). **Sampling** digitizes the **coordinates**: where to measure. **Quantization** digitizes the **amplitude**: each value is rounded to one of L levels. Tiny run, 3 bits so $L = 8$ (levels 0–7), rule level $=$ round$(7v)$:

sample v	0.12	0.47	0.83	0.95	0.50
$7v$	0.84	3.29	5.81	6.65	3.5
level	1	3	6	7	4

Each kind of rounding has its own resolution. **Spatial resolution** = smallest visible detail, in line pairs or dots per unit distance (**dpi**); you need the dpi *and* the print size, not just the pixel count (L2 p34). **Intensity resolution** = smallest visible change in intensity, in **bits per pixel**, since k bits give $L = 2^k$ levels (L2 p35): 1 bit → 2, 3 → 8, 8 → 256, 9 → 512, 16 → 65,536. Too few levels breaks smooth shading into bands, **false contouring** (the skulls on L2 p36).

Don't mix up two look-alikes (L2 p33): **dynamic range** is a **ratio**, max measurable ÷ min detectable intensity (a property of the system); **contrast** is a **difference**, highest − lowest intensity in this image (values 40…200 → contrast 160).

🧭 **Like copying a photo as a mosaic.** Sampling decides how many tiles there are; quantization decides how many shades of tile are in your box, so each tile gets the nearest shade you own. *Where it breaks:* a real sensor averages light over each cell instead of reading one point.

Szeliski Ch 2.3 covers the sensor side: szeliski.org/Book.

🖍 PEN & PAPER · ⏱ 9 min · Tier A

3 · How many bits, and how long to send them

> IN PLAIN WORDS
> ① Sheet 3 Q1–Q2 give a resolution (rows × columns), the number of intensity levels L, and grey (1 channel) or RGB (3 channels, red/green/blue); find the size. Q3–Q4 send an image over a modem whose **baud rate** is bits per second, in **packets** of start bit + data + stop bit; find the time.

> WARM-UP BY HAND
> ② A 100×50 RGB image with 16 levels: $\log_2 16 = 4$ bits × 3 channels = 12 bits per pixel; $5000 \times 12 = 60{,}000$ bits $= 7{,}500$ bytes. Sent in 8-bit packets, that is 7,500 packets × 10 line bits = 75,000 bits; at 9,600 baud, $75{,}000/9{,}600 = 7.81$ s.

> THE ONE IDEA

③ Size (bits) = rows × columns × channels × $\log_2 L$. On the wire, every packet costs 2 extra bits, so count packets first, then multiply by (data bits + 2).

BUILD IT

④ on the slides' camera (L2 p37–40): 4 MP = 2680 × 1520, RGB, 256 levels.

1. Pixels: 2680 × 1520 = 4,073,600.
   Why: one pixel per grid cell.
2. Bits per pixel: $\log_2 256 = 8$ per channel × 3 = 24.
   Why: RGB stores one 8-bit value per colour.
3. 4,073,600 × 24 = 97,766,400 bits; ÷ 8 = 12,220,800 bytes.
   Why: 1 byte = 8 bits.
4. ÷ 1,048,576 = **11.655 MB**.
   Why: the slides take 1 MB = 2^{20} bytes.

CHECK

⑤ Dividing by 10^6 instead gives 12.22 MB: wrong unit, not wrong maths. Forgetting the 3 channels gives a third of the answer. More: the guide's Chapter 2, Steps 4–5 (vision-study.html).

EXAM PROBLEM

Your turn — Sheet 3 Q1. A camera with resolution 4 Mega Pixel (2680 × 1520 pixel). What is the size of the output RGB image if it has 512 intensity levels?
Q2. A camera with resolution 2 Mega Pixel (1920 × 1080 pixel). What is the size of the output grey image if it has 128 intensity levels?
Q3. Transmission is in packets of a start bit, a byte (8 bits) of information and a stop bit. (a) How many minutes to transmit a 1024 × 1024 image with 256 intensity levels using a 56K baud modem? (b) The time at 3000K baud (a medium-speed DSL line)?
Q4. Now packets carry a start bit, 2 bytes (16 bits) of information and a stop bit. (a) How many minutes for a 1024 × 1024 image with 65,536 intensity levels at 70K baud? (b) The time at 4000K baud and 256 intensity levels?

✍ Show the full exam answer: S2.1 (end of this lecture)

TRY ALONE

⑦ A 512 × 512 grey image with 64 levels: size in bits and in MB?

✍ Answers: S2.2 (end of this lecture)

🖍 PEN & PAPER · ⏱ 5 min · Tier B

4 · A movie is just frames × bits per frame

IN PLAIN WORDS

① Sheet 3 Q5–Q6 fix the number of horizontal TV **lines** (the image height in pixels) and the **aspect ratio** width:height; pixels per line = lines × ratio. Each pixel has 24 bits (8 each for R, G, B). Find the bits for a movie of given length at 30 frames per second.

WARM-UP BY HAND

② 90 lines at 16:9 → $90 \times 16/9 = 160$ pixels per line. One frame = $160 \times 90 \times 24 = 345{,}600$ bits; one second at 30 fps = 10,368,000 bits.

THE ONE IDEA

③ **Total bits = (lines × lines·ratio) × 24 × fps × seconds.** "Interlaced" (two half-fields of 1/60 s) changes nothing: two fields make one frame and the question already gives frames per second.

BUILD IT

④ : 720 lines, 4:3, 24 bits, a 10-second clip at 25 fps.

1. Width = $720 \times 4/3 = 960$ pixels per line.
   Why: width/height = 4/3 and height = 720.
2. Per frame = $960 \times 720 \times 24 = 16{,}588{,}800$ bits.
   Why: pixels × bits per pixel.
3. Frames = $25 \times 10 = 250$.
   Why: fps × seconds.
4. Total = $16{,}588{,}800 \times 250 = 4{,}147{,}200{,}000$ bits.
   Why: every frame is stored in full (no compression in these questions).

CHECK

⑤ Hours become seconds: 2 h = 7,200 s, not 120. If the width isn't whole (1000 × 4/3), keep the fraction and say so. More: the guide's Chapter 2, Step 6 (vision-study.html).

EXAM PROBLEM

Your turn — Sheet 3 Q5. HDTV with 1125 horizontal lines, interlaced (two fields of 1/60 s each), aspect ratio 16:9. Pixels per line are in proportion to the vertical resolution, the proportion being the width-to-height ratio. Each pixel has 24 bits (8 per R, G, B). How many bits to store a 2-hour HDTV movie at 30 frames per second?

Q6. The same with 1000 lines, aspect ratio 4:3, and a 1-hour movie at 30 frames per second.

✍ Show the full exam answer: S2.3 (end of this lecture)

TRY ALONE

⑦ 1080 lines at 16:9, 24 bits, one minute at 30 fps: how many bits?

✍ Answers: S2.4 (end of this lecture)

📖 **READ & UNDERSTAND · ⏱ 3 min · Tier B**

5 · The rest of the lecture in five short facts

Key stages (L2 p19), in the slide's arrow order: acquisition → enhancement → restoration → morphological processing → segmentation → object recognition → representation & description, with colour processing and compression underneath, used wherever needed. One line each: **enhancement**

makes it look better (subjective); **restoration** undoes a known degradation such as noise or blur (objective, model-based); **segmentation** partitions the image into regions; **recognition** attaches a label to an object; **compression** shrinks storage.

Image sources (L2 p9–13): EM spectrum, acoustic, ultrasonic, electron beams, synthetic. Gamma rays: PET, where the isotope emits **positrons** that annihilate with electrons and the detectors catch the resulting **gamma rays**. X-rays: CT (CAT). Satellite imagery: GIS.

Simultaneous contrast (L2 p27): equal grey squares look darker on lighter backgrounds, so perceived brightness does not depend *simply* on intensity. Like the viral "what colour is the dress" photo: same pixels, different perception.

skimage code (L2 p41–61): `io.imread` gives a uint8 array 0–255 of shape rows × columns × 3; `img_as_float` rescales to 0.0–1.0 (÷255), `img_as_ubyte` back to 0–255; `img*0.3` on the float image darkens it; a 2-D array needs `plt.imshow(a, 'gray')` or it shows in false colour. Slicing is `img[rows, cols, channel]`: `img[:50, :, :] = [255, 0, 0]` paints the top 50 rows red, `img[:, :50, :]` is the left 50 columns.

All five with figures and tap-through code: the guide's Chapters 2–3 (vision-study.html).

RECALL DRILL · ⏱ 4 min · Tier A

6 · Sheet 1 in one pass

Cover the answers and say yours aloud; the questions are shortened, the options are the sheet's.

1. Digital image processing is: (A) computer algorithms on digital images (B) analog → digital (C) physical filters (D) printing.
2. The order of the key stages: options A–D (C follows acquisition → enhancement → restoration → morphology → segmentation → recognition → representation → compression → colour).
3. NOT an acquisition device: (A) cameras (B) range sensors (C) tomography (D) image enhancement techniques.
4. Purpose of enhancement: (A) produce images with sensors (B) improve visual quality (C) go to the frequency domain (D) operate on the Fourier transform.
5. Enhancement working directly on pixels: (A) spatial domain (B) transform (C) Fourier (D) frequency.
6. Works on the Fourier transform and back: (A) spatial (B) transform domain (C) frequency domain (D) pixel-based.
7. Goal of restoration: (A) enhance quality (B) remove noise from the image (C) segment (D) extract shape components.
8. Common causes of noise: (A) camera shake and low light (B) motion blur and mis-focus (C) corrupted files, pixelation (D) over/under-exposure.
9. Edge detection belongs to: (A) enhancement (B) restoration (C) morphology (D) segmentation.
10. Goal of segmentation: (C) partition the image into sets of pixels, or another option?
11. Local vs global segmentation: (A) local uses small windows on the image (B) global has fewer pixels (C) edges vs noise (D) large vs small images.
12. Primary goal of segmentation: (a) complicate (b) discard information (c) simplify/change the representation for easier analysis (d) complicate analysis.

13. Object recognition: (c) perceive an object's properties and attach semantic attributes, or another?
14. Representing an object by its boundary: (a) Huffman (b) run-length (c) lossy compression (d) global segmentation.
15. Lossy compression: (a) exact reconstruction (b) accepts losing some information (c) visible spectrum (d) enhances details.
16. What segmentation aims to achieve (same options as Q12).

✍️ Answers: S2.5 (end of this lecture)

⚠️ Traps

- L2 p10 says the isotope "emits positrons (Gamma rays)" and calls a bone scan a PET example. A positron is not a gamma ray: the detectors see the two gamma rays from positron–electron annihilation, and a bone scan is plain gamma-ray imaging, not PET.
- L2 p27 says perceived brightness "does not depend on its intensity". Correct version: it does not depend *simply* on intensity; the background changes it too.
- L2 p33 attaches dynamic range to "an image"; it is really a property of the imaging system (a ratio), while contrast is a difference within one image.
- Levels → bits is $\log_2 L$: 256 levels is 8 bits, not 256. RGB multiplies by 3; grey doesn't.
- MB in these slides = 2^{20} bytes (11.655 MB on L2 p40; the slide's last line rounds it to 11.66). Dividing by 10^6 gives 12.22.
- Transmission: each packet adds a start and a stop bit, so a byte costs 10 line bits (18 for a 16-bit packet). Sheet 3 Q4(b) is ambiguous: two 8-bit pixels per packet gives 2.36 s, one per packet 4.72 s.
- uint8 arithmetic wraps: in NumPy, uint8 250 + 10 = 4, not 260 or 255. Convert with `img_as_float` before doing maths.
- Sheet 1: Q14 has no correct option, Q6 has two defensible ones, Q8 mixes blur with noise, Q16 repeats Q12.

Solutions and answers — Lecture 2

S2.1 · 3 · How many bits, and how long to send them — Show the full exam answer

Every size question is the same chain: **pixels → bits → bytes → MB**. Every sending question is: **pixels → packets → bits on the line → seconds → minutes**. First write down what you are given, then do one multiplication or division per step.

Q1 · 4 MP colour camera, 512 levels. "Levels → bits" means: which power of 2 is it? $512 = 2^9$, so 9 bits per channel.

rows	cols	channels	bits per channel
1520	2680	3 (R, G, B)	9

Step 1 · count the pixels

2680 cols × 1520 rows = **4,073,600 pixels**.

Step 2 · bits in one pixel

9 bits × 3 channels = **27 bits per pixel**. Colour stores three numbers per pixel.

Step 3 · pixels → bits

4,073,600 pixels × 27 bits = **109,987,200 bits**.

Step 4 · bits → bytes

109,987,200 bits ÷ 8 = **13,748,400 bytes**. 8 bits make 1 byte.

Step 5 · bytes → MB

13,748,400 bytes ÷ 1,048,576 = **≈ 13.11 MB**. The slides use 1 MB = 2^{20} = 1,048,576 bytes.

Q2 · 2 MP grey camera, 128 levels. $128 = 2^7$, so 7 bits. Grey means one number per pixel.

rows	cols	channels	bits per channel
1080	1920	1 (grey)	7

Step 1 · count the pixels

1920 cols × 1080 rows = **2,073,600 pixels**.

Step 2 · pixels → bits

2,073,600 pixels × 7 bits = **14,515,200 bits** (1 channel, so nothing to multiply by 3).

Step 3 · bits → bytes

14,515,200 bits ÷ 8 = **1,814,400 bytes**.

Step 4 · bytes → MB

1,814,400 bytes ÷ 1,048,576 = **≈ 1.73 MB**.

Q3 · 1024 × 1024 grey image, 256 levels, sent in packets. $256 = 2^8$, so 8 bits per pixel: one pixel fills exactly the one data byte of a packet. A **packet** is what actually travels down the wire:

start: 1 bit	data: 8 bits (one pixel)	stop: 1 bit

So each pixel costs 1 + 8 + 1 = 10 bits on the line. **Baud** = bits per second the modem sends; 56K baud = 56,000 bits per second.

rows	cols	channels	bits per pixel	bits per packet
1024	1024	1 (grey)	8	10

Step 1 · count the pixels
1024 × 1024 = **1,048,576 pixels**.

Step 2 · pixels → packets
One pixel per packet, so **1,048,576 packets**.

Step 3 · packets → bits on the line
1,048,576 packets × 10 bits = **10,485,760 bits**.

Step 4 (a) · bits → seconds
10,485,760 bits ÷ 56,000 bits per second = **187.2 seconds**.

Step 5 (a) · seconds → minutes
187.2 s ÷ 60 = **≈ 3.12 minutes**.

Step 4 (b) · faster line
3000K baud = 3,000,000 bits per second. 10,485,760 bits ÷ 3,000,000 = **≈ 3.50 seconds**.

Q4 · packets now carry 2 bytes (16 bits) of data.

start: 1 bit	data: 16 bits	stop: 1 bit

So every packet is 1 + 16 + 1 = 18 bits on the line.

	rows	cols	channels	bits per pixel	bits per packet	baud
(a)	1024	1024	1	16 (65,536 = 2^{16})	18	70,000
(b)	1024	1024	1	8 (256 = 2^{8})	18	4,000,000

(a) Step 1 · pixels → packets
A 16-bit pixel fills one packet exactly: 1,048,576 pixels = **1,048,576 packets**.

(a) Step 2 · packets → bits on the line
1,048,576 packets × 18 bits = **18,874,368 bits**.

(a) Step 3 · bits → seconds
18,874,368 bits ÷ 70,000 bits per second = **269.6 seconds**.

(a) Step 4 · seconds → minutes
269.6 s ÷ 60 = **≈ 4.49 minutes**.

(b) Now a pixel is only 8 bits, and the packet has room for 16, so **two pixels share one packet**:

start: 1	pixel A: 8 bits	pixel B: 8 bits	stop: 1

(b) Step 1 · pixels → packets
1,048,576 pixels ÷ 2 = **524,288 packets**.

(b) Step 2 · packets → bits on the line
524,288 packets × 18 bits = **9,437,184 bits**.

(b) Step 3 · bits → seconds
9,437,184 bits ÷ 4,000,000 bits per second = **≈ 2.36 seconds**.

The question doesn't say whether two pixels share a packet. If you put one pixel per packet (half of each packet wasted), it is 1,048,576 packets × 18 bits ÷ 4,000,000 = 4.72 s. Write down which one you assumed.

What to write in the exam: For each: given table (rows, cols, channels, bits) → pixels → bits → bytes → MB (÷ 8, then ÷ 1,048,576). For sending: packet = start + data + stop bits; packets × bits per packet = bits on line; ÷ baud = seconds; ÷ 60 = minutes. Q4(b): state "2 pixels per packet".

Q1 109,987,200 bits ≈ 13.11 MB · Q2 14,515,200 bits ≈ 1.73 MB · Q3 (a) ≈ 3.12 min (b) ≈ 3.50 s · Q4 (a) ≈ 4.49 min (b) ≈ 2.36 s (4.72 s with one pixel per packet)

↩ back to the card

S2.2 · 3 · How many bits, and how long to send them — Answers

$64 = 2^6$, so 6 bits per pixel. Grey, so one channel.

rows	cols	channels	bits per channel
512	512	1 (grey)	6

Step 1 · count the pixels

512 × 512 = **262,144 pixels**.

Step 2 · pixels → bits

262,144 pixels × 6 bits = **1,572,864 bits**.

Step 3 · bits → bytes

1,572,864 bits ÷ 8 = **196,608 bytes**.

Step 4 · bytes → MB

196,608 bytes ÷ 1,048,576 = **0.1875 MB**.

What to write in the exam: 1,572,864 bits = 196,608 bytes = 0.1875 MB.

↩ back to the card

S2.3 · 4 · A movie is just frames × bits per frame — Show the full exam answer

A movie is a stack of still frames. So: **size of one frame** (same chain as Q1) × **how many frames**. "Lines" = rows. The aspect ratio width:height gives the columns: cols = rows × width ÷ height. "Interlaced" changes nothing here: two half-fields make one full frame, and you're already told 30 frames per second.

Q5 · HDTV, 1125 lines, 16:9, 2-hour movie.

rows	cols	channels	bits per channel	frames per second	length
1125	1125 × 16 ÷ 9 = 2000	3 (R, G, B)	8	30	2 hours

Step 1 · pixels in one frame

2000 cols × 1125 rows = **2,250,000 pixels**.

Step 2 · pixels → bits per frame

2,250,000 pixels × 24 bits (8 × 3 channels) = **54,000,000 bits per frame**.

Step 3 · hours → seconds

2 hours × 3600 = **7200 seconds**.

Step 4 · seconds → frames

7200 seconds × 30 frames per second = **216,000 frames**.

Step 5 · frames → bits

216,000 frames × 54,000,000 bits = 1.1664×10^{13} **bits**.

Q6 · 1000 lines, 4:3, 1-hour movie. Here 1000 × 4 ÷ 3 = 1333.33..., not a whole number. Keep it as the fraction 4000/3; it cancels later.

rows	cols	channels	bits per channel	frames per second	length
`1000`	`1000 × 4 ÷ 3 = 4000/3`	`3 (R, G, B)`	`8`	`30`	`1 hour`

Step 1 · pixels in one frame

4000/3 cols × 1000 rows = **4,000,000/3 ≈ 1,333,333 pixels**.

Step 2 · pixels → bits per frame

4,000,000/3 pixels × 24 bits = **32,000,000 bits per frame** (the 3 cancels: 24 ÷ 3 = 8).

Step 3 · hours → seconds → frames

1 hour = 3600 seconds; 3600 × 30 = **108,000 frames**.

Step 4 · frames → bits

108,000 frames × 32,000,000 bits = 3.456×10^{12} **bits**.

If you round to 1333 whole pixels per line you get ≈ 3.455×10^{12} bits. Either is fine; say which you used.

What to write in the exam: cols = lines × ratio; bits per frame = rows × cols × 24; frames = 30 × seconds; total = bits per frame × frames. Q5: 54,000,000 × 216,000 = 1.1664×10^{13} bits. Q6: 32,000,000 × 108,000 = 3.456×10^{12} bits.

Q5: 1.1664×10^{13} **bits · Q6:** 3.456×10^{12} **bits**

↩ back to the card

S2.4 · 4 · A movie is just frames × bits per frame — Answers

rows	cols	channels	bits per channel	frames per second	length
`1080`	`1080 × 16 ÷ 9 = 1920`	`3`	`8`	`30`	`60 seconds`

Step 1 · pixels in one frame

1920 × 1080 = **2,073,600 pixels**.

Step 2 · pixels → bits per frame

2,073,600 pixels × 24 bits = **49,766,400 bits per frame**.

Step 3 · seconds → frames

60 seconds × 30 = **1800 frames**.

Step 4 · frames → bits

1800 frames × 49,766,400 bits = **89,579,520,000 bits**.

What to write in the exam: 89,579,520,000 ≈ 8.96×10^{10} bits.

↩ back to the card

S2.5 · 6 · Sheet 1 in one pass — Answers

1. **A.** "Digital image processing" just means running computer programs on images that are already numbers.
2. **C.** It is the only option that follows the arrows on the slide (L2 p19): acquire, enhance, restore, morphology, segment, recognise, represent.
3. **D.** Cameras, range sensors and scanners *capture* pictures; enhancement only changes a picture you already have.
4. **B.** Enhancement exists to make the picture look better to a person.
5. **A.** "Spatial domain" means you change the pixel values themselves, $g(x, y) = T[f(x, y)]$.
6. **B or C.** You go to the Fourier transform, edit there, and come back; the lecture calls that the transform domain, and frequency domain is the same idea, so the question is flawed; write B and mention C.
7. **B.** Restoration undoes damage such as noise, while making it prettier (A) is enhancement.
8. **A** (intended). Low light makes a camera produce noise; camera shake actually makes blur, so the question is a bit loose.
9. **D.** Finding edges is how you find the borders between regions, which is segmentation's job.
10. **C.** Segmentation cuts the image into groups of pixels (regions), and that is exactly what C says.
11. **A.** Local methods look at one small window at a time, while global methods look at the whole image at once.
12. **c.** You segment so the image becomes simpler (a few regions instead of millions of pixels) and easier to analyse.
13. **c.** Recognition means looking at an object's properties and giving it a name ("car", "digit 7").
14. **None.** All four options are compression or segmentation words; describing an object by its outline really uses chain codes or polygons.
15. **b.** "Lossy" literally means you accept losing some information to make the file smaller.
16. **c.** It is the same question as Q12, so the same answer.

↩ back to the card

LECTURE 3 (PP. 1–19)

Pixel relationships: neighbours, adjacency, paths, regions, distances

Tier A · must ⏱ 35 min Slides: Lec 3 pp. 1–19 · Sheet 2 Q1–Q4 ↗ Szeliski Ch 3

Lecture 2 turned a picture into a grid of numbers. This part asks which numbers **belong together**: which pixels touch, whether you can walk from one to another through pixels of the right value, and how far apart two pixels are. Every later idea that talks about "an object" or "a region" (and the 3×3 neighbourhoods of filtering in Lecture 6) stands on these definitions.

What the exam can ask from this lecture

- Define N_4, N_D, N_8, and 4-, 8- and m-adjacency for a value set V.
- Decide whether two pixels, or two image subsets, are 4-, 8- or m-adjacent.
- Find (draw) the shortest 4-, 8- and m-paths between p and q and give their lengths, or explain why a path does not exist.
- Count the connected components (regions) of a binary image under 4-, 8- and m-adjacency.
- Compute the Euclidean, city-block (D_4) and chessboard (D_8) distances between two pixels.

⏳ **Only 20 minutes?** Do cards 1 and 2, then the drill (card 5).

📖 READ & UNDERSTAND · ⏱ 9 min · Tier A

1 · Who touches whom: neighbours, adjacency, paths, regions

Put your finger on one pixel p at row x, column y. Four pixels share an edge with it: $(x \pm 1, y)$ and $(x, y \pm 1)$. They are its **4-neighbours**, $N_4(p)$. Four more touch it only at a corner, $(x \pm 1, y \pm 1)$: the **diagonal neighbours** $N_D(p)$. Together they make the **8-neighbours**, $N_8(p) = N_4(p) \cup N_D(p)$ (L3 p2, p5). A pixel on the image border has fewer, because some of them fall outside.

	×	
×	p	×
	×	

×		×
	p	
×		×

×	×	×
×	p	×
×	×	×

Neighbours are about position only. **Adjacency** adds a condition on the values: pick a set V of allowed grey levels (for a binary image $V = \{1\}$; $V = \{2, 3\}$ means "only pixels whose value is 2 or 3 count"). Two pixels p, q *both with values in* V are (L3 p5):

- **4-adjacent** if $q \in N_4(p)$;
- **8-adjacent** if $q \in N_8(p)$;
- **m-adjacent** (mixed) if $q \in N_4(p)$, *or* $q \in N_D(p)$ and $N_4(p) \cap N_4(q)$ contains no pixel with a value in V.

That last line reads badly but works simply. A diagonal pair sits in a 2×2 square; the other two cells of that square are $N_4(p) \cap N_4(q)$, the **two corner pixels**. If either corner is in V, refuse the diagonal (you can already walk round through that corner). If both corners are outside V, the diagonal is allowed. Slide toy (L3 p6), $V = \{1\}$:

0	1	1
0	1	0
0	0	1

The top-right 1 and the centre 1 are diagonal; their corners are the 1 on the top row and the 0 on the right, and that 1 is in V, so under m the diagonal is refused (8-adjacency would allow it, giving two routes between the same pixels). The bottom-right 1 and the centre are diagonal with corners 0 and 0, so that diagonal is kept. **m-adjacency exists to remove the double routes that 8-adjacency creates.**

🧭 **Like a chess king (8) versus a rook moving one square (4).** m is a king that may only step diagonally when it can't go round by two straight steps. *Where it breaks:* a king may stand on any square; here every square on the route must hold a value in V.

A **path** from p to q is a chain of pixels, each adjacent to the next (4-, 8- or m-path depending on the adjacency used). Its **length** is the number of **steps**, i.e. pixels − 1: a path through 5 pixels has length 4. Two pixels are **connected** in a set S if a path inside S joins them, and a **connected component** (region) is a largest group of mutually connected pixels (L3 p11). The slide's 5×4 binary image has three 1-blocks touching only at corners: **3 regions** under 4-adjacency, **1** under 8. m always gives the same *regions* as 8 (it only drops a diagonal when the corner already connects the two pixels); it changes the *paths*.

Background reading: Gonzalez & Woods §2.5, or the interactive guide's Tutorial 4 (vision-study.html), which lights up every path and region in this section.

✏️ **PEN & PAPER · ⏱ 13 min · Tier A**

2 · Shortest 4-, 8- and m-paths, and when none exists

IN PLAIN WORDS

① Given an image, a value set V, a start pixel p and an end pixel q, find the fewest steps from p to q when every pixel on the way has a value in V and each step is a 4-, 8- or m-adjacency. Coordinates here are (row, column) counted from 1, top-left.

WARM-UP BY HAND

② $V = \{1\}$, p bottom-left, q top-right of $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. 4-path: up, right, up, right = **4**. 8-path: two diagonals through the centre = **2**. m-path: both diagonals have a 1 in a corner, so they are refused and m falls back to the 4-route = **4**.

The slides do the same on bigger grids. L3 p7–p9, $V = \{2, 3\}$: no 4-path, shortest 8-path 3 (straight up the diagonal), the only m-path has length 5. Figure 5 (L3 p10, $V = \{2, 3, 5\}$): no 4-path, 8-path 4, m-path 5.

THE ONE IDEA

③ Walk outward from p one step at a time and stop the first time you reach q; for every diagonal step, check its two corner pixels before you take it under m.
A 4-path is impossible as soon as p or q has no 4-neighbour in V.

BUILD IT

④ (the warm-up, m-path).

1. Check the ends: p's 4-neighbour above is a 1, so p is not stuck.
   Why: if an end has no neighbour in V, stop and write "no path".
2. Try the diagonal $p \rightarrow$ centre: corners are the 1 above p and the 0 to its right. A corner is in $V \rightarrow$ refused.
   Why: m rule from card 1.
3. Go round: $p \rightarrow (2,1) \rightarrow (2,2)$, then centre $\rightarrow q$ is refused the same way, so $(2,2) \rightarrow (1,2) \rightarrow (1,3)$.
   Why: the refused diagonal is replaced by its two straight steps.
4. Count steps: 4 (five pixels).
   Why: length = pixels − 1.

CHECK

⑤ Always 8-length ≤ m-length, and m-length ≤ 4-length whenever a 4-path exists: each extra rule can only make the route longer. The guide's Tutorial 4 (vision-study.html) has an explorer that runs these searches on any grid.

EXAM PROBLEM

Your turn — Sheet 2 Q2. Consider the image segment shown. (a) Let $V = \{0, 1\}$ and compute the lengths of the shortest 4-, 8- and m-path between p and q. If a particular path does not exist between these two points, explain why. (b) Repeat for $V = \{1, 2\}$.

3	1	2	1 (q)
2	2	0	2
1	2	1	1
(p) 1	0	1	2

Sheet 2 Q3. According to the following intensity values of a 5×5 image, show the paths by drawing dashed lines for 4-adjacency, 8-adjacency and m-adjacency. Let $V = \{1, 2, 3\}$. From p to q.

5	8	3	4	2 (q)
4	8	7	1	3
3	6	1	8	3
8	2	3	6	1
(p) 2	4	6	9	3

✍ Show the full exam answer: S3.1 (end of this lecture)

TRY ALONE

⑦ (a) A path visits 7 pixels; its length? (b) A diagonal pair's corners hold 0 and 4, $V = \{2, 4\}$: m-adjacent? (c) Could an m-path ever be shorter than the 8-path between the same pixels?

✍ Answers: S3.2 (end of this lecture)

✎ PEN & PAPER · ⏱ 7 min · Tier A

3 · Adjacent subsets and counting regions

IN PLAIN WORDS

① Two **subsets** S_1, S_2 of an image are adjacent if *some* pixel of S_1 is adjacent to *some* pixel of S_2; you only need to look along their shared border. Counting regions means grouping all pixels in V into connected components and counting the groups.

WARM-UP BY HAND

② $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, $V = \{1\}$. Under 4-adjacency the right-hand pair of 1s is one region and the three diagonal 1s are alone: **4 regions**. Under 8 (and m) the diagonals join everything: **1 region**.

THE ONE IDEA

③ **Edge contact joins under all three adjacencies; corner-only contact joins under 8 and m but never under 4.** So for regions, compare only 4 against 8; m always matches 8.

BUILD IT

④ on the same warm-up grid, split into subsets S_1 = columns 1–2, S_2 = columns 3–4.

1. 1s next to the border (columns 2 and 3): $(2, 2)$ in S_1, $(3, 3)$ in S_2.
   Why: only pixels next to the border can be adjacent across it.
2. $(2, 2)$ and $(3, 3)$ meet at a corner: not 4-adjacent, but 8-adjacent.
   Why: diagonal contact.
3. Corners $(2, 3) = 0$, $(3, 2) = 0$: both outside V, so also m-adjacent.
   Why: the m rule allows a diagonal with two empty corners.

CHECK

⑤ Across a shared border, 8-adjacent subsets are always m-adjacent too: a corner pixel in V would itself lie in S_1 or S_2 and be 4-adjacent across the border. The guide's Tutorial 4 (vision-study.html) colours each region of Q4 when you tap a pixel.

EXAM PROBLEM

Your turn — Sheet 2 Q1. Consider the two image subsets S_1 and S_2 in the figure. For $V = \{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent. (S_1 = rows 1–4, columns 2–5; S_2 = rows 1–4, columns 6–9; the dashed boxes on the sheet.)

0	0	0	0	0	0	0	1	1	0
1	0	0	1	0	0	1	0	0	1

1	0	0	1	0	1	1	0	0	0
0	0	1	1	1	0	0	0	0	0
0	0	1	1	1	0	0	1	1	1

Sheet 2 Q4. Find the regions in the binary matrix below, where $V = \{1\}$ [4-adjacency, 8-adjacency and m-adjacency].

0	0	1	1	1	1	0	0	0	0	0	0
0	1	0	0	0	0	1	0	0	0	0	0
1	0	1	0	0	1	0	1	0	0	0	1
1	0	0	0	0	0	0	1	0	0	1	1
1	0	1	0	0	1	0	1	0	1	1	1
1	0	0	1	1	0	0	1	0	1	1	1
0	1	0	0	0	0	1	0	1	1	1	1
0	0	1	1	1	1	0	0	0	1	0	1
0	0	0	0	0	0	0	0	0	1	0	0
1	1	1	1	1	0	0	0	1	0	0	0
1	0	1	0	1	0	0	0	1	0	0	0
1	0	1	0	1	0	0	0	0	0	0	0

✍ Show the full exam answer: S3.3 (end of this lecture)

TRY ALONE

⑦ (a) Two 1s touch only at a corner and both corners are 0: same 4-region? same m-region? (b) Can going from 4- to 8-adjacency ever increase the number of regions?

✍ Answers: S3.4 (end of this lecture)

📖 **READ & UNDERSTAND · ⏱ 3 min · Tier A**

4 · Three ways to measure distance between pixels

How far is $q = (s, t)$ from $p = (x, y)$? It depends on how you are allowed to move (L3 p13–p16):

name	formula	moves like	pixels with $D \le r$ form
Euclidean D_e	$\sqrt{(x-s)^2 + (y-t)^2}$	**a straight line**	**a disc**
city-block D_4	$\lvert x-s\rvert + \lvert y-t\rvert$	**a taxi on a street grid (4-steps)**	**a diamond (13 pixels for $r = 2$)**
chessboard D_8	$\max(\lvert x-s\rvert, \lvert y-t\rvert)$	**a chess king (8-steps)**	**a square (5×5 for $r = 2$)**

Sibling: $p = (2, 5)$, $q = (6, 2)$, so the row and column gaps are 4 and 3. $D_4 = 4 + 3 = 7$, $D_8 = \max(4, 3) = 4$, $D_e = \sqrt{16 + 9} = 5$. **Always** $D_8 \le D_e \le D_4$**.**

The slides' examples: L3 p17, $q = (1, 1)$, $p = (2, 2)$: $D_e = \sqrt{2} \approx 1.41$, $D_4 = 2$, $D_8 = 1$. L3 p19, $p = (4, 2)$, $q = (1, 4)$: $D_4 = \lvert 1-4\rvert + \lvert 2-4\rvert = 3 + 2 = 5$, $D_8 = \max(3, 2) = 3$, $D_e = \sqrt{9 + 4} = \sqrt{13} \approx 3.6$. These ignore pixel values; the length of a 4- or m-path (card 2) depends on them, so it can be longer than D_4 or not exist at all.

The guide's Tutorial 4 (vision-study.html) draws all three distance maps around a pixel.

🔁 RECALL DRILL · ⏱ 3 min · Tier A

5 · Quick recall

1. How many pixels are in $N_8(p)$ for an interior pixel? In $N_4(p)$ for a corner pixel of the image?
2. State the m-adjacency rule for a diagonal pair in one sentence.
3. A path passes through 9 pixels. Length?
4. $p = (3, 3)$, $q = (1, 6)$: D_4, D_8, D_e?
5. Why do 8- and m-adjacency always give the same number of regions?

✍️ Answers: S3.5 (end of this lecture)

⚠️ Traps

- Path length counts steps, not pixels: 5 pixels = length 4.
- Under m, check the two corner pixels of *every* diagonal step; a corner value in V refuses the diagonal (Sheet 2 Q2 and Q3 both have refused diagonals).
- "No 4-path" needs a reason: name the end pixel whose 4-neighbours are all outside V (Sheet 2 Q2(a): q; Q3: p).
- m changes paths, never regions: Sheet 2 Q4 gives 3 regions under both 8 and m.
- L3 p19 redraws the 4×4 example with p at $(4, 2)$ (row 4 printed as 2 2 0 1), whereas L3 p7 puts p at $(4, 1)$ (row 4 = 2 0 0 1). Use the coordinates each slide marks; the distances only depend on the coordinates.

Solutions and answers — Lecture 3 (pp. 1–19)

S3.1 · 2 · Shortest 4-, 8- and m-paths, and when none exists — Show the full exam answer

Q2 (a) · V = {0, 1}

Step 1 · what you are given

	c1	c2	c3	c4
r1	3	1	2	1_q
r2	2	2	0	2
r3	1	2	1	1
r4	1_p	0	1	2

Rows r1–r4 are counted from the top, columns c1–c4 from the left. p is bottom-left (r4,c1), q is top-right (r1,c4).

Step 2 · mark the walls

	c1	c2	c3	c4
r1	■	1	■	1_q
r2	■	■	0	■
r3	1	■	1	1
r4	1_p	0	1	■

V = {0, 1} means you may only stand on cells holding 0 or 1. Every 2 and the 3 becomes a wall (dark). Now it is a maze: walk from p to q on the white cells.

Step 3 · 4-path: only ↑ ↓ ← →

	c1	c2	c3	c4
r1	■	1	~~2~~	1_q
r2	■	■	0	~~2~~
r3	1	■	1	1
r4	1_p	0	1	■

q sits in the corner, so the last step must come from its left (r1,c3) or from below (r2,c4). Both are walls (crossed out). Nothing can step into q, so **no 4-path exists**.

Step 4 · 8-path: diagonals ↗ allowed too

	c1	c2	c3	c4
r1	■	1	■	1_q
r2	■	■	0	■
r3	1	■	1	1
r4	1_p	0	1	■

p → (r4,c2) ↗ (r3,c3) ↑ (r2,c3) ↗ q. Four arrows, so **length 4**. Three is impossible: the only 3-step route is ↗↗↗, and its first cell (r3,c2) is a wall.

Step 5 · m-path: check the two corners of every diagonal

	c2	c3
r3	■	1
r4	0	1

Diagonal (r4,c2) ↗ (r3,c3). Corner (r4,c3) holds 1, so it is open → **refused**. Go round: → then ↑.

	c3	c4
r1	■	1_q
r2	0	■

Diagonal (r2,c3) ↗ q. Both corners are walls → **allowed**.

The rule in words: a diagonal move is allowed only if both corner cells of its little 2×2 square are walls. If a corner is open, you must walk round through it.

Step 6 · the m-path

	c1	c2	c3	c4
r1	■	1	■	1_q
r2	■	■	0	■
r3	1	■	1	1
r4	1_p	0	1	■

p → → ↑ ↑ ↗ q. Five arrows, so **length 5**: one more than the 8-path, because one diagonal was refused.

What to write in the exam: No 4-path: q's only 4-neighbours (r1,c3) = 2 and (r2,c4) = 2 are not in V, so no step can enter q. Shortest 8-path = 4: (4,1)→(4,2)→(3,3)→(2,3)→(1,4). Shortest m-path = 5: (4,1)→(4,2)→(4,3)→(3,3)→(2,3)→(1,4); the diagonal (4,2)→(3,3) is refused because its corner (4,3) = 1 is in V.

Q2 (b) · V = {1, 2}

Step 1 · new walls

	c1	c2	c3	c4
r1	■	1	2	1_q
r2	2	2	■	2
r3	1	2	1	1
r4	1_p	■	1	2

Same image, new V. Now every 1 and 2 is open; the 3 at (r1,c1) and the two 0s at (r2,c3) and (r4,c2) are the walls.

Step 2 · 4-path

	c1	c2	c3	c4
r1	■	1	2	1_q
r2	2	2	■	2
r3	1	2	1	1
r4	1_p	■	1	2

q is 3 rows up and 3 columns right. A 4-step moves one row or one column, so you need at least 3 + 3 = 6 steps. p ↑ ↑ → ↑ → → q works: **length 6**.

Step 3 · 8-path

	c1	c2	c3	c4
r1	■	1	2	1_q
r2	2	2	■	2
r3	1	2	1	1
r4	1_p	■	1	2

p ↑ (r3,c1) ↗ (r2,c2) ↗ (r1,c3) → q: **length 4**. The straight ↗↗↗ is blocked by the wall at (r2,c3).

Step 4 · m: check both diagonals

	c1	c2
r2	~~2~~	2
r3	1	~~2~~

(r3,c1) ↗ (r2,c2): both corners are 2s, open → **refused**.

	c2	c3
r1	~~1~~	2
r2	2	■

(r2,c2) ↗ (r1,c3): corner (r1,c2) = 1 is open → **refused**. One open corner is enough.

Same rule: a diagonal is allowed only when both corners are walls.

Step 5 · the m-path

	c1	c2	c3	c4
r1	■	1	2	1_q
r2	2	2	■	2
r3	1	2	1	1
r4	1_p	■	1	2

Replace each refused diagonal by its two straight steps and you get the 4-path back: **length 6**.

What to write in the exam: 4-path = 6: (4,1)→(3,1)→(2,1)→(2,2)→(1,2)→(1,3)→(1,4). 8-path = 4: (4,1)→(3,1)→(2,2)→(1,3)→(1,4). m-path = 6: both diagonals of the 8-path have a corner in V, so they are refused and the path goes round, the same cells as the 4-path.

Q2 (a) $V = \{0, 1\}$: 4-path none (q has no 4-neighbour in V), 8-path 4, m-path 5. (b) $V = \{1, 2\}$: 4-path 6, 8-path 4, m-path 6.

Q3 · V = {1, 2, 3}

Step 1 · what you are given

	c1	c2	c3	c4	c5
r1	5	8	3	4	2_q
r2	4	8	7	1	3
r3	3	6	1	8	3
r4	8	2	3	6	1
r5	2_p	4	6	9	3

Rows from the top, columns from the left. p is bottom-left (r5,c1), q is top-right (r1,c5).

Step 2 · mark the walls

	c1	c2	c3	c4	c5
r1	■	■	3	■	2_q
r2	■	■	■	1	3
r3	3	■	1	■	3
r4	■	2	3	■	1
r5	2_p	■	■	■	3

Only 1, 2 and 3 are allowed. Every 4, 5, 6, 7, 8, 9 becomes a wall. Now it is a maze.

Step 3 · 4-path: only ↑ ↓ ← →

	c1	c2	c3	c4	c5
r1	■	■	3	■	2_q
r2	■	■	■	1	3
r3	3	■	1	■	3
r4	~~8~~	2	3	■	1
r5	2_p	~~4~~	■	■	3

p is in the corner, so its only 4-moves are ↑ to (r4,c1) and → to (r5,c2). Both are walls. p cannot take even one step, so **no 4-path exists**.

Step 4 · 8-path

	c1	c2	c3	c4	c5
r1	■	■	3	■	2_q
r2	■	■	■	1	3
r3	3	■	1	■	3
r4	■	2	3	■	1
r5	2_p	■	■	■	3

Straight up the diagonal: ↗ ↗ ↗ ↗, **length 4**. That is the least possible: q is 4 rows up and a step climbs at most one row.

Step 5 · m: check all four diagonals

	c1	c2
r4	■	2
r5	2_p	■

① both corners walls → allowed.

	c2	c3
r3	■	1
r4	2	~~3~~

② corner (r4,c3) = 3 is open → refused. Go → then ↑.

	c3	c4
r2	■	1
r3	1	■

③ both corners walls → allowed.

	c4	c5
r1	■	2_q
r2	1	~~3~~

④ corner (r2,c5) = 3 is open → refused. Go → then ↑.

A diagonal is allowed only if both corner cells of its 2×2 square are walls.

Step 6 · the m-path

	c1	c2	c3	c4	c5
r1	■	■	3	■	2_q
r2	■	■	■	1	3
r3	3	■	1	■	3
r4	■	2	3	■	1
r5	2_p	■	■	■	3

p ↗ → ↑ ↗ → ↑ q: six arrows, **length 6**. It is the only m-path in this grid.

What to write in the exam: Draw the dashed lines on the grid. 4-adjacency: no path, because p's 4-neighbours (4,1) = 8 and (5,2) = 4 are not in V. 8-adjacency: (5,1)→(4,2)→(3,3)→(2,4)→(1,5), length 4. m-adjacency: (5,1)→(4,2)→(4,3)→(3,3)→(2,4)→(2,5)→(1,5), length 6; the diagonals (4,2)→(3,3) and (2,4)→(1,5) are refused because their corners (4,3) and (2,5) are in V.

Q3: no 4-path (p has no 4-neighbour in V); shortest 8-path length 4 along the anti-diagonal; m-path length 6 (unique), detouring through $(4,3)$ and $(2,5)$.

↩ back to the card

S3.2 · 2 · Shortest 4-, 8- and m-paths, and when none exists — Answers

(a) 7 pixels

1	2	3	4	5	6	7

Count the gaps, not the cells: 7 cells in a row have 6 gaps between them, so 6 steps. **Length 6.**

(b) corners 0 and 4, V = {2, 4}

	q
p	4

The 0 is a wall, but the 4 is in V, so that corner is open. One open corner is enough → **not m-adjacent** (refused).

(c) m shorter than 8?

No. Every m-step is also an allowed 8-step, so every m-path is an 8-path too. The shortest 8-path can only be the same length or shorter.

What to write in the exam: (a) 6. (b) No: the corner 4 is in V, so the diagonal is refused. (c) No: the shortest 8-path is never longer than the shortest m-path.

↩ back to the card

S3.3 · 3 · Adjacent subsets and counting regions — Show the full exam answer

Q1 · V = {1}

Step 1 · what you are given

	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10
r1	0	0	0	0	0	0	0	1	1	0
r2	1	0	0	1	0	0	1	0	0	1
r3	1	0	0	1	0	1	1	0	0	0
r4	0	0	1	1	1	0	0	0	0	0
r5	0	0	1	1	1	0	0	1	1	1

Blue = S_1 (rows r1–r4, columns c2–c5). Amber = S_2 (rows r1–r4, columns c6–c9). Only the 1s count, because V = {1}.

Step 2 · only the border matters

	c4	c5	c6	c7
r1				
r2	1			1
r3	1		1	1
r4	1	1		
r5	1	1		

The two boxes meet along the line between c5 and c6. Zoom in there, 0s as walls. The only 1 of S_1 next to the line is (r4,c5); the only 1 of S_2 next to it is (r3,c6).

Step 3 · 4- and 8-adjacent?

	c5	c6
r3		1
r4	1	

(r4,c5) and (r3,c6) touch only at a corner, never along an edge. 4-adjacency needs an edge → **not 4-adjacent**. 8-adjacency also counts corners → **8-adjacent**.

Step 4 · m-adjacent?

	c5	c6
r3		1
r4	1	

Look at the other two cells of this 2×2 square: (r3,c5) = 0 and (r4,c6) = 0. Both are walls, so the diagonal is allowed → **m-adjacent**.

What to write in the exam: (a) Not 4-adjacent: the only 1s on the border, (4,5) in S1 and (3,6) in S2, touch only diagonally. (b) 8-adjacent: (3,6) is a diagonal neighbour of (4,5). (c) m-adjacent: their two shared 4-neighbours (3,5) and (4,6) are both 0, not in V.

Q1: (a) not 4-adjacent, (b) 8-adjacent, (c) m-adjacent.

Q4 · V = {1}

Step 1 · turn it into a maze

	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12
r1			1	1	1	1						
r2		1					1					
r3	1		1			1		1				1
r4	1							1			1	1
r5	1		1			1		1		1	1	1
r6	1			1	1			1		1	1	1
r7		1					1		1	1	1	1
r8			1	1	1	1				1		1
r9										1		
r10	1	1	1	1	1				1			
r11	1		1		1				1			
r12	1		1		1							

Every 0 is a wall (dark); the 55 white cells are the 1s. A region is a group of 1s you can walk between without leaving the 1s.

Step 2 · 4-adjacency: only ↑ ↓ ← →

	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12
r1	■	■	A	A	A	A	■	■	■	■	■	■
r2	■	B	■	■	■	■	C	■	■	■	■	■
r3	D	■	E	■	■	F	■	G	■	■	■	H
r4	D	■	■	■	■	■	■	G	■	■	H	H
r5	D	■	I	■	■	J	■	G	■	H	H	H
r6	D	■	■	K	K	■	■	G	■	H	H	H
r7	■	L	■	■	■	■	M	■	H	H	H	H
r8	■	■	N	N	N	N	■	■	■	H	■	H
r9	■	■	■	■	■	■	■	■	■	H	■	■
r10	O	O	O	O	O	■	■	■	P	■	■	■
r11	O	■	O	■	O	■	■	■	P	■	■	■
r12	O	■	O	■	O	■	■	■	■	■	■	■

Give each group of 1s joined by shared edges its own letter. You get **16 regions**, A to P. Many are single cells: the strokes of the ring touch each other only at corners, and 4-moves cannot cross a corner.

Step 3 · a corner contact

	c2	c3
r1	■	1
r2	1	■

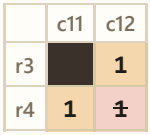
Top-left: (r1,c3) and (r2,c2) touch only at a corner. Under 4 that is no link (A and B are separate). Under 8 it is a link.

Step 4 · 8-adjacency: corners join too

	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	c11	c12
r1	■	■	A	A	A	A	■	■	■	■	■	■
r2	■	A	■	■	■	■	A	■	■	■	■	■
r3	A	■	A	■	■	A	■	A	■	■	■	A
r4	A	■	■	■	■	■	■	A	■	■	A	A
r5	A	■	B	■	■	B	■	A	■	A	A	A
r6	A	■	■	B	B	■	■	A	■	A	A	A
r7	■	A	■	■	■	■	A	■	A	A	A	A
r8	■	■	A	A	A	A	■	■	■	A	■	A
r9	■	■	■	■	■	■	■	■	■	A	■	■
r10	C	C	C	C	C	■	■	■	A	■	■	■
r11	C	■	C	■	C	■	■	■	A	■	■	■
r12	C	■	C	■	C	■	■	■	■	■	■	■

All those corner contacts now join up. The ring, the stroke in column c8 and the big blob on the right become one region A (40 cells). The small arc inside the ring is B (4 cells). The comb at the bottom-left is C (11 cells). **3 regions**.

Step 5 · m-adjacency: same regions



m refuses the diagonal (r3,c12) ↙ (r4,c11) because its corner (r4,c12) is a 1. But you can still walk round through that 1. That is always the case: m only refuses a diagonal when a corner 1 already links the two cells, so the regions never change. **3 regions**.

What to write in the exam: 4-adjacency: 16 regions. 8-adjacency: 3 regions (40, 4 and 11 pixels). m-adjacency: the same 3 regions as 8, because m only drops a diagonal when the two pixels are already joined through a corner pixel in V.

Q4: 4-adjacency 16 regions; 8-adjacency 3 regions (40, 4 and 11 pixels); m-adjacency 3 regions (same as 8).

↩ back to the card

S3.4 · 3 · Adjacent subsets and counting regions — Answers

(a) two 1s touching at a corner



They share no edge, so a 4-move cannot get from one to the other: **different 4-regions**. 8 counts corners: **same 8-region**. Both corners are walls (0), so m allows the diagonal: **same m-region**.

(b) 4 → 8: more regions?

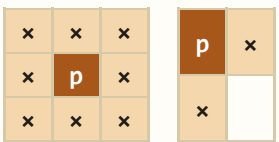
No. 8 keeps every 4-link and adds the diagonal ones. Adding links can only glue regions together, never split one.

What to write in the exam: (a) Different 4-regions; the same 8- and m-region. (b) No: 8-adjacency only adds connections, so regions can merge but never split.

↩ back to the card

S3.5 · 5 · Quick recall — Answers

1 · how many neighbours



Inside the image: all 8 cells around p → **8**. At the top-left corner of the image, p has only a right and a lower edge-neighbour → N_4 has **2**.

2 · the m rule



A diagonal pair p, q is m-adjacent only if **both** corner cells c of their 2×2 square are outside V (walls).

3 · 9 pixels

Length counts steps = pixels − 1. 9 pixels → **8**.

4 · p = (3,3), q = (1,6)

2 rows	3 cols

Row gap 2, column gap 3. $D_4 = 2 + 3 =$ **5**. D_8 = the bigger gap = **3**. $D_e = \sqrt{(4 + 9)} = \sqrt{13} \approx$ **3.61**.

5 · why 8 and m give the same regions

m only refuses a diagonal when a corner cell is in V, and then you can walk round through that corner. The two pixels stay connected, so no region splits.

What to write in the exam: 1) 8; 2. 2) Diagonal allowed only if neither shared corner pixel is in V. 3) 8. 4) D4 = 5, D8 = 3, De = √13 ≈ 3.61. 5) m drops a diagonal only when a corner pixel in V already links the two pixels.

↩ back to the card

LECTURE 3 (PP. 20–42)

Interpolation and operations on whole images

Tier A · must ⏱ 25 min Slides: Lec 3 pp. 20–42 · Sheet 4 Q1–Q3 ↗ Szeliski Ch 3

The first half of Lecture 3 was about single pixels and their neighbours. The second half does two jobs with whole images: **resizing** them (inventing values at positions that never existed) and **combining** them (adding, multiplying, masking, testing whether an operator is linear). Sheet 4 Q1–Q3 come straight from the second job, and each takes a few minutes once the idea is clear.

What the exam can ask from this part

- Zoom a small image by nearest neighbour (pixel replication) or bilinear interpolation; shrink by row/column deletion; bicubic = 16 neighbours.
- Array product vs matrix product of two images.
- Prove an operator is linear or not (sum: linear; max, gamma: not), with a counter-example.
- Image arithmetic: the 0–510 and −255–255 ranges, clipping; adding noise; ROI masks (Sheet 4 Q2–Q3).
- Set and logical operations on binary images; point vs local vs global operations.

⌛ **Only 20 minutes?** Do cards 3, 4 and 5 (the three sheet questions), then card 2.

📖 READ & UNDERSTAND · ⏱ 5 min · Tier A

1 · Zooming means inventing pixels: copy the nearest, or blend the four nearest

How do you make a 2×2 image into a 4×4 one when only four values exist? You estimate the rest. That is **interpolation**: using known pixel values to estimate values where there is no pixel yet (L3 p20). Zooming has two steps: create the new grid, then give each new pixel a grey level (L3 p21).

Nearest neighbour copies the closest original pixel, so a 2× zoom turns each pixel into a 2×2 block (**pixel replication**). The slide's example (L3 p24):

50	100
180	250

50	50	100	100
50	50	100	100
180	180	250	250
180	180	250	250

It is fast but blocky: the slides call this the **checkerboard effect** (L3 p22). **Shrinking** by a whole factor runs it backwards: delete rows and columns instead of copying them (L3 p25).

Bilinear blends the 4 nearest pixels with $v(x, y) = ax + by + cxy + d$ (L3 p26), x = row, y = column; the four neighbours give four equations for a, b, c, d. On the same image (made for this guide; the slides have no numbers), corners $(0,0) = 50$, $(0,1) = 100$, $(1,0) = 180$, $(1,1) = 250$:

1. $d = 50$; $b + d = 100 \Rightarrow b = 50$; $a + d = 180 \Rightarrow a = 130$; $a + b + c + d = 250 \Rightarrow c = 20$.
2. Centre $(0.5, 0.5)$: $130(0.5) + 50(0.5) + 20(0.25) + 50 = 145$, the average of the four corners.
3. Shortcut: interpolate along the top and bottom edges, then between them. Zoom to 3×3: $\begin{bmatrix} 50 & 75 & 100 \\ 115 & 145 & 175 \\ 180 & 215 & 250 \end{bmatrix}$.

4. New in-between values (75, 145, …) make it smoother than nearest neighbour.

✍ Show both zooms as pictures: S3b.1 (end of this lecture)

Bicubic (L3 p28) uses the 16 nearest pixels and $v = \sum_{i=0}^{3}\sum_{j=0}^{3} a_{ij}x^i y^j$: 16 coefficients, smoothest and slowest, the photo-editor default.

🧭 **Like zooming into an old phone photo.** A cheap viewer makes each pixel a bigger square (nearest neighbour); a better one fades smoothly between neighbouring colours (bilinear, bicubic). *Where it breaks:* neither adds real detail; both only guess from what is already there. More: Wikipedia: Bilinear interpolation.

📖 READ & UNDERSTAND · ⏱ 6 min · Tier A

2 · Combining images: pixel by pixel, inside 0–255

"Multiply two images" means the **array product**: each pixel times the pixel at the same place (L3 p29), not the matrix product. $\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix} \odot \begin{bmatrix}5 & 6\\7 & 8\end{bmatrix} = \begin{bmatrix}5 & 12\\21 & 32\end{bmatrix}$, whereas the matrix product is $\begin{bmatrix}19 & 22\\43 & 50\end{bmatrix}$.

An **operator** H turns an image f into $g = H[f]$. It is **linear** if $H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$ for all same-size images and all constants (L3 p30). The sum of pixels is linear (L3 p31). The max is not (L3 p32): with $f_1 = \begin{bmatrix}0 & 2\\2 & 3\end{bmatrix}$, $f_2 = \begin{bmatrix}6 & 5\\4 & 7\end{bmatrix}$, $a_1 = 1$, $a_2 = -1$, the left side is $\max \begin{bmatrix}-6 & -3\\-2 & -4\end{bmatrix} = -2$ and the right is $3 - 7 = -4$. One counter-example settles it.

Arithmetic ($+, -, \times, \div$) works pixel by pixel on images of the same size (L3 p33). Two 8-bit images sum to 0…510 and differ by −255…255, so the slides **clip**: negatives → 0, above 255 → 255 (L3 p35). Example: $\begin{bmatrix}200 & 50\\30 & 120\end{bmatrix} - \begin{bmatrix}100 & 90\\30 & 10\end{bmatrix} = \begin{bmatrix}100 & -40\\0 & 110\end{bmatrix} \to \begin{bmatrix}100 & 0\\0 & 110\end{bmatrix}$. (Gonzalez instead scales $f - \min$ to 0–255, giving $\begin{bmatrix}238 & 0\\68 & 255\end{bmatrix}$; it keeps −40 and 0 apart.) A **mask** of 1s inside a **region of interest (ROI)** and 0s elsewhere, multiplied in, keeps only the ROI (L3 p34).

A binary image can be read as the **set** of its 1-pixels, so set operations become logical ones: union $A \cup B$ = OR, intersection $A \cap B$ = AND, complement A^c = NOT, difference $A - B = A \cap B^c$ = A AND NOT B (L3 p36–37). **Slide error (L3 p36):** "elements belonging to either A, B or both" is printed next to $D = A \cap B$; that sentence describes the **union**. The intersection is the elements in **both**.

Finally, classify an operation by how much input one output pixel sees (L3 p38): **point** (only its own pixel: negative, threshold, gamma), **local** (a neighbourhood: 3×3 averaging), **global** (the whole image: the image mean, the equalization mapping). The Lec 3 code (L3 p39–42) does all three kinds of step at once: threshold the grey rabbit at 0.5 (point), NOT it by swapping 0 and 1, then `img = BWimageMask * BWimage` (an array product with a mask).

🧭 **Like a stencil over a spray-painted wall.** Where the stencil is open (1) the paint shows, where it is solid (0) the wall stays bare: that is the ROI mask. *Where it breaks:* a real stencil is all-or-nothing; a mask could hold 0.5 and dim a region instead. More: Szeliski Ch 3.1.

✏ PEN & PAPER · ⏱ 6 min · Tier A

3 · Is gamma a linear operator? (Sheet 4 Q1)

IN PLAIN WORDS

① The gamma transformation $s = c\,r^{\gamma}$ raises each pixel value r to the power γ and multiplies by c. Linear means $H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$ for **every** pair of images and constants (card 2).

WARM-UP BY HAND

② $s = r^2$, one-pixel images $f_1 = f_2 = 1$, $a_1 = a_2 = 1$: mix then square, $(1+1)^2 = 4$; square then mix, $1^2 + 1^2 = 2$. Four is not two.

THE ONE IDEA

③ **One counter-example is a full proof of "not linear"**, because linearity must hold for every input.

BUILD IT

④ 1. Left side: $c\,(a_1 f_1 + a_2 f_2)^{\gamma}$.

Why: apply H to the mixed image.

2. Right side: $a_1 c f_1^{\gamma} + a_2 c f_2^{\gamma}$.

Why: apply H to each, then mix.

3. Pick numbers that make them differ.

Why: a power doesn't spread over a sum, $(x+y)^2 \neq x^2 + y^2$.

CHECK

⑤ With $\gamma = 1$ both sides are $c\,(a_1 f_1 + a_2 f_2)$: equal. So gamma is linear only when $\gamma = 1$; say so in the answer.

EXAM PROBLEM

Your turn — Sheet 4 Q1. The gamma transformation is $s = c\,r^{\gamma}$, with constants c and γ. Is it a linear operator? Prove your answer.

✍ Show the full exam answer: S3b.2 (end of this lecture)

TRY ALONE

⑦ Is the negative $s = 255 - r$ linear?

✍ Answers: S3b.3 (end of this lecture)

🖍 PEN & PAPER · ⏱ 4 min · Tier A

4 · Making a noisy image (Sheet 4 Q2)

IN PLAIN WORDS

① A "corrupted" image is the clean image with noise added on top, pixel by pixel: $c(x,y) = b(x,y) + a(x,y)$, both images the same size (L3 p33).

WARM-UP BY HAND

② Clean $\begin{bmatrix} 200 & 0 \\ 0 & 60 \end{bmatrix}$ plus noise $\begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix} = \begin{bmatrix} 210 & 20 \\ 30 & 100 \end{bmatrix}$.

THE ONE IDEA

③ **Additive noise is an array sum**, kept inside 0–255.

BUILD IT

④ 1. Check the sizes match.

Why: pixel (x, y) pairs with pixel (x, y).

2. Add each pair. 3. Clip anything above 255 to 255 (L3 p35).

Why: an 8-bit pixel can't store 256 or more.

CHECK

⑤ The largest possible sum is $255 + 255 = 510$, so scan for values above 255 before you finish.

EXAM PROBLEM

Your turn — Sheet 4 Q2. Generate a corrupted image c from the noisy image a and the noiseless image b, and explain the process.

a		
250	0	0
250	250	0
0	0	0

b		
1	2	3
4	5	6
7	8	9

✍ Show the full exam answer: S3b.4 (end of this lecture)

TRY ALONE

⑦ Add noise $\begin{bmatrix} 250 & 0 \\ 0 & 0 \end{bmatrix}$ to $\begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix}$. What is the top-left pixel?

✍ Answers: S3b.5 (end of this lecture)

✍ PEN & PAPER · ⏱ 4 min · Tier A

5 · Keeping a region of interest (Sheet 4 Q3)

IN PLAIN WORDS

① A **mask** is an image of 1s and 0s the same size as yours. The array product keeps every pixel under a 1 and blacks out every pixel under a 0; what survives is the **ROI** (L3 p34).

WARM-UP BY HAND

② $\begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 0 \\ 6 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix}$: the top-left 2×2 survives.

THE ONE IDEA

③ **Times 1 keeps, times 0 erases**, using the array product, never the matrix product.

BUILD IT

④ 1. Line the mask up on the image. 2. Multiply each pair.

Why: the mask's 1s mark the ROI (the dental X-ray on L3 p34).

CHECK

⑤ Count the 1s in the mask: that is the most non-zero pixels the result can have.

EXAM PROBLEM

Your turn — Sheet 4 Q3. Generate the ROI c from image a and mask b; describe the method.

a			
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

b			
0	0	0	0
0	1	1	0
0	1	1	0
0	0	0	0

✍ Show the full exam answer: S3b.6 (end of this lecture)

TRY ALONE

⑦ Which mask would keep only the four corners of a?

✍ Answers: S3b.7 (end of this lecture)

⚠ Traps

- "Not linear" needs one counter-example; "linear" needs the general algebra. Don't prove linearity with a single example.
- Multiplying two images (noise, masks) means the array product, never the matrix product.
- Arithmetic results outside 0–255 must be clipped (the slides) or scaled (Gonzalez) before you call the result an 8-bit image.
- L3 p36 labels "in A, B or both" as $A \cap B$: that is the union $A \cup B$; the intersection is "in both".
- L3 p21 says "two techniques" for zooming, then lists three (nearest, bilinear, bicubic).
- Bilinear's x is the row and y the column here, so $(1, 0)$ is the bottom-left pixel (180), not the top-right.

Solutions and answers — Lecture 3 (pp. 20–42)

S3b.1 · 1 · Zooming means inventing pixels: copy the nearest, or blend the four nearest — Show both zooms as pictures

Nearest neighbour, 2×2 → 4×4. Each input pixel gets its own colour. Rows counted from the top, columns from the left.

Step 1 · the input (2×2)

	c1	c2
r1	50	100
r2	180	250

Four pixels, four colours.

Step 2 · the output (4×4)

	c1	c2	c3	c4
r1	50	50	100	100
r2	50	50	100	100
r3	180	180	250	250
r4	180	180	250	250

Every input pixel is simply copied into a 2×2 block of the same colour. No new values appear, which is why it looks blocky.

Bilinear, 2×2 → 3×3. The rule in words: a new pixel halfway between two known pixels gets their average. Do it in two steps: first along the rows, then down the columns.

Step 1 · place the 4 known corners

	c1	c2	c3
r1	50		100
r2			
r3	180		250

The original pixels go to the corners; the blank cells are new.

Step 2 · along the top and bottom rows

	c1	c2	c3
r1	50	75	100
r2			
r3	180	215	250

Top: $(50 + 100)/2 = 75$. Bottom: $(180 + 250)/2 = 215$.

Step 3 · down each column

	c1	c2	c3
r1	50	75	100
r2	115	145	175
r3	180	215	250

Middle row = average of the cell above and below: $(50 + 180)/2 = 115$, $(75 + 215)/2 = 145$, $(100 + 250)/2 = 175$.

What to write in the exam: Nearest neighbour: each pixel copied into a 2×2 block (4×4 grid above). Bilinear: $v = ax + by + cxy + d$ with $a = 130$, $b = 50$, $c = 20$, $d = 50$; 3×3 result rows 50 75 100 / 115 145 175 / 180 215 250.

↩ back to the card

S3b.2 · 3 · Is gamma a linear operator? (Sheet 4 Q1) — Show the full exam answer

The rule in words: an operator is linear if "mix first, then apply it" always gives the same answer as "apply it to each, then mix". If you find even one case where the two orders disagree, it is not linear.

What you're given: $H[f] = c\,f^{\gamma}$ on every pixel. Pick small numbers: $c = 1$, $\gamma = 2$ (so H = "square it"), two one-pixel images $f_1 = 2$ and $f_2 = 3$, mix weights $a_1 = a_2 = 1$ (plain adding).

Way A · mix first, then apply

mix	2 + 3 = 5
square	$5^2 = 25$

Add the images, then square the result.

Way B · apply to each, then mix

square	$2^2 = 4$, $3^2 = 9$
mix	4 + 9 = 13

Square each image first, then add.

$25 \neq 13$: the two orders disagree, so the rule fails. The reason: a square doesn't spread over a sum, $(2 + 3)^2 \neq 2^2 + 3^2$. Scaling fails too: $H[3 \cdot 2] = 6^2 = 36$ but $3 \cdot H[2] = 3 \cdot 4 = 12$.

Only when $\gamma = 1$ ($s = c\,r$, a plain multiply) do the two ways always agree.

What to write in the exam: Linear needs $H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$ for all inputs. Counter-example $c = 1, \gamma = 2, a_1 = a_2 = 1, f_1 = 2, f_2 = 3$: left $(2 + 3)^2 = 25$, right $2^2 + 3^2 = 13$, $25 \neq 13$, so gamma is not linear (except $\gamma = 1$).

The gamma transformation is not linear (for $\gamma \neq 1$); it is linear only in the special case $\gamma = 1$, $s = c\,r$.

↩ back to the card

S3b.3 · 3 · Is gamma a linear operator? (Sheet 4 Q1) — Answers

Same test: two one-pixel images $f_1 = 0$, $f_2 = 0$, plain adding ($a_1 = a_2 = 1$).

Way A · mix first, then apply

mix	0 + 0 = 0
negative	255 − 0 = 255

Way B · apply to each, then mix

negative	255 − 0 = 255, 255 − 0 = 255
mix	255 + 255 = 510

$255 \neq 510$, so it is not linear. The fixed "255" gets counted once in Way A but twice in Way B.

What to write in the exam: Not linear: $f_1 = f_2 = 0$ gives left 255, right 510.

↩ back to the card

S3b.4 · 4 · Making a noisy image (Sheet 4 Q2) — Show the full exam answer

The rule in words: corrupted image = clean image + noise, added **cell by cell**: each cell of b plus the cell in the same place in a. Anything above 255 becomes 255.

Step 1 · clean b + noise a = c

b (clean)				a (noise)				c (corrupted)		
1	2	3	+	250	0	0	=	251	2	3
4	5	6		250	250	0		254	255	6
7	8	9		0	0	0		7	8	9

Blue = the noisy cells; dark = where the result changed. Where the noise is 0 the clean value just passes through.

Step 2 · one cell in full

Top-left: $b = 1$, $a = 250$, so $c = 1 + 250 = 251$. Middle: $5 + 250 = 255$. The other cells the same way.

Step 3 · check for clipping

The biggest value is 255, which still fits in 8 bits, so nothing needs clipping.

What to write in the exam: $c(x, y) = b(x, y) + a(x, y)$, element by element (same size, 3×3), clip above 255. Result below; no value exceeds 255.

$$c = \begin{bmatrix} 251 & 2 & 3 \\ 254 & 255 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

↩ back to the card

S3b.5 · 4 · Making a noisy image (Sheet 4 Q2) — Answers

10	5	+	250	0	=	~~260~~	5	→	255	5
5	5		0	0		5	5		5	5

Top-left: $10 + 250 = 260$. An 8-bit pixel can't hold more than 255, so 260 is crossed out and clipped to 255.

What to write in the exam: $10 + 250 = 260 > 255 \rightarrow$ clipped to **255**.

↩ back to the card

S3b.6 · 5 · Keeping a region of interest (Sheet 4 Q3) — Show the full exam answer

The rule in words: multiply **cell by cell** (each cell of a times the cell in the same place in b). Anything × 1 stays the same; anything × 0 becomes 0.

Step 1 · image a × mask b = result c

a (image)			
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

×

b (mask)			
	1	1	
	1	1	

=

c (ROI)			
0	0	0	0
0	6	7	0
0	10	11	0
0	0	0	0

Amber = the cells the mask keeps (its 1s). Dark = the mask's 0s, which black out everything under them.

Step 2 · two cells in full

(r2,c2): $6 \times 1 = 6$, kept. (r1,c1): $1 \times 0 = 0$, erased. The other cells the same way.

Step 3 · check

The mask has four 1s, and the result has four non-zero cells. Good.

What to write in the exam: Array (element-wise) multiplication $c(x, y) = a(x, y) \cdot b(x, y)$, not matrix multiplication. Mask = 1 inside the ROI (central 2×2), 0 outside, so ROI pixels are kept and the rest become 0.

$$c = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 6 & 7 & 0 \\ 0 & 10 & 11 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

↩ back to the card

S3b.7 · 5 · Keeping a region of interest (Sheet 4 Q3) — Answers

Put a 1 wherever you want to keep a pixel, a 0 everywhere else.

a			
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

×

mask			
1			1
1			1

=

result			
1	0	0	4
0	0	0	0
0	0	0	0
13	0	0	16

What to write in the exam: Mask with 1s at the four corners (r1,c1), (r1,c4), (r4,c1), (r4,c4), 0 elsewhere; the result keeps 1, 4, 13, 16.

↩ back to the card

LECTURE 4

Image enhancement: point transforms and histograms

Tier A · must ⏱ 35 min Slides: Lec 4-Image Enhancement.pdf (38 p) · Sheet 4 Q4–Q6 are in Lecture 5 ↗ Szeliski Ch 3

Enhancement means making a picture more useful: pulling detail out of the shadows, boosting contrast, or making it easier on the eye (L4 p3). This lecture does it the simplest way, one pixel at a time: each grey level r goes through a fixed curve and comes out as s. The histogram at the end is the bridge to Lecture 5, where the curve is read off the image itself.

What the exam can ask from this lecture

- Write and apply the negative, log and gamma transforms; say what each does to dark and bright values (L4 pp. 8–16).
- Min-max contrast stretching of a small image to 0–255, and the piecewise function's special cases (L4 pp. 18–21).
- Intensity-level slicing, both versions, on a small image (L4 p25).
- Histogram equalization by table on the 4 × 8 slide image (L4 pp. 31–32).

⏳ **Only 20 minutes?** Do cards 2 and 4, in that order.

📖 READ & UNDERSTAND · ⏱ 7 min · Tier A

1 · Every pixel through the same curve

The slides frame enhancement as $g(x, y) = T[f(x, y)]$ (L4 p4): f is the input grey value at position (x, y), g the output, and T an **operator** that looks at a **neighbourhood** (a small patch of pixels around (x, y)). Shrink the neighbourhood to the pixel alone and T becomes a curve $s = T(r)$ from input level r to output level s: a **point operation**. (The slides also mention working in a **transform domain**, e.g. Fourier: transform, edit, transform back, L4 pp. 5–6.)

🧭 **Like** a phone photo's brightness slider: every pixel is looked up in the same table, so two pixels with the same grey always end up the same. *Where it breaks:* blur and sharpen need the neighbours too, so they are not point operations (Lecture 6).

Negative (p9): $s = L - 1 - r$, where L is the number of grey levels (256 for 8-bit). In 8-bit, $40 \to 215$; in a 4-bit image ($L = 16$), $3 \to 12$. It helps **white or grey detail buried in large black regions**.

Log (p11): $s = c \log(1 + r)$ spreads a **narrow band of dark inputs over a wide band of outputs** and squeezes the brights. With $c = 255 / \ln 256 = 45.99$, input $10 \to 45.99 \ln 11 = 110.27 \to 110$. Used on Fourier spectra, whose values span millions (p12).

Gamma (p14): $s = c\, r^{\gamma}$ on intensities scaled to 0–1. $\gamma < 1$ **brightens the darks**, $\gamma > 1$ darkens. With $\gamma = 0.4$, $64 \to 255\,(64/255)^{0.4} = 146.69 \to 147$; with $\gamma = 2.5$, $64 \to 8$.

Slide code (L4 pp. 13, 29). `np.log(1 + img)` on a uint8 image wraps 255 + 1 to 0, and $\log 0 = -\infty$; convert first with `np.log1p(img.astype(float))`. The code also reads `my_image` but transforms an undefined `img_gray`.

↗ Guide Tutorial 7, steps 1–3 ↗ Szeliski §3.1

🖍 PEN & PAPER · ⏱ 9 min · Tier A

2 · Min-max stretching: the dullest pixel to 0, the brightest to 255

IN PLAIN WORDS

① A low-contrast image uses only part of the grey scale, say 60 to 150. **Contrast stretching** (the slides also call it **normalization**, p19) maps that part onto the whole range with a straight line (L4 p21): $I_{new} = (I - Min)\,\dfrac{NewMax - NewMin}{Max - Min} + NewMin$. I is the input pixel, Min/Max the smallest and largest input values, $NewMin/NewMax$ the target range (0 and 255). Round to the nearest integer, halves up.

WARM-UP BY HAND

② Image $\begin{bmatrix}10 & 30\\ 20 & 50\end{bmatrix}$: $Min = 10$, $Max = 50$, factor $255/40 = 6.375$. Then $10 \to 0$, $20 \to 63.75 \to 64$, $30 \to 127.5 \to 128$, $50 \to 255$. Output $\begin{bmatrix}0 & 128\\ 64 & 255\end{bmatrix}$.

THE ONE IDEA

③ **Subtract Min first, then multiply by one fixed factor.** The order of pixels never changes; only the gaps between them grow. Lecture 4's general version (p18) is a three-piece line through (r_1, s_1) and (r_2, s_2) with $r_1 \le r_2$, $s_1 \le s_2$: if $r_1 = s_1$ and $r_2 = s_2$ it is the identity (no change); if $r_1 = r_2$, $s_1 = 0$, $s_2 = L - 1$ it is **thresholding**, which outputs a binary image.

BUILD IT

④ 1. Find Min, Max.

Why: they pin the line's two ends.

2. Factor $= 255/(Max - Min)$.

Why: $Max - Min$ input steps must cover 255 output steps.

3. $(I - Min)\times$ factor for each pixel.

Why: the darkest pixel must land on 0.

4. Round, halves up.

Why: the slide rounds 124.67 to 125, not down to 124.

CHECK

⑤ Min must give 0 and Max must give 255. A value outside 0–255 means you subtracted the wrong end.

EXAM PROBLEM

Your turn — Lec 4 p21. An 8-bit image has range [60, 150]. Stretch it to [0, 255]:

100	120	125
104	110	150
60	130	140

✍ Show the full exam answer: S4.1 (end of this lecture)

TRY ALONE

⑦ (a) Stretch 130 from [60, 150] to [0, 100]. (b) What does stretching do to an image that already has a 0 and a 255?

✍ Answers: S4.2 (end of this lecture)

Slide wording (L4 p7). It describes the S-shaped curve (darks pushed darker, brights lighter: contrast stretching) and then calls it a thresholding function. The S-curve is contrast stretching; the step function beside it is thresholding.

↗ Guide Tutorial 7, step 4 ↗ Szeliski §3.1

🖍 PEN & PAPER · ⏱ 6 min · Tier A

3 · Slicing: light up one band of greys

IN PLAIN WORDS

① **Intensity-level slicing** highlights every pixel whose level lies in a chosen band $[A, B]$ by setting it to 255. Version (a) **preserves the background**: pixels outside the band keep their value. Version (b) is **binary**: pixels outside become 0.

WARM-UP BY HAND

② Band 20–40 on $\begin{bmatrix} 10 & 25 \\ 40 & 60 \end{bmatrix}$: (a) $\begin{bmatrix} 10 & 255 \\ 255 & 60 \end{bmatrix}$, (b) $\begin{bmatrix} 0 & 255 \\ 255 & 0 \end{bmatrix}$.

THE ONE IDEA

③ **Each pixel asks one yes/no question: am I inside the band?** Yes → 255 in both versions; no → keep (a) or 0 (b). Decide the boundary values (is B itself inside?) before you start, and write it down.

BUILD IT

④ 1. Mark every pixel inside the band.

Why: those are the only ones that become 255.

2. (a) copy the rest unchanged; (b) set the rest to 0.

Why: that is the only difference between the versions.

CHECK

⑤ Version (b) contains only 0 and 255, and its 255s sit exactly where version (a)'s do.

EXAM PROBLEM

Your turn — Lec 4 p25. Apply intensity-level slicing: (a) intensities between 50 and 100 → 255, preserve the background; (b) between 50 and 100 → 255, everything else → 0.

110	120	130	135
100	94	98	200
30	60	70	30
28	29	25	27

✍ Show the full exam answer: S4.3 (end of this lecture)

Slide boundary (L4 p25). The question says "between 50 and 100", yet both printed answers treat the pixel 100 as outside (it stays 100 in (a), becomes 0 in (b)). That only fits $50 \le r < 100$. In the exam, state which reading you used.

TRY ALONE

⑦ (a) Version (b), band 50 ≤ r < 100: what happens to 49? (b) Slides pp. 23–24 are titled "Contrast stretching: Example". What do they really show?

✍ Answers: S4.4 (end of this lecture)

↗ Guide Tutorial 7, step 5 (slicing)

🖍 PEN & PAPER · ⏱ 9 min · Tier A

4 · Histogram equalization by table

IN PLAIN WORDS

① The **histogram** counts pixels per grey level: $h(r_k) = n_k$, with r_k the k-th level $(0 \dots L-1)$ and n_k its count (L4 p26). Divided by the pixel count MN it becomes the **normalized histogram** $p_r(r_k) = n_k/MN$, which sums to 1. Its shape tells the story (p27): piled left = dark image, piled right = light, narrow in the middle = low contrast, spread wide = high contrast. **Equalization** (p31) maps each level to $s_k = (L-1)\sum_{j=0}^{k} p_r(r_j)$, rounded, so the levels spread over the whole range.

WARM-UP BY HAND

② 2-bit image $(L=4)$ $\begin{bmatrix}0 & 0 & 1 & 1\\ 0 & 0 & 0 & 3\end{bmatrix}$, $MN = 8$, $L-1 = 3$.

r_k	n_k	running $\sum n$	$3 \times \sum n/8$	s_k
0	5	5	1.875	2
1	2	7	2.625	3
2	0	7	2.625	3
3	1	8	3	3

New image $\begin{bmatrix}2 & 2 & 3 & 3\\ 2 & 2 & 2 & 3\end{bmatrix}$.

THE ONE IDEA

③ **A level's new position is the fraction of pixels at or below it, times $L-1$.** A crowded level makes the running sum jump, so it is pushed away from its neighbours; the top level always lands on $L-1$.

🧭 **Like** ranking students by percentile instead of raw mark: a cluster of near-identical marks gets spread over many percentiles. *Where it breaks:* everyone with the same mark must share one percentile, so the result is spread out, never perfectly flat.

BUILD IT

④ 1. Count n_k for every level, empty ones too.

Why: an empty level still repeats the running sum.

2. Running sum of n_k, divide by MN.

Why: that is $\sum p_r$, the cumulative fraction.

3. Multiply by $L-1$ and round, halves up.
Why: it rescales 0–1 to the level range.

4. Replace every pixel r_k by s_k.
Why: it is still a point operation.

CHECK

⑤ The last occupied level gives exactly $L-1$, and s_k never decreases as k grows.

EXAM PROBLEM

Your turn — Lec 4 p32. Apply histogram equalization to the 3-bit image ($L=8$, $MN = 4 \times 8 = 32$) whose four rows all read

0	1	1	2	2	3	4	4

✍ Show the full exam answer: S4.5 (end of this lecture)

Slide error (L4 p32, row $r=1$). The slide writes 7(0.357) = 2 and prints rows 1 2 2 4 4 5 7 7. Its own running sum is 0.375, and 7 × 0.375 = 2.625 → **3**. Correct rows: 1 3 3 4 4 5 7 7.

TRY ALONE

⑦ (a) Gonzalez Example 3.5 (L4 pp. 33–34): 3-bit, $MN = 4096$, $n = 790, 1023, 850, 656, 329, 245, 122, 81$. Find s_0 and s_1. (b) What does `plt.hist(img)` draw for a 2-D image (L4 p28)?

✍ Answers: S4.6 (end of this lecture)

Matching (specification) and Sheet 4 Q4–Q6 continue in Lecture 5.

↗ Guide Tutorial 8 (equalization stepper) ↗ Wikipedia: Histogram equalization

🔁 **RECALL DRILL · ⏱ 4 min · Tier A**

5 · Quick recall

1. Negative of 200 in an 8-bit image? Of 5 in a 3-bit image?
2. An image is too dark overall. Which helps: $\gamma = 0.5$ or $\gamma = 3$?
3. Piecewise stretching with $r_1 = r_2$, $s_1 = 0$, $s_2 = L-1$ gives what?
4. A histogram with all its mass between 100 and 140 (of 0–255) means what kind of image?
5. In an equalization table, what is s_k for the highest occupied level, and why?

✍ Answers: S4.7 (end of this lecture)

⚠ Traps

- Using 255 in the negative of a non-8-bit image: it is $L-1$ (15 for 4-bit, 7 for 3-bit).
- Truncating instead of rounding in stretching: the slide gets 125 from 124.67; `np.uint8` gives 124.
- Copying the slide's equalization row 1 2 2 4 4 5 7 7: the correct row is 1 3 3 4 4 5 7 7.
- Leaving the slicing boundary unstated: the slide's "between 50 and 100" keeps 100 outside.
- Dividing by the number of levels instead of the pixel count MN when normalizing a histogram.
- Running the slide's log code on uint8: $1 + 255$ wraps to 0. Convert to float first.

Solutions and answers — Lecture 4

S4.1 · 2 · Min-max stretching: the dullest pixel to 0, the brightest to 255 — Show the full exam answer

Step 1 · what you're given

	c1	c2	c3
r1	100	120	125
r2	104	110	150
r3	60	130	140

A 3×3 image. Rows counted from the top, columns from the left. Every number is one pixel's grey value.

Step 2 · find the darkest and brightest pixel

	c1	c2	c3
r1	100	120	125
r2	104	110	150
r3	60	130	140

The smallest value is 60, so $Min = 60$. The largest is 150, so $Max = 150$. The rule: 60 must become 0 and 150 must become 255.

Step 3 · the stretch factor

The image uses 150 − 60 = 90 grey steps. You want it to use 255.

So every input step becomes $255/90 = 2.8333$ output steps. This number is the same for every pixel.

Step 4 · one pixel in full: 104 at (r2,c1)

	c1	c2	c3
r1	100	120	125
r2	104	110	150
r3	60	130	140

Subtract Min: $104 - 60 = 44$ (how far above the darkest pixel it is).

Multiply by the factor: $44 \times 255/90 = 11220/90 = 124.67$.

Round to the nearest whole number: $124.67 \rightarrow \mathbf{125}$.

Step 5 · the other pixels the same way

pixel I	60	100	104	110	120	125	130	140	150
$I - 60$	0	40	44	50	60	65	70	80	90
$\times 255/90$	0	113.33	124.67	141.67	170	184.17	198.33	226.67	255
rounded	0	113	125	142	170	184	198	227	255

Read each column top to bottom: subtract 60, multiply, round. Check: 60 gave 0 and 150 gave 255.

Step 6 · put each answer back in its place

	c1	c2	c3
r1	**113**	**170**	**184**
r2	**125**	**142**	**255**
r3	**0**	**198**	**227**

Same positions as the input. The 125 you computed in step 4 sits where 104 was.

What to write in the exam: $Min = 60$, $Max = 150$. $I_{new} = (I - 60) \times 255/90$, rounded. Show one pixel (104 → 44 × 2.8333 = 124.67 → 125), then the table and the output grid.

$$I_{new} = \text{round}\big((I - 60) \cdot 255/90\big) \rightarrow \begin{bmatrix} 113 & 170 & 184 \\ 125 & 142 & 255 \\ 0 & 198 & 227 \end{bmatrix}$$ **(matches the slide)**

If you code it: `np.uint8(...)` chops off the decimals instead of rounding, so it gives 124, 141 and 226 for 104, 110 and 140. Round first.

↩ back to the card

S4.2 · 2 · Min-max stretching: the dullest pixel to 0, the brightest to 255 — Answers

(a) Step 1 · subtract the old minimum

Old range 60 to 150, new range 0 to 100. $130 - 60 = 70$: the pixel is 70 steps above the darkest.

(a) Step 2 · new factor, then round

The new range is only 100 wide, so the factor is $100/90$, not $255/90$.

$70 \times 100/90 = 77.78 \rightarrow$ round $\rightarrow$ **78**. Add $NewMin = 0$: still 78.

(b) · already uses the full range

0	**90**
140	**255**

Here $Min = 0$ and $Max = 255$, so the factor is $255/255 = 1$ and you subtract 0. Every pixel stays the same.

What to write in the exam: (a) $(130 - 60) \times 100/90 = 77.78 \rightarrow 78$. (b) Nothing changes: factor = 1 and $Min = 0$, so $I_{new} = I$. Equalization (Lecture 5) can still help.

↩ back to the card

S4.3 · 3 · Slicing: light up one band of greys — Show the full exam answer

Step 1 · what you're given

	c1	c2	c3	c4
r1	110	120	130	135
r2	100	94	98	200
r3	30	60	70	30
r4	28	29	25	27

A 4×4 image, rows from the top, columns from the left. The band is "50 to 100".

Step 2 · fix the edges of the band first

The rule: a pixel is **inside** if $50 \leq r < 100$. So 50 is in, 100 is out.

Why 100 is out: the slide's own answer keeps the pixel 100 unchanged. Say this reading in the exam.

Step 3 · ask each pixel "am I inside?"

	c1	c2	c3	c4
r1	110	120	130	135
r2	~~100~~	94	98	200
r3	30	60	70	30
r4	28	29	25	27

Amber = inside the band: 94, 98, 60, 70. The crossed-out 100 is the edge case: outside. Everything else is far outside (too dark or too bright).

Step 4 · version (a): amber → 255, keep the rest

	c1	c2	c3	c4
r1	110	120	130	135
r2	100	255	255	200
r3	30	255	255	30
r4	28	29	25	27

"Preserve the background" means: pixels outside the band are copied unchanged.

Step 5 · version (b): amber → 255, the rest → 0

	c1	c2	c3	c4
r1	0	0	0	0
r2	0	255	255	0
r3	0	255	255	0
r4	0	0	0	0

"Binary" means only two values. Check: the 255s are in exactly the same spots as in (a).

What to write in the exam: Band taken as $50 \leq r < 100$ (100 is outside, as on the slide). In-band pixels 94, 98, 60, 70 → 255. (a) the rest unchanged; (b) the rest → 0. Draw both 4×4 grids.

The four pixels 94, 98, 60, 70 → 255. (a) the rest unchanged; (b) the rest → 0. If "between 50 and 100" includes 100, the pixel at row 2, column 1 also becomes 255 in both.

↩ back to the card

S4.4 · 3 · Slicing: light up one band of greys — Answers

(a) · is 49 inside?

value	~~49~~	50	…	99	~~100~~

The band starts at 50, and 49 is one below it, so it is outside. Version (b) sends every outside pixel to **0**.

(b) · what pp. 23–24 really show

The cameraman photo, with pixels between 100 and 150 set to 255 and the rest set to 0 (binary) or kept (background preserved).

That is slicing, not stretching: nothing is multiplied by a factor, pixels just answer "inside or not?".

What to write in the exam: (a) 49 → 0 (outside the band). (b) Intensity-level slicing of the band 100–150, despite the "contrast stretching" title.

↩ back to the card

S4.5 · 4 · Histogram equalization by table — Show the full exam answer

Step 1 · what you're given

	c1	c2	c3	c4	c5	c6	c7	c8
r1	0	1	1	2	2	3	4	4
r2	0	1	1	2	2	3	4	4
r3	0	1	1	2	2	3	4	4
r4	0	1	1	2	2	3	4	4

4 rows × 8 columns = 32 pixels ($MN = 32$). 3-bit means levels 0 to 7 ($L = 8$, so $L - 1 = 7$). Only 0–4 are used: the image is squeezed into the dark half.

Step 2 · count each level (n_k)

level r_k	count n_k
0	4
1	8
2	8
3	4
4	8
5, 6, 7	0

Each row has one 0, so 4 rows give 4 zeros. Each row has two 1s → 8. Same for the others. Check: 4 + 8 + 8 + 4 + 8 = 32.

Step 3 · divide by 32 (p_r)

r_k	n_k	$n_k/32$
0	4	0.125
1	8	0.25
2	8	0.25
3	4	0.125
4	8	0.25
5–7	0	0

This is the fraction of the image at each level: $4/32 = 0.125$, $8/32 = 0.25$.

Step 4 · running total ($\sum p_r$)

r_k	n_k	$n_k/32$	running total
0	4	0.125	0.125
1	8	0.25	0.375
2	8	0.25	0.625
3	4	0.125	0.75
4	8	0.25	1
5–7	0	0	1

Add each new fraction to the one above it: $0.125 + 0.25 = 0.375$, $0.375 + 0.25 = 0.625$, ... It means "fraction of pixels at this level or darker". It must end at 1.

Step 5 · multiply by $L - 1 = 7$

r_k	running total	$\times 7$
0	0.125	0.875
1	0.375	2.625
2	0.625	4.375
3	0.75	5.25
4	1	7
5–7	1	7

This stretches the 0-to-1 fraction onto the level range 0 to 7. Example: $7 \times 0.375 = 2.625$.

Step 6 · round to a whole level (s_k)

r_k	$\times 7$	new level s_k
0	0.875	1
1	2.625	3
2	4.375	4
3	5.25	5
4	7	7
5–7	7	7

Nearest whole number. So the lookup is $0 \to 1$, $1 \to 3$, $2 \to 4$, $3 \to 5$, $4 \to 7$.

Step 7 · replace every pixel using the lookup

	c1	c2	c3	c4	c5	c6	c7	c8
r1	**1**	**3**	**3**	**4**	**4**	**5**	**7**	**7**
r2	**1**	**3**	**3**	**4**	**4**	**5**	**7**	**7**
r3	**1**	**3**	**3**	**4**	**4**	**5**	**7**	**7**
r4	**1**	**3**	**3**	**4**	**4**	**5**	**7**	**7**

Every 0 becomes 1, every 1 becomes 3, and so on. The image now spans 1–7 instead of 0–4: more contrast.

What to write in the exam: the full table with columns r_k, n_k, $n_k/32$, $\sum p_r$, $7\sum p_r$, s_k (one row per level 0–7), then the output image: every row 1 3 3 4 4 5 7 7. New histogram: $n(1) = 4$, $n(3) = 8$, $n(4) = 8$, $n(5) = 4$, $n(7) = 8$.

$s = 1, 3, 4, 5, 7$ **for** $r = 0 \ldots 4$**; each row →** **1 3 3 4 4 5 7 7**

↩ back to the card

S4.6 · 4 · Histogram equalization by table — Answers

(a) Step 1 · the counts you're given

r_k	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
n_k	**790**	**1023**	**850**	**656**	**329**	**245**	**122**	**81**

The histogram is already counted for you. $MN = 4096$ pixels, $L - 1 = 7$.

(a) Step 2 · s_0: only level 0 so far

r_k	**0**	**1**
n_k	**790**	**1023**

Running total = 790. $7 \times 790/4096 = 1.35 \rightarrow$ round $\rightarrow$ **1**.

(a) Step 3 · s_1: levels 0 and 1 together

r_k	**0**	**1**
n_k	**790**	**1023**

Running total = 790 + 1023 = 1813. $7 \times 1813/4096 = 3.10 \rightarrow$ round $\rightarrow$ **3**.

(b) · `plt.hist` on a 2-D image

It treats each column as its own data set: one histogram per column, 10 bins each, in many colours. Flatten first: `plt.hist(img.ravel(), bins=256)`.

What to write in the exam: (a) $s_0 = 7 \times 790/4096 = 1.35 \rightarrow 1$; $s_1 = 7 \times 1813/4096 = 3.10 \rightarrow 3$. (Full map 1, 3, 5, 6, 6, 7, 7, 7. The book prints 1.33, 3.08 because it rounds p_r first; same levels.) (b) One histogram per column; use `img.ravel()`.

↩ back to the card

S4.7 · 5 · Quick recall — Answers

1 · negative = flip around $L-1$

8-bit: $L-1=255$, so $255-200=55$. 3-bit: $L-1=7$, so $7-5=2$.

2 · too dark → $\gamma<1$

$\gamma=0.5$ lifts the darks: $64 \to 255\,(64/255)^{0.5}=127.75 \to 128$. $\gamma=3$ would make it darker.

3 · $r_1=r_2$, $s_1=0$, $s_2=L-1$

The line becomes a vertical jump: below $r_1 \to 0$, above $\to L-1$. That is thresholding; the output is binary.

4 · all mass in 100–140

A narrow pile in the middle of 0–255: low contrast (everything is a similar grey).

5 · highest occupied level

Its running total has counted every pixel, so $\sum p_r=1$ and $s_k=(L-1)\times 1=L-1$.

What to write in the exam: 1. 55; 2. 2. $\gamma=0.5$. 3. Thresholding (binary image). 4. Low contrast. 5. $L-1$, because $\sum p_r=1$.

↩ back to the card

LECTURE 5

Histogram processing: equalization and matching

Tier A · must ⏱ 45 min Slides: Lec 5-Histogram Processing.pdf (19 p) · Sheet 4 Q4–Q6, Sheet 5 Q1–Q2

↗ Wikipedia: Histogram matching

Lecture 4 changed each pixel with a formula you chose (negative, log, gamma, stretching). Lecture 5 lets the image choose: the formula is read off the image's own histogram. Equalization spreads the grey levels so the picture uses the whole range; matching bends them until the histogram looks like one you specify. Equalization was taught in Lecture 4 (the slide example on L5 p3 repeats Lec 4 p32, with the same 0.357 slip), so the cards go straight to the sheet problems.

What the exam can ask from this lecture

- Contrast-stretch a small image to 0–255 (Sheet 4 Q4).
- Equalize from a histogram or from an image: the full p_r, running-sum, s_k table, the transformation, the new image and its histogram (Sheet 4 Q5–Q6, Lec 5 p3).
- Histogram matching: s_k, $G(z_q)$, the $s \to z$ lookup and the final $r \to z$ map (Sheet 5 Q1–Q2, Lec 5 pp. 10–15).
- Explain why equalization can wash out a dark image and how matching fixes it (Lec 5 pp. 16–18).

⌛ **Only 20 minutes?** Do cards 2 and 5, in that order.

PEN & PAPER · ⏱ 6 min · Tier A

1 · Stretching: pull the darkest pixel to 0 and the brightest to 255 (Sheet 4 Q4)

IN PLAIN WORDS

① **Histogram (contrast) stretching** is Lec 4's straight-line map: $s = (r - r_{\min}) \cdot \dfrac{255}{r_{\max} - r_{\min}}$, rounded with halves up, where $r_{\min}$, $r_{\max}$ are the image's smallest and largest values. The sheet gives no bit depth, so take Lec 4's 8-bit target 0–255.

WARM-UP BY HAND

② $\begin{bmatrix} 2 & 5 \\ 4 & 8 \end{bmatrix}$: $r_{\min} = 2$, $r_{\max} = 8$, slope $255/6 = 42.5$.

$2 \to 0$, $4 \to 2 \times 42.5 = 85$, $5 \to 127.5 \to 128$, $8 \to 255$, so the output is $\begin{bmatrix} 0 & 128 \\ 85 & 255 \end{bmatrix}$.

THE ONE IDEA

③ **Stretching is linear: it moves the histogram's ends to 0 and 255 and keeps its shape.** Equalization reshapes it.

BUILD IT

④ 1. Find $r_{\min}$, $r_{\max}$.

Why: they pin the line's ends.

2. Slope $= 255/(r_{\max} - r_{\min})$.

Why: the input span must cover 255 output steps.

3. $s = (r - r_{\min}) \times$ slope, rounded halves up.

Why: subtract first so the darkest pixel lands on 0; Lec 4 p21 rounds this way.

CHECK

⑤ The minimum comes out 0, the maximum 255, and the pixel order never changes.

EXAM PROBLEM

Your turn — Sheet 4 Q4. Apply histogram stretching to the image

7	12	8
20	9	6
10	15	1

✍ Show the full exam answer: S5.1 (end of this lecture)

TRY ALONE

⑦ Stretch 35 from [10, 60] to [0, 255].

✍ Answers: S5.2 (end of this lecture)

↗ Guide Tutorial 7, step 4 (contrast stretching, worked 3 × 3)

🖍 PEN & PAPER · ⏱ 7 min · Tier A

2 · Equalizing when you are handed only the counts (Sheet 4 Q5)

IN PLAIN WORDS

① The exam may give only the histogram: n_k pixels at level r_k, MN pixels in all, L levels. Equalization (Lec 4, L5 p2) needs nothing else: $s_k = (L-1)\sum_{j\le k} p_r(r_j)$, rounded halves up. Sheet 4 Q5 prints a histogram and no question; the natural task, solved here, is "equalize it".

WARM-UP BY HAND

② 2-bit ($L-1=3$), $n = 2, 1, 3, 2$, $MN = 8$. Running counts 2, 3, 6, 8.
$\times 3/8$: 0.75, 1.125, 2.25, 3 $\rightarrow s = 1, 1, 2, 3$; new histogram $n_1 = 3$, $n_2 = 3$, $n_3 = 2$.

THE ONE IDEA

③ **Work with running counts and divide once at the end:** $s_k = \frac{L-1}{MN}\sum_{j\le k} n_j$**.** It avoids rounding the p_r column, and the last row must give exactly $L-1$.

BUILD IT

④ 1. MN = sum of counts.

Why: divide by pixels, not by levels.

2. Running counts.

Why: $T(r_k)$ counts every pixel at or below level k.

3. Multiply by $(L-1)/MN$ and round.

Why: scale 0–1 to the level range, integers only.

4. Merge levels that land together.

Why: that gives the new histogram.

CHECK

⑤ The last s is $L-1$, the column never decreases, the new counts still total MN.

EXAM PROBLEM

Your turn — Sheet 4 Q5. An image has 3 bits/pixel, so values run 0 to 7, with this histogram. Equalize it: give the transformation and the new histogram.

Grey level	0	1	2	3	4	5	6	7
Pixels	10	8	9	2	14	1	5	2

✍ Show the full exam answer: S5.3 (end of this lecture)

TRY ALONE

⑦ Why is $s_3 = 4$ the same as s_2 in Q5?

✍ Answers: S5.4 (end of this lecture)

↗ Wikipedia: Histogram equalization (worked 8 × 8 example)

✎ PEN & PAPER · ⏱ 7 min · Tier A

3 · From image to histogram to new image (Sheet 4 Q6)

IN PLAIN WORDS

① Now the input is the image itself. (a) Count each level and draw bars (level across, count up). (b) The **transformation** is the lookup table $r_k \to s_k$. (c) Rewrite the image through that lookup, pixel by pixel.

WARM-UP BY HAND

② 2-bit $\begin{bmatrix}0 & 0 & 3\\ 1 & 1 & 3\end{bmatrix}$: $n = 2, 2, 0, 2$, $MN = 6$, running 2, 4, 4, 6.
$\times 3/6 \to s = 1, 2, 2, 3$; new image $\begin{bmatrix}1 & 1 & 3\\ 2 & 2 & 3\end{bmatrix}$.

THE ONE IDEA

③ **Count carefully, and give empty levels a row too**: a level with $n_k = 0$ repeats the running sum above it. Most lost marks here are miscounts, not formula slips.

BUILD IT

④ 1. Tally each level, check the tally sums to MN.
Why: one missed pixel shifts every later running sum.
2. Fill the table, empty levels included.
Why: T must say where every level goes.
3. Replace each pixel r by s_r.
Why: equalization is a point operation; neighbours don't matter.

CHECK

⑤ The new image's counts are the old counts moved to new levels (warm-up: two 1s, two 2s, two 3s).

EXAM PROBLEM

Your turn — Sheet 4 Q6. A 3-bit 4 × 4 image:

0	0	0	4
1	1	1	5
1	2	2	7
2	2	2	7

(a) Plot the histogram of this image. (b) Define the transformation that will equalize the pixel histogram. (c) Find the values of the new 4 × 4 histogram-equalized image.

✍ Show the full exam answer: S5.5 (end of this lecture)

TRY ALONE

⑦ The Q6 image already spans 0–7. Would stretching change it? Does equalization?

✍ Answers: S5.6 (end of this lecture)

↗ Guide Tutorial 8, step 2 (the slide's example, done right)

📖 READ & UNDERSTAND · ⏱ 4 min · Tier A

4 · Matching: equalize both histograms, then meet in the middle

What if a flat histogram is the wrong goal? Lec 5 pp. 16–18 show Mars's moon Phobos: mostly black sky, a huge spike near 0. Equalization hands that spike a big share of the range and the picture comes out **washed out**. **Histogram matching** (specification) lets you choose the target shape $p_z(z)$ yourself. The method (L5 pp. 7–9) runs two equalizations and joins them; input levels are r, output levels are z.

1. Equalize the input: $s_k = T(r_k) = (L-1)\sum_{j \le k} p_r(r_j)$, rounded. 2. Equalize the **target** the same way: $G(z_q) = (L-1)\sum_{i \le q} p_z(z_i)$, rounded. 3. Both are now on the same "flat" scale, so set $G(z) = s$ and solve: $z = G^{-1}(s)$. In a table, that means: **for each s_k, find the z_q whose $G(z_q)$ is closest to s_k**. 4. Chain them: $r_k \to s_k \to z_q$.

A tiny number: if $s_0 = 1$ and the G row reads 0, 0, 0, 1, 2, 5, 6, 7, then $G(z_3) = 1$ matches exactly and every pixel at r_0 becomes 3. If $s = 3$, no G equals 3; the nearest is $G(z_4) = 2$ (distance 1, against 2 for $G(z_5) = 5$), so $s = 3 \to z = 4$.

🧭 **Like** converting a grade between two schools through a percentile: "top 30% at my school" is looked up as "whatever mark is top 30% at yours". s and G are the percentiles. *Where it breaks:* marks are whole levels, so the exact percentile often doesn't exist and you take the nearest one, which is why the output histogram only approximates the target.

Two rules the slides use without saying: both s_k and $G(z_q)$ are rounded to integers before comparing (p11: "$G(z_3) = 1.05 \to 1$"), and when several z share the same G value, take the smallest z (Gonzalez's convention).

Slide error (L5 pp. 11, 13–15). The slides list $s_4 = 7$ and so map $r_4 \to 7$. Their own equalization slide (p4) has $s_4 = 6.23 \to 6$, and the correct running sum is $7 \times 3648/4096 = 6.23$. With $s_4 = 6$, $r_4 \to z_6$. Proof that 6 is right: only then do you get the slide's own "Actual p_z" column (p10): $z_6 = 0.16 + 0.08 = 0.24$ and $z_7 = 0.06 + 0.03 + 0.02 = 0.11$.

↗ Wikipedia: Histogram matching ↗ Guide Tutorial 8, step 5 (matching table)

PEN & PAPER · ⏱ 9 min · Tier A

5 · The full matching table, 3-bit (Sheet 5 Q1)

IN PLAIN WORDS

① Table A is the input histogram (n_k pixels at level r_k); table B is the **reference** $p_z(z_q)$ you want the output to have. Give the map $r_k \to z_q$ and the histogram the output actually ends up with.

WARM-UP BY HAND

② 2-bit. Input $n = 2, 1, 4, 1$ ($MN = 8$) → 3× running/8 = 0.75, 1.125, 2.625, 3 → $s = 1, 1, 3, 3$.
Target $p_z = 0, 0.1, 0.3, 0.6$ → 3× running = 0, 0.3, 1.2, 3 → $G = 0, 0, 1, 3$.
$s = 1$ finds $G(z_2) = 1$, $s = 3$ finds $G(z_3) = 3$: $r = 0, 1, 2, 3 \to z = 2, 2, 3, 3$.

THE ONE IDEA

③ Two equalization columns, then a lookup from s into G; the answer is the z where you land, never the G value.

BUILD IT

④ 1. s_k from the input counts.
Why: puts the input on the flat scale.
2. $G(z_q)$ = running $p_z \times (L-1)$, rounded.
Why: puts the target on the same scale.
3. For each s_k, take the z with the nearest G.
Why: $z = G^{-1}(s)$ by table lookup.
4. Add the counts landing on each z.
Why: that is the actual output histogram.

CHECK

⑤ The map never decreases; levels whose G is 0 get no pixels unless some $s_k = 0$.

EXAM PROBLEM

Your turn — Sheet 5 Q1. Compute the histogram matching from the tables below; table A is the original image and table B is the reference (3-bit, $L = 8$).

level	0	1	2	3	4	5	6	7
A: count	790	1023	850	656	329	245	122	81
A: PDF	0.19	0.25	0.21	0.16	0.08	0.06	0.03	0.02
B: PDF	0	0	0	0.15	0.2	0.3	0.2	0.15

✍ Show the full exam answer: S5.7 (end of this lecture)

TRY ALONE

⑦ In Q1, which output levels can never appear, and why?

✍ Answers: S5.8 (end of this lecture)

↗ Szeliski Ch 3.1 (histogram equalization)

PEN & PAPER · ⏱ 9 min · Tier A

6 · Sixteen levels and a tie every few rows (Sheet 5 Q2)

IN PLAIN WORDS

① Card 5's method on a 4-bit image ($L = 16$, multiplier $L - 1 = 15$). New: the **tie**, where s sits exactly between two G values (e.g. $s = 4$ between $G = 2$ and $G = 6$). The slides never meet one, so state your rule; here **ties go to the smaller** z, the direction of Gonzalez's "smallest z" rule.

WARM-UP BY HAND

② 2-bit, $n = 2, 2, 2, 2 \to s = 1, 2, 2, 3$; $p_z = 0.1, 0.2, 0, 0.7 \to 3\times$ running 0.3, 0.9, 0.9, 3 $\to G = 0, 1, 1, 3$.
$s = 1$: $G = 1$ at z_1 and $z_2 \to z_1$. $s = 2$: $G = 1$ and $G = 3$ both 1 away → smaller, z_1. $s = 3 \to z_3$.
Map $r = 0, 1, 2, 3 \to z = 1, 1, 1, 3$ (ties upward would give $1, 3, 3, 3$).

THE ONE IDEA

③ **Write your tie rule in one line at the top of the answer, then apply it every time.** A marker can follow a stated rule, not a coin toss.

BUILD IT

④ 1. Running counts × $15/MN$, rounded.
Why: two-decimal PDFs put level 1 at 1.50 instead of 1.51, a hair from a different rounding.
2. Running p_z × 15, rounded → G.
Why: the slides round G before matching.
3. Nearest G; same G or equal distance → smaller z.
Why: one fixed rule makes the map well defined.
4. Assemble $r \to z$.
Why: that is the requested matching.

CHECK

⑤ The map never decreases and the top level goes to 15.

EXAM PROBLEM

Your turn — Sheet 5 Q2. Compute the histogram matching for a 4-bit image (table A original, table B reference).

level	0	1	2	3	4	5	6	7
A: count	2596	4096	3926	6692	3596	3396	3796	3196
B: PDF	0	0	0	0.05	0.07	0	0	0.25
level	8	9	10	11	12	13	14	15
A: count	4296	4616	4746	3896	3796	4596	4396	4996
B: PDF	0.17	0.15	0.07	0.09	0.03	0.02	0.04	0.06

✍ Show the full exam answer: S5.9 (end of this lecture)

TRY ALONE

⑦ In Q2, which z levels with nonzero p_z still receive no pixels?

✍ Answers: S5.10 (end of this lecture)

↗ Wikipedia: Histogram matching (the algorithm)

🔁 **RECALL DRILL · ⏱ 3 min · Tier B**

7 · Ten-second answers

1. You equalize the Sheet 4 Q5 result a second time. What changes?
2. Why did equalization wash out the Phobos photo (L5 p17), and what fixed it?
3. In matching, why can s and G be compared at all?

✍ Answers: S5.11 (end of this lecture)

⚠ Traps

- Dividing by the number of levels instead of the pixel count MN.
- Skipping empty levels in the table: their row repeats the running sum, and leaving them out shifts every s_k after them.
- Rounding halves to even (as Python's `round(2.5) = 2` does): the slides always round halves up, e.g. 127.5 → 128, 13.5 → 14.
- Copying the slides' 3-bit example (row $r = 1$ gives 3, not 2) or the matching slides' $s_4 = 7$ (it is 6, so $r_4 \to z_6$).
- Answering a matching question with the G value instead of the z index where that G sits.
- Leaving the tie rule unstated in a matching answer; Sheet 5 Q2 has four ties.

Solutions and answers — Lecture 5

S5.1 · 1 · Stretching: pull the darkest pixel to 0 and the brightest to 255 (Sheet 4 Q4) — Show the full exam answer

Step 1 · what you're given

	c1	c2	c3
r1	7	12	8
r2	20	9	6
r3	10	15	1

Rows from the top, columns from the left. Find the darkest and brightest pixel (blue): smallest = 1 at (r3,c3), largest = 20 at (r2,c1).

Step 2 · the rule in words

"Slide the darkest pixel down to 0, then stretch so the brightest lands on 255."

Slide: subtract 1. Stretch: the gap 20 − 1 = 19 must become 255, so multiply by $255/19 = 13.4211$. Round halves up.

$s = (r - 1) \times 13.4211$

Step 3 · one cell in full

	c1	c2	c3
r1	7	12	8
r2	20	9	6
r3	10	15	1

The orange cell (r1,c1) holds 7.
$7 - 1 = 6$, then $6 \times 13.4211 = 80.53$, rounded → **81**.

Step 4 · every value the same way

r	1	6	7	8	9	10	12	15	20
$r - 1$	0	5	6	7	8	9	11	14	19
× 13.4211	0	67.11	80.53	93.95	107.37	120.79	147.63	187.89	255
s	0	67	81	94	107	121	148	188	255

Each grey value appears once, so you only compute nine numbers.

Step 5 · put them back in place

	c1	c2	c3
r1	81	148	94
r2	255	107	67
r3	121	188	0

Check: the 1 became 0, the 20 became 255, and bigger inputs stayed bigger.

What to write in the exam: $r_{\min} = 1$, $r_{\max} = 20$, $s = (r-1) \cdot 255/19$ rounded. Show the r → s table, then the new image.

$$s = \text{round}\big((r-1) \cdot 255/19\big) \rightarrow \begin{bmatrix} 81 & 148 & 94 \\ 255 & 107 & 67 \\ 121 & 188 & 0 \end{bmatrix}$$

↩ back to the card

S5.2 · 1 · Stretching: pull the darkest pixel to 0 and the brightest to 255 (Sheet 4 Q4) — Answers

One cell, same rule

Darkest = 10, so slide down: $35 - 10 = 25$.

The gap $60 - 10 = 50$ must become 255: multiply by $255/50 = 5.1$.

$25 \times 5.1 = 127.5$, halves round up → **128**.

Sanity check: 10 → 0, and 35 is half-way between 10 and 60, so it lands half-way up, near 128.

What to write in the exam: $s = (35 - 10) \cdot 255/50 = 127.5 \rightarrow 128$.

↩ back to the card

S5.3 · 2 · Equalizing when you are handed only the counts (Sheet 4 Q5) — Show the full exam answer

Step 1 · what you're given

level r	pixels n
0	10
1	8
2	9
3	2
4	14
5	1
6	5
7	2

3 bits → levels 0 to 7, so the top level is 7. Add the counts: 51 pixels in all.

Step 2 · add a "running total" column

r	n	running
0	10	10
1	8	18
2	9	27
3	2	29
4	14	43
5	1	44
6	5	49
7	2	51

Each row = the row above + this row's pixels (10, 10+8 = 18, 18+9 = 27 …). It says "how many pixels are this dark or darker". The last one must be 51.

Step 3 · scale it: × 7 ÷ 51

r	n	running	× 7/51
0	10	10	1.37
1	8	18	2.47
2	9	27	3.71
3	2	29	3.98
4	14	43	5.90
5	1	44	6.04
6	5	49	6.73
7	2	51	7.00

One cell in full: $10 \times 7/51 = 1.37$. The other rows the same way. The fraction "running ÷ 51" goes 0 → 1; times 7 spreads it over levels 0 → 7.

Step 4 · round → the new level s

r	n	running	× 7/51	s
0	10	10	1.37	1
1	8	18	2.47	2
2	9	27	3.71	4
3	2	29	3.98	4
4	14	43	5.90	6
5	1	44	6.04	6
6	5	49	6.73	7
7	2	51	7.00	7

Round to the nearest whole level (halves up). This last column **is** the transformation: every pixel at level r becomes s.

Step 5 · the new histogram

new level s	0	1	2	3	4	5	6	7
pixels	~~0~~	10	8	~~0~~	9+2 = 11	~~0~~	14+1 = 15	5+2 = 7

Move each old count to its new level. Two old levels landing on the same s add up (2 and 3 both → 4). Nobody lands on 0, 3, 5 (red). Total still 51 ✓.

What to write in the exam: $MN = 51$, $s_k = \frac{7}{51} \times$(running count), rounded. The 5-column table, then: transformation $0, 1, \ldots, 7 \to 1, 2, 4, 4, 6, 6, 7, 7$; new histogram 0, 10, 8, 0, 11, 0, 15, 7.

$s = 1, 2, 4, 4, 6, 6, 7, 7$ for $r = 0 \ldots 7$; equalized histogram $n_s = 0, 10, 8, 0, 11, 0, 15, 7$ for $s = 0 \ldots 7$.

↩ back to the card

S5.4 · 2 · Equalizing when you are handed only the counts (Sheet 4 Q5) — Answers

Look at the two rows

r	n	× 7/51	s
2	9	3.71	4
3	2	3.98	4

Level 3 has only 2 of the 51 pixels. So the scaled running total only grows by $7 \times 2/51 = 0.27$: from 3.71 to 3.98. Both round to 4.

In words: a level with very few pixels barely moves the total, so it gets glued to its neighbour.

What to write in the exam: Level 3 adds only 2/51 of the pixels, +0.27, so 3.71 and 3.98 both round to 4.

↩ back to the card

S5.5 · 3 · From image to histogram to new image (Sheet 4 Q6) — Show the full exam answer

Step 1 · what you're given

	c1	c2	c3	c4
r1	0	0	0	4
r2	1	1	1	5
r3	1	2	2	7
r4	2	2	2	7

Rows from the top, columns from the left. 3 bits → levels 0 to 7. 4 × 4 = 16 pixels.

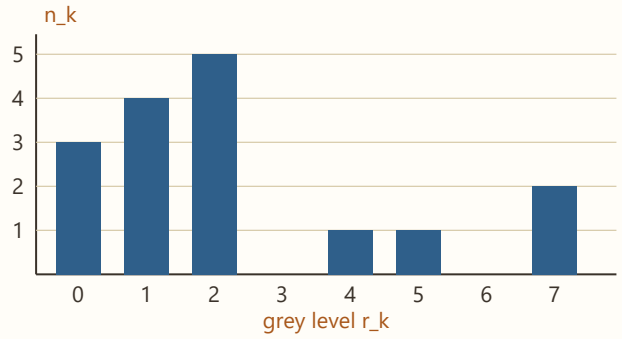
Step 2 · (a) count each level

	c1	c2	c3	c4
r1	0	0	0	4
r2	1	1	1	5
r3	1	2	2	7
r4	2	2	2	7

Example: the blue cells are the 1s — there are 4. Do the same for every level, empty ones too:

r	0	1	2	3	4	5	6	7
n	3	4	5	0	1	1	0	2

Sum 16 ✓. Draw one bar per level, height = count:



Step 3 · (b) add the running total

r	n	running
0	3	3
1	4	7
2	5	12
3	0	12
4	1	13
5	1	14
6	0	14
7	2	16

Row above + this row's count. An empty level (3, 6) just repeats the number above. Last row = 16.

Step 4 · scale: × 7 ÷ 16

r	n	running	× 7/16
0	3	3	1.3125
1	4	7	3.0625
2	5	12	5.25
3	0	12	5.25
4	1	13	5.6875
5	1	14	6.125
6	0	14	6.125
7	2	16	7

One cell in full: $3 \times 7/16 = 21/16 = 1.3125$. The other rows the same way.

Step 5 · round → the transformation

r	n	running	× 7/16	s
0	3	3	1.3125	1
1	4	7	3.0625	3
2	5	12	5.25	5
3	0	12	5.25	5
4	1	13	5.6875	6
5	1	14	6.125	6
6	0	14	6.125	6
7	2	16	7	7

The $r \to s$ columns are the answer to (b): "every pixel at level r becomes s".

Step 6 · (c) rewrite the image

	c1	c2	c3	c4
r1	1	1	1	6
r2	3	3	3	6
r3	3	5	5	7
r4	5	5	5	7

Look up each pixel in Step 5. Orange cell (r1,c1) was 0, and $0 \to 1$, so it becomes 1. Every $1 \to 3$, every $2 \to 5$, 4 and $5 \to 6$, $7 \to 7$.

Step 7 · new histogram (check)

s	0	1	2	3	4	5	6	7
pixels	~~0~~	3	~~0~~	4	~~0~~	5	2	2

The same piles of pixels, moved to new levels and spread further apart. Total 16 ✓.

What to write in the exam: (a) the counts 3, 4, 5, 0, 1, 1, 0, 2 and the bar chart. (b) the 5-column table; $T : 0, 1, \ldots, 7 \to 1, 3, 5, 5, 6, 6, 6, 7$. (c) the new 4×4 image.

T: $(0, 1, 2, 3, 4, 5, 6, 7) \to (1, 3, 5, 5, 6, 6, 6, 7)$; equalized image rows: 1 1 1 6 / 3 3 3 6 / 3 5 5 7 / 5 5 5 7.

↩ back to the card

S5.6 · 3 · From image to histogram to new image (Sheet 4 Q6) — Answers

Stretching · nothing moves

Darkest = 0, brightest = 7, target 0 to 7. Slide by 0, multiply by $7/7 = 1$. Every pixel stays itself.

Equalization · the crowd spreads out

old r	0	1	2	4	5	7
new s	1	3	5	6	6	7

12 of the 16 pixels sit at 0–2 (blue). Equalization gives that crowd more room: 0, 1, 2 → 1, 3, 5. The few bright pixels get squeezed into 6–7.

What to write in the exam: Stretching: no change (min 0, max 7, slope 1). Equalization: yes; dark levels 0, 1, 2 spread to 1, 3, 5, bright ones squeeze into 6–7.

↩ back to the card

S5.7 · 5 · The full matching table, 3-bit (Sheet 5 Q1) — Show the full exam answer

The idea in one line: put the input **and** the target on the same "equalized" scale, then for each input level find the target level that sits at the same spot. Input levels are called r, output levels z. 4096 pixels, top level 7.

Step 1 · equalize the input A → s

r	0	1	2	3	4	5	6	7
count	790	1023	850	656	329	245	122	81
running	790	1813	2663	3319	3648	3893	4015	4096
× 7/4096	1.35	3.10	4.55	5.67	6.23	6.65	6.86	7.00
s (rounded)	1	3	5	6	6	7	7	7

Exactly card 2: running total, × 7 ÷ 4096, round. One cell: $790 \times 7/4096 = 1.35 \rightarrow 1$. Note $r = 4$ gives 6.23 → **6** (the slides wrongly print 7).

Step 2 · equalize the target B → G

z	0	1	2	3	4	5	6	7
B: PDF	0	0	0	0.15	0.2	0.3	0.2	0.15
running	0	0	0	0.15	0.35	0.65	0.85	1
× 7	0	0	0	1.05	2.45	4.55	5.95	7.00
G (rounded)	0	0	0	1	2	5	6	7

Same recipe on the target. B already gives fractions, so there's nothing to divide: running total × 7, round. One cell: $0.15 \times 7 = 1.05 \rightarrow 1$.

Step 3 · for each s, find the nearest G

z	0	1	2	3	4	5	6	7
G	0	0	0	1	2	5	6	7
s lands here				1 ↑	3 ↑	5 ↑	6 ↑	7 ↑

Take each s from Step 1 (1, 3, 5, 6, 7) and look along the G row for the same number. 1, 5, 6, 7 are there exactly. 3 is not: the nearest are 2 (1 away) and 5 (2 away), so 3 goes to the 2, i.e. $z = 4$.

The answer is the z **above** the G, not the G itself.

Step 4 · chain them: $r \to s \to z$

r	0	1	2	3	4	5	6	7
s	1	3	5	6	6	7	7	7
z	3	4	5	6	6	7	7	7

Then move the counts. Output histogram:

z	0	1	2	3	4	5	6	7
pixels	0	0	0	790	1023	850	656+329 = 985	245+122+81 = 448
÷ 4096	0	0	0	0.19	0.25	0.21	0.24	0.11

Close to the target 0.15, 0.2, 0.3, 0.2, 0.15 but not equal: you can't split one input level into two.

What to write in the exam: Rule: round s and G; each s goes to the z with the nearest G (smallest z if tied). Table 1 (s), table 2 (G), then $r = 0 \dots 7 \to z = 3, 4, 5, 6, 6, 7, 7, 7$ and the output histogram 0, 0, 0, 0.19, 0.25, 0.21, 0.24, 0.11.

$r = 0 \dots 7 \to z = 3, 4, 5, 6, 6, 7, 7, 7$**; output PDF** $p_z = 0, 0, 0, 0.19, 0.25, 0.21, 0.24, 0.11$**. (The slides'** $4 \to 7$ **is a slip:** $s_4 = 6.23 \to 6$**.)**

↩ back to the card

S5.8 · 5 · The full matching table, 3-bit (Sheet 5 Q1) — Answers

Look at the lookup row

z	0	1	2	3
G	0	0	0	1
s lands here				1 ↑

Only an s of 0 could land on a G of 0. The smallest s in Q1 is 1, and it finds an exact $G = 1$ at $z = 3$. So $z = 0, 1, 2$ never get a pixel.

What to write in the exam: z_0, z_1, z_2: their G is 0, and no s_k is 0 (smallest is 1), so the lookup never picks them.

↩ back to the card

S5.9 · 6 · Sixteen levels and a tie every few rows (Sheet 5 Q2) — Show the full exam answer

Same four steps as card 5, just 16 levels (top level 15) and 66 632 pixels. The only new thing is the **tie rule**; write it first: "if two z are equally near, or share the same G, take the smaller z."

Step 1 · equalize the input A → s

r	running	× 15/66632	s
0	2596	0.58	1
1	6692	1.51	2
2	10618	2.39	2
3	17310	3.90	4
4	20906	4.71	5
5	24302	5.47	5
6	28098	6.33	6
7	31294	7.04	7
8	35590	8.01	8
9	40206	9.05	9
10	44952	10.12	10
11	48848	11.00	11
12	52644	11.85	12
13	57240	12.89	13
14	61636	13.88	14
15	66632	15.00	15

Running total of the counts, × 15 ÷ 66632, round. One cell: $2596 \times 15/66632 = 0.58 \to 1$.

Step 2 · equalize the target B → G

z	running PDF	× 15	G
0	0	0	0
1	0	0	0
2	0	0	0
3	0.05	0.75	1
4	0.12	1.80	2
5	0.12	1.80	2
6	0.12	1.80	2
7	0.37	5.55	6
8	0.54	8.10	8
9	0.69	10.35	10
10	0.76	11.40	11
11	0.85	12.75	13
12	0.88	13.20	13
13	0.90	13.50	14
14	0.94	14.10	14
15	1	15	15

Running total of B's PDF, × 15, round. One cell: $0.05 \times 15 = 0.75 \to 1$.

Step 3 · for each s, find the nearest G

z	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
G	0	0	0	1	2	2	2	6	8	10	11	13	13	14	14	15
s lands here				1 ↑	2, 4 ↑			5, 6, 7 ↑	8, 9 ↑	10 ↑	11, 12 ↑	13 ↑		14 ↑		15 ↑

Take each s (1, 2, 4, 5 … 15) and find the same number in the G row; if it isn't there, the nearest one.

Same G twice (e.g. $G = 2$ at $z = 4, 5, 6$): take the leftmost, $z = 4$. Red cells are these skipped copies.

The four ties (equally near on both sides): $s = 4$ sits between $G = 2$ and $G = 6$, both 2 away → left, $z = 4$. Same for $s = 7$ (6 | 8 → $z = 7$), $s = 9$ (8 | 10 → $z = 8$), $s = 12$ (11 | 13 → $z = 10$).

$s = 5$ is not a tie: 6 is 1 away, 2 is 3 away → $z = 7$.

Step 4 · chain $r \to s \to z$, then count

r	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
s	1	2	2	4	5	5	6	7	8	9	10	11	12	13	14	15
z	3	4	4	4	7	7	7	7	8	8	9	10	10	11	13	15

Output histogram (add the counts that land on each z):

z	3	4	7	8	9	10	11	13	15
pixels	2596	14714	13984	8912	4746	7692	4596	4396	4996

Every other z gets 0. Total 66 632 ✓.

What to write in the exam: State the tie rule (smaller z). Table of s, table of G, then $r = 0 \ldots 15 \to z = 3, 4, 4, 4, 7, 7, 7, 7, 8, 8, 9, 10, 10, 11, 13, 15$ and the output counts.

$r = 0 \ldots 15 \to z = 3, 4, 4, 4, 7, 7, 7, 7, 8, 8, 9, 10, 10, 11, 13, 15$ **(ties to the smaller z). If ties are broken upward, only four entries change:** $r_3 \to 7$, $r_7 \to 8$, $r_9 \to 9$, $r_{12} \to 11$.

↩ back to the card

S5.10 · 6 · Sixteen levels and a tie every few rows (Sheet 5 Q2) — Answers

Look for repeated G values

z	11	12	13	14
B: PDF	0.09	0.03	0.02	0.04
G	13	~~13~~	14	~~14~~

$z = 12$ and $z = 14$ want some pixels (PDF not 0). But their G is a copy of the one just left of them, and the rule "same G → smaller z" always picks the left one. So they get nothing.

What to write in the exam: z_{12} and z_{14}: their G (13, 14) equals that of z_{11} and z_{13}, and the smaller z wins.

↩ back to the card

S5.11 · 7 · Ten-second answers — Answers

1 · equalize twice

level	1	2	4	6	7
pixels	10	8	11	15	7
→ new level	1	2	4	6	7

Run card 2's recipe on the Q5 result: every used level maps to itself. Nothing changes.

2 · Phobos washed out

Most pixels are black sky, a huge spike near 0. Equalization gives each level room in proportion to its pixels, so the spike grabs a big share of the range and everything is pushed up to light grey. Matching to a chosen histogram (curve 2, L5 p18) keeps dark pixels dark.

3 · why s and G compare

Both are equalized onto the same 0 … $L-1$ scale. "$s = 3$" and "$G = 3$" both mean "this far up the pixel ranking", like a percentile.

What to write in the exam: 1. Nothing; each level maps to itself. 2. The dark spike takes most of the range → washed out; matching keeps darks dark. 3. Both are equalized onto the same flat scale.

↩ back to the card

LECTURE 6

Spatial filtering: correlation, convolution, smoothing

Tier A · must ⏱ 55 min Slides: Lec 6-Spatial Filtering.pdf (31 p) · Sheet 5 Q3–Q5, Sheet 6 Q1–Q6 ↗ Szeliski Ch 3

Lectures 4 and 5 changed each pixel on its own: a lookup table, a histogram. Now every new pixel is computed from a small **neighbourhood** of old pixels, which is how blur, noise removal and (next lecture) sharpening all work. Almost every exam number in this lecture is a sum of nine products or a sorted list of nine values, so the skill is careful bookkeeping.

What the exam can ask from this lecture

- Turn a correlation kernel into the convolution kernel that gives the same output (rotate 180°), in 1-D and 2-D.
- Apply a filter to a small sub-image: the output size, then every output value.
- Pad an image for a given filter size (how many rows/columns, what values).
- Blur with the 3×3 averaging filter, with and without zero padding.
- 3×3 median filter with zero padding; why the median beats the mean on salt-and-pepper noise.
- Definitions: kernel/mask, linear vs nonlinear (order-statistic) filter, box vs weighted average, low-pass.

⏳ **Only 20 minutes?** Do cards 2, 5 and 6, in that order.

📖 READ & UNDERSTAND · ⏱ 8 min · Tier A

1 · A filter is a small grid of weights that slides over the image

Picture a 3×3 window on the image. The nine pixels under it are the pixel's **neighbourhood**. A **spatial filter** (also **mask**, **kernel**, template, window, L6 p3) is that neighbourhood plus an operation; the result goes into the output at the window's **centre**, then the window moves one pixel.

"Multiply by weights and add" makes the filter **linear** (L6 p4):

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)\, f(x+s,\ y+t), \qquad a = \tfrac{m-1}{2},\ b = \tfrac{n-1}{2}$$

f is the input, g the output, w the $m \times n$ kernel with $(0,0)$ at its centre. Kernels have **odd sizes** (smallest 3×3) so there is a centre to write into. Tiny example: patch 1 2 0 / 3 1 1 / 0 2 4 with kernel 1 0 1 / 0 2 0 / 1 0 1 gives $1 \cdot 1 + 2 \cdot 1 + 1 \cdot 4 = 7$ (only the non-zero weights count). Sorting the nine values and keeping one (min, max, **median**) is a **nonlinear**, order-statistic filter (L6 p7, p18).

A **low-pass** (smoothing) filter blurs: it reduces noise and small details (L6 p10–p11). Its weights are all positive and **sum to 1**, so flat regions keep their brightness. Two masks: the **box filter** $\frac{1}{9}\times$(all ones), where every neighbour counts equally, and the **weighted average** $\frac{1}{16}\times$ 1 2 1 / 2 4 2 / 1 2 1, which trusts the centre most and blurs less.

The slides' example (L6 p13–p15) is a 4×4 image, left half 10 and right half 200, with border padding. Column 2 sees six 10s and three 200s: $\frac{660}{9} = 73.3 \to$ **73**; column 3: $\frac{1230}{9} = 136.7 \to$ **137**. The step becomes a ramp 10, 73, 137, 200. The weighted mask gives $\frac{920}{16} = 57.5 \to 58$ and $\frac{2440}{16} = 152.5 \to 153$. The slides **round to the nearest integer**, halves up. The zero-padded "?" on L6 p16–p17 works out to box: 4, 49, 91, 89 (top/bottom rows) and 7, 73, 137, 133 (middle rows); weighted: 6, 43, 114, 113 and 8, 58, 153, 150. The borders go dark because the zeros are averaged in.

⚠️ **Slide error (L6 p15, p17).** These slides print the weighted mask with $\frac{1}{9}$. Its weights sum to 16, so the factor must be $\frac{1}{16}$, as on L6 p11 and p13. The slides' own 58 and 153 only come out with $\frac{1}{16}$.

🧭 **Like a magnifying glass sliding over a page.** At each stop you see one small patch and write one summary of it. *Where it breaks:* the glass only shows the patch, while a filter also weights each cell, and those weights decide whether it blurs or sharpens.

More: Wikipedia: Kernel (image processing), Gaussian blur (the 1-2-1 mask approximates one).

✏️ PEN & PAPER · ⏱ 9 min · Tier A

2 · Correlation and convolution differ by one 180° turn

IN PLAIN WORDS

① Card 1's formula is **correlation** ($w \star f$ with a hollow star): slide the kernel as printed. **Convolution** (solid star) first **rotates the kernel 180°** (flip left–right and up–down), shown by the minus signs in $\sum\sum w(s,t)\, f(x-s, y-t)$ (L6 p8). Sheet 6 Q1–Q2 give a correlation kernel and ask for the convolution kernel that gives the same output.

WARM-UP BY HAND

② Kernel [2 0 6] on an impulse [0 0 1 0 0] (zero padded). Correlation gives [0 6 0 2 0], the kernel backwards. Convolution gives [0 2 0 6 0], a true copy (2-D version: L6 p9).

THE ONE IDEA

③ Convolution with w = correlation with w rotated 180°. Hand the convolution the rotated kernel and the two rotations cancel.

BUILD IT

④ on the sibling $\begin{bmatrix} 2 & 0 & 1 \\ 3 & 5 & 4 \\ 9 & 6 & 8 \end{bmatrix}$.

1. Flip each row left–right: $\begin{bmatrix} 1 & 0 & 2 \\ 4 & 5 & 3 \\ 8 & 6 & 9 \end{bmatrix}$.
   Why: first half of the turn (in 1-D this is the whole turn: [2 0 6] → [6 0 2]).
2. Reverse the row order: $\begin{bmatrix} 8 & 6 & 9 \\ 4 & 5 & 3 \\ 1 & 0 & 2 \end{bmatrix}$.
   Why: two mirror flips make a 180° rotation, and the centre stays put.

CHECK

⑤ On $f = [3\ 1\ 4\ 1\ 5]$, correlation with [2 0 6] and convolution with [6 0 2] both give [30 8 38]. A kernel that a half-turn leaves unchanged (box, weighted) gives the same output either way. More: Wikipedia: Convolution.

🧭 **Like a rubber stamp.** The carved letters are mirrored, so pressing the stamp (convolution) prints them the right way round. *Where it breaks:* a stamp mirrors only left–right, while the kernel also flips

up–down.

EXAM PROBLEM

Your turn — Sheet 6 Q1. Given the correlation matrix below, convert it into a convolution matrix so the output is the same as the correlation, and specify the method you use.

1	7	3

Sheet 6 Q2. Do the same for the 3×3 correlation matrix:

1	7	3
8	8	7
1	4	5

✍ Show the full exam answer: S6.1 (end of this lecture)

TRY ALONE

⑦ Rotate the column $[4; 5; 6]$ by 180°.

✍ Answers: S6.2 (end of this lecture)

🖍 PEN & PAPER · ⏱ 10 min · Tier A

3 · Applying a filter: first the size, then the sums

IN PLAIN WORDS

① Sheet 6 Q3–Q4 give a **sub-image** and a filter and ask for the new size and the output. No padding is mentioned, so the window only sits where it fits completely ("ignore the edges", L6 p6) and the output **shrinks**. "Applying a filter" means the slides' formula (L6 p4), which is **correlation**.

WARM-UP BY HAND

② $[2\ 1\ 0\ 3]$ with filter $[1\ 0\ 2]$: the window fits twice. $2 \cdot 1 + 1 \cdot 0 + 0 \cdot 2 = 2$ and $1 \cdot 1 + 0 \cdot 0 + 3 \cdot 2 = 7$ give $[2\ 7]$: four inputs, two outputs.

THE ONE IDEA

③ **Output size $= n - k + 1$ per direction (input length n, kernel length k). With p zeros on each side it becomes $n + 2p - k + 1$.** The slides only say "smaller"; the formula comes from counting where the window fits.

BUILD IT

④ on a 3×4 sibling $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 \end{bmatrix}$ with kernel $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

1. Size $(3 - 3 + 1) \times (4 - 3 + 1) = 1 \times 2$.
   Why: the window fits in one vertical position and two horizontal ones.
2. Columns 1–3, using only the non-zero weights: $1 \cdot 1 + 2 \cdot 2 + 1 \cdot 1 + 1 \cdot 1 = 7$.
   Why: zero weights wipe out their pixel, so skip them.
3. Columns 2–4: $1 \cdot 2 + 2 \cdot 0 + 1 \cdot 3 + 1 \cdot 1 = 6$, so the output is $[7\ 6]$.

Why: the same four kernel positions on new pixels.

CHECK

⑤ Two values for a 1×2 output. Convolution would give [3 8], a different answer, so name the operation you used. Szeliski Ch 3.2 (szeliski.org/Book) works the same sums.

🧭 **Like a stencil on a tiled floor.** A 3-tile stencil on a 5-tile row fits in only 3 places, so you get 3 readings. *Where it breaks:* padding lets the stencil hang over the edge, which a real floor can't.

EXAM PROBLEM

Your turn — Sheet 6 Q3. Given the sub-image and filter below, what is the new size of the sub-image after applying the filter? Display the output.

1	0	1	0	1

filter

3	4	5

Sheet 6 Q4. Same question for

1	0	1	0	1
0	0	1	0	0
1	0	1	0	1
0	1	1	1	0

filter

3	4	5
1	0	1
0	4	0

✍️ Show the full exam answer: S6.3 (end of this lecture)

TRY ALONE

⑦ What is the output size for a 6-long row and a 5-long filter, with 2 zeros on each side?

✍️ Answers: S6.4 (end of this lecture)

🖍 **PEN & PAPER · ⏱ 6 min · Tier A**

4 · Padding: give the window something to stand on at the border

IN PLAIN WORDS

① On a border pixel the window hangs outside the image (L6 p6). **Padding** adds a frame of extra pixels so that every pixel can be a centre and the output keeps the input's size. Sheet 6 Q5–Q6 ask for the padding for filters of size 3 and 3×3.

WARM-UP BY HAND

② [3 1 4] with a 3-long filter becomes [0 3 1 4 0]. With a 5-long filter it becomes [0 0 3 1 4 0 0].

THE ONE IDEA

③ Pad $p = \frac{k-1}{2}$ on *every* side for a $k \times k$ filter. The output size $n + 2p - k + 1$ then equals n.

BUILD IT

④ on $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ with a 3×3 filter.

1. $p = (3-1)/2 = 1$.
   Why: the window reaches one pixel past its centre.
2. Add a ring of zeros, going from 2×2 to 4×4: $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.
   Why: $+2p$ per dimension; the original values stay untouched.

CHECK

⑤ A 3×3 filter over the 4×4 array gives $4 - 3 + 1 = 2$, so the output is back to 2×2. The slides also use **border padding**, which copies the edge pixel outward (L6 p14, p21): [3 1 4] → [3 3 1 4 4]. Zero padding darkens the borders, the "unwanted artifacts" of L6 p6. Other border rules are under "Edge handling" in Wikipedia: Kernel (image processing).

🧭 **Like the mount around a framed photo.** It adds a border without covering the picture. *Where it breaks:* a mount is never averaged into the photo, but zero padding is.

EXAM PROBLEM

Your turn — Sheet 6 Q5. Given the sub-image below, calculate its padding if the filter size is 3.

1	2	1	2	1	2	1

Sheet 6 Q6. Calculate the padding if the filter size is 3×3:

1	2	1	2	1
2	1	2	1	2
1	2	1	2	1
2	1	2	1	2
1	2	1	2	1

✍️ Show the full exam answer: S6.5 (end of this lecture)

TRY ALONE

⑦ A 10×10 image padded for a 5×5 filter: what size is the padded image?

✍️ Answers: S6.6 (end of this lecture)

🖍 PEN & PAPER · ⏱ 10 min · Tier A

5 · Box blur by hand, without padding and with zeros

IN PLAIN WORDS

① In Sheet 5 Q3 ("blur using a 3×3 filter") no padding is mentioned, so the output is the 3×3 interior (card 3). Q4 pads with zeros, so the output stays 5×5 (card 4). The filter is the slides' **box filter**: each output is the sum of the nine pixels under the window ÷ 9, rounded to the nearest integer.

WARM-UP BY HAND

② 4×4 sibling 1 3 2 0 / 4 2 1 3 / 0 1 5 2 / 2 2 0 1. The four window sums are 19, 19, 17, 17; ÷ 9 gives 2.11 and 1.89, all rounding to 2. With zero padding, the top-left corner holds only $1 + 3 + 4 + 2 = 10$, and $10/9 \to 1$.

THE ONE IDEA

③ **Add column sums, not nine separate numbers.** For each band of three rows, write each column's 3-pixel sum once. Each window sum is then three neighbouring column sums added together.

BUILD IT

④ on the sibling.

1. Rows 1–3 column sums: 5, 6, 8, 5.
   Why: each column is shared by up to three windows.
2. Window sums: $5 + 6 + 8 = 19$ and $6 + 8 + 5 = 19$, then $19/9 = 2.11 \to 2$.
   Why: the box mask weights every cell by $\frac{1}{9}$.
3. Rows 2–4: column sums 6, 5, 6, 6 give windows 17, 17, and $17/9 = 1.89 \to 2$.
   Why: slide the band down one row and repeat.

CHECK

⑤ Each blurred value lies between its window's min and max. With zero padding the border values come out lower than the interior ones. The box filter is the first entry in Wikipedia's kernel table.

Like blurry CCTV. Each dot on screen is the average of a patch of the scene, so a face becomes a smooth blob. *Where it breaks:* CCTV loses detail because it has few dots; here the pixel count stays the same and the detail is averaged away.

EXAM PROBLEM

Your turn — Sheet 5 Q3. Given the sub-image below, blur it using a 3×3 filter.

5	2	4	2	1
2	1	2	4	2
5	4	5	2	1
2	1	4	5	2
1	2	1	2	5

Sheet 5 Q4. Blur the same sub-image with padding of 0s using a 3×3 filter.

Show the full exam answer: S6.7 (end of this lecture)

TRY ALONE

⑦ In Q4, which output is the smallest, and why there?

✍ Answers: S6.8 (end of this lecture)

🖍 PEN & PAPER · ⏱ 8 min · Tier A

6 · The median filter: sort nine values, keep the middle one

IN PLAIN WORDS

① Sheet 5 Q5 asks for a 3×3 **median filter** with zero padding. For each pixel, list the nine values under the window (padding zeros included), sort them, and write down the 5th one. There are no weights and no division: it is an order-statistic (nonlinear) filter (L6 p18).

WARM-UP BY HAND

② Take the window 10 12 11 / 13 **255** 12 / 11 10 12, where 255 is a "salt" pixel. The mean is 346/9 → 38, a grey smudge. The median is 10 10 11 11 **12** 12 12 13 255 → 12, a value that was really there.

THE ONE IDEA

③ **Sort the nine values and take position 5.** One extreme pixel only ever moves to an end of the list, so the median removes **salt-and-pepper** noise while keeping edges (L6 p18–p20).

BUILD IT

④ on $\begin{bmatrix} 1 & 9 & 2 \\ 3 & 0 & 4 \\ 5 & 6 & 7 \end{bmatrix}$ with zero padding.

1. Centre: 0 1 2 3 **4** 5 6 7 9 → 4.
   Why: the window is fully inside the image.
2. Top-left: 1, 9, 3, 0 plus five zeros: 0 0 0 0 **0** 0 1 3 9 → 0.
   Why: five padding zeros fill sorted positions 1–5.
3. Top-middle: 0 0 0 0 **1** 2 3 4 9 → 1. All nine: $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 4 & 2 \\ 0 & 3 & 0 \end{bmatrix}$.
   Why: an edge window has three padding zeros.

CHECK

⑤ Every output is one of its nine window values, and with zero padding all four corners come out 0. The slides' own example (L6 p21–p23) uses border padding, which avoids those dark corners. More: Wikipedia: Median filter.

🧭 **Like dropping the highest and lowest judges' scores.** One crazy score can't move the result. *Where it breaks:* the median keeps only the single middle score instead of averaging the rest.

EXAM PROBLEM

Your turn — Sheet 5 Q5. Given the sub-image below, blur it with padding of 0s using a 3×3 median filter.

5	2	4	2	1
2	1	2	4	2
5	4	5	2	1
2	1	4	5	2
1	2	1	2	5

✍ Show the full exam answer: S6.9 (end of this lecture)

TRY ALONE

⑦ What is the median of 7 3 3 8 1 0 3 9 2?

✍ Answers: S6.10 (end of this lecture)

🔁 **RECALL DRILL · ⏱ 4 min · Tier A**

7 · Quick recall before you move on

1. Correlation or convolution: which one rotates the kernel by 180° first?
2. A 7×7 image with a 3×3 filter and no padding: what is the output size? And with $p = 1$?
3. Box vs weighted average: which one blurs more, and why?
4. Salt-and-pepper noise: mean filter or median filter? Answer in one sentence.

✍ Answers: S6.11 (end of this lecture)

⚠ Traps

- Rotating 180° is *both* flips. Flipping only left–right on a 3×3 kernel gives $\begin{bmatrix} 3 & 7 & 1 \\ 7 & 8 & 8 \\ 5 & 4 & 1 \end{bmatrix}$ for Sheet 6 Q2, which is wrong.
- "Applying a filter" in these slides means correlation (L6 p4). If you convolve, say so: Sheet 6 Q4 gives a different second row.
- No padding mentioned → the output shrinks ($n - k + 1$). Don't silently pad.
- Box blur divides by 9 even when some of the nine are padding zeros; that is why zero-padded borders come out dark.
- The weighted mask 1 2 1 / 2 4 2 / 1 2 1 divides by 16, not 9 (L6 p15 and p17 print $\frac{1}{9}$ by mistake).
- Median: count the padding zeros as values. A corner window has five of them.

Solutions and answers — Lecture 6

S6.1 · 2 · Correlation and convolution differ by one 180° turn — Show the full exam answer

The rule in words. Convolution secretly turns its kernel upside down (a 180° turn) before it slides. Correlation does not. So if you want convolution to give the same output, hand it a kernel that is *already* turned 180°: its own turn puts it back to the original. A 180° turn = flip left–right, then flip up–down.

Q1 · the 1-D kernel (one row, read left to right):

Step 1 · what you're given

	c1	c2	c3
r1	**1**	**7**	**3**

Follow the amber 1 so you can see it move.

Step 2 · flip left–right

3	**7**	**1**

Read the row backwards. The 1 jumps to the right end.

Step 3 · flip up–down

3	**7**	**1**

There is only one row, so turning it upside down changes nothing. Done: [3 7 1].

Check on a test row $f = [3\ 1\ 4\ 1\ 5]$: correlation with [1 7 3] gives $1 \cdot 3 + 7 \cdot 1 + 3 \cdot 4 = 22$, then 32, then 26. Convolution with [3 7 1] turns it back into [1 7 3] first, so it gives the same [22 32 26].

Q2 · the 3×3 kernel (rows counted from the top, columns from the left):

Step 1 · what you're given

	c1	c2	c3
r1	**1**	**7**	**3**
r2	**8**	**8**	**7**
r3	**1**	**4**	**5**

The amber 1 starts in the top-left corner.

Step 2 · flip left–right

3	**7**	**1**
7	**8**	**8**
5	**4**	**1**

Write every row backwards (1 7 3 → 3 7 1, and so on). Row order stays. The 1 is now top-right.

Step 3 · flip up–down

5	**4**	**1**
7	**8**	**8**
3	**7**	**1**

Now swap the top and bottom rows; the middle row stays. The 1 ends bottom-right: the opposite corner, as a 180° turn should do.

Quick self-check: the centre 8 never moves, and every corner swaps with the opposite corner.

What to write in the exam: Method: rotate the correlation kernel 180° (flip left–right, then up–down); convolution rotates it back, so the output is the same. Q1: [3 7 1]. Q2: $\begin{bmatrix} 5 & 4 & 1 \\ 7 & 8 & 8 \\ 3 & 7 & 1 \end{bmatrix}$.

Q1: convolution kernel [3 7 1]**. Q2: convolution kernel** $\begin{bmatrix} 5 & 4 & 1 \\ 7 & 8 & 8 \\ 3 & 7 & 1 \end{bmatrix}$**. Method: rotate the correlation kernel by 180° (flip horizontally and vertically).**

↩ back to the card

S6.2 · 2 · Correlation and convolution differ by one 180° turn — Answers

Step 1 · the kernel

r1	**4**
r2	**5**
r3	**6**

One column, read top to bottom.

Step 2 · flip left–right

4
5
6

Each row has one number, so nothing changes.

Step 3 · flip up–down

6
5
4

Top and bottom swap. The 4 goes to the bottom.

What to write in the exam: [6; 5; 4] (a column only flips top to bottom).

↩ back to the card

S6.3 · 3 · Applying a filter: first the size, then the sums — Show the full exam answer

The rule in words. "Apply the filter" = lay the filter on the image, multiply each pair of numbers that sit on top of each other, add them up. That sum is **one** output number. Then slide the filter one cell and repeat. No padding is mentioned, so the filter may only sit where it fits completely inside the image (this is correlation, the slides' formula, L6 p4).

Q3 · 1-D (image 1 0 1 0 1, filter 3 4 5):

Step 1 · window at the first position

c1	c2	c3	c4	c5
1	0	1	0	1

The filter covers the blue cells c1–c3. Pair them up with 3 4 5:

$3 \cdot 1 + 4 \cdot 0 + 5 \cdot 1 = 8$

8	?	?

That is the first output cell.

Step 2 · slide one cell right

c1	c2	c3	c4	c5
1	0	1	0	1

$3 \cdot 0 + 4 \cdot 1 + 5 \cdot 0 = 4$

8	4	?

Step 3 · count the positions = the size

8	4	8

The 3-wide window can start at c1, c2 or c3 (starting at c4 it would stick out). **3 positions → 3 outputs**, so 1×3. Formula: $5 - 3 + 1 = 3$. The last one (c3–c5 = 1 0 1) is 8 again.

What to write in the exam: No padding → size $5 - 3 + 1 = 3$, new sub-image 1×3. Output [8 4 8] (e.g. first: $3 \cdot 1 + 4 \cdot 0 + 5 \cdot 1 = 8$).

Q3: size 1×3, output [8 4 8].

(Convolution gives the same [8 4 8] here because the image reads the same backwards. With zero padding the size would stay 1×5: [4 8 4 8 4].)

Q4 · 2-D (rows counted from the top, columns from the left). The filter is

3	4	5
1	0	1
0	4	0

Step 1 · window at the first position

	c1	c2	c3	c4	c5
r1	1	0	1	0	1
r2	0	0	1	0	0
r3	1	0	1	0	1
r4	0	1	1	1	0

Put the filter on the blue 3×3 block (r1–r3, c1–c3). Multiply cell by cell, row by row, filter number first:

$3 \cdot 1 + 4 \cdot 0 + 5 \cdot 1 \; + \; 1 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 \; + \; 0 \cdot 1 + 4 \cdot 0 + 0 \cdot 1 = 9$

9	?	?
?	?	?

Step 2 · slide one column right

	c1	c2	c3	c4	c5
r1	1	0	1	0	1
r2	0	0	1	0	0
r3	1	0	1	0	1
r4	0	1	1	1	0

Now the block is r1–r3, c2–c4 (0 1 0 / 0 1 0 / 0 1 0):

$3 \cdot 0 + 4 \cdot 1 + 5 \cdot 0 \; + \; 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 0 \; + \; 0 \cdot 0 + 4 \cdot 1 + 0 \cdot 0 = 8$

9	8	?
?	?	?

Step 3 · the other cells the same way

block	sum
r1–r3, c3–c5	9
r2–r4, c1–c3	11
r2–r4, c2–c4	8
r2–r4, c3–c5	9

Shortcut: the 0s in the filter wipe out their cell, so only the 3, 4, 5, the two 1s and the bottom 4 matter.

Step 4 · finished output and its size

9	8	9
11	8	9

Across: the window can start at c1, c2, c3 → **3 positions** $(5 - 3 + 1)$. Down: it can start at r1, r2 → **2 positions** $(4 - 3 + 1)$. So the output is 2×3.

What to write in the exam: Correlation, no padding → size $(4 - 3 + 1) \times (5 - 3 + 1) = 2 \times 3$. Show one sum in full (the 9 above), then the output $\begin{bmatrix} 9 & 8 & 9 \\ 11 & 8 & 9 \end{bmatrix}$.

Q4: size 2×3, output $\begin{bmatrix} 9 & 8 & 9 \\ 11 & 8 & 9 \end{bmatrix}$ (correlation, no padding).

If the examiner means convolution, first turn the filter 180° to $\begin{bmatrix} 0 & 4 & 0 \\ 1 & 0 & 1 \\ 5 & 4 & 3 \end{bmatrix}$, then do exactly the same steps: the output becomes $\begin{bmatrix} 9 & 8 & 9 \\ 9 & 16 & 11 \end{bmatrix}$. The size is 2×3 either way.

↩ back to the card

S6.4 · 3 · Applying a filter: first the size, then the sums — Answers

Step 1 · pad the row

0	0	x	x	x	x	x	x	0	0

6 real cells (x) plus 2 zeros on each side: $6 + 2 + 2 = 10$ cells.

Step 2 · window at the first position

0	0	x	x	x	x	x	x	0	0

The 5-wide window covers cells 1–5.

Step 3 · count the positions

0	0	x	x	x	x	x	x	0	0

The last position is cells 6–10. Starts 1, 2, 3, 4, 5, 6 → **6 positions** = 6 outputs.

What to write in the exam: $6 + 2 \cdot 2 - 5 + 1 = 6$, the same length as the input.

↩ back to the card

S6.5 · 4 · Padding: give the window something to stand on at the border — Show the full exam answer

The rule in words. A 3-wide window reaches **one** cell past its centre on each side. To let the window sit centred on the very first and very last pixel, you add that one missing cell outside the image, filled with 0. In general: pad $p = \frac{k-1}{2}$ on every side (k = filter size).

Q5 · the row (filter size 3):

Step 1 · what you're given

c1	c2	c3	c4	c5	c6	c7
1	2	1	2	1	2	1

7 pixels.

Step 2 · how much padding

$p = \frac{3-1}{2} = 1$ zero on each end.

Step 3 · the padded row

0	1	2	1	2	1	2	1	0

Grey = padding. Length $7 + 2 = 9$.

Step 4 · check it works

0	1	2	1	2	1	2	1	0

Now the window (blue) can sit centred on c1. It fits at $9 - 3 + 1 = 7$ places → 7 outputs, the original length.

What to write in the exam: $p = \frac{k-1}{2} = 1$ on each side → [0 1 2 1 2 1 2 1 0], length 9; output stays 7 long.

Q5: $p = 1$ **on each side** → [0 1 2 1 2 1 2 1 0] **(length 9).**

Q6 · the 5×5 image (filter 3×3):

Step 1 · what you're given

	c1	c2	c3	c4	c5
r1	1	2	1	2	1
r2	2	1	2	1	2
r3	1	2	1	2	1
r4	2	1	2	1	2
r5	1	2	1	2	1

Rows from the top, columns from the left.

Step 2 · add a ring of zeros

		c1	c2	c3	c4	c5	
	0	0	0	0	0	0	0
r1	0	1	2	1	2	1	0
r2	0	2	1	2	1	2	0
r3	0	1	2	1	2	1	0
r4	0	2	1	2	1	2	0
r5	0	1	2	1	2	1	0
	0	0	0	0	0	0	0

$p = 1$ on *every* side: one row on top, one below, one column left, one right. $5 + 2 = 7$, so 7×7. The image itself is untouched.

Step 3 · check it works

		c1	c2	c3	c4	c5	
	0	0	0	0	0	0	0
r1	0	1	2	1	2	1	0
r2	0	2	1	2	1	2	0
r3	0	1	2	1	2	1	0
r4	0	2	1	2	1	2	0
r5	0	1	2	1	2	1	0
	0	0	0	0	0	0	0

The 3×3 window (blue) can now sit centred on the corner pixel (r1, c1). It fits at $7 - 3 + 1 = 5$ places across and 5 down → 5×5 output, same as the image.

What to write in the exam: $p = \frac{3-1}{2} = 1$ row/column of zeros on every side → 5×5 becomes 7×7 (draw it); a 3×3 filter then gives a 5×5 output.

Q6: $p = 1$ → 7×7 array, a ring of zeros around the 5×5 checkerboard; a 3×3 filter then returns a 5×5 output.

If border (replicate) padding is wanted instead, copy the edge values: Q5 becomes [1 1 2 1 2 1 2 1 1], and Q6's top padded row becomes 1 1 2 1 2 1 1 (the first image row with its end values repeated).

↩ back to the card

S6.6 · 4 · Padding: give the window something to stand on at the border — Answers

Step 1 · how much padding

A 5×5 window reaches **two** cells past its centre, so $p = \frac{5-1}{2} = 2$ on every side.

Step 2 · one row of the padded image

0	0	x	x	…	x	x	0	0

10 real pixels plus 2 zeros on each end: $2 + 10 + 2 = 14$. Same for every column.

What to write in the exam: $p = 2$, padded size 14×14 (and the 5×5 filter gives back $14 - 5 + 1 = 10$, i.e. 10×10).

↩ back to the card

S6.7 · 5 · Box blur by hand, without padding and with zeros — Show the full exam answer

The rule in words. The 3×3 blur ("box") filter is nine 1s times $\frac{1}{9}$ (L6 p11). So "apply it" just means: **add the nine pixels under the window, divide by 9, round to the nearest whole number** (halves go up, as on L6 p13). That is the average of the window.

Q3 · no padding (rows from the top, columns from the left):

Step 1 · window at the first position

	c1	c2	c3	c4	c5
r1	5	2	4	2	1
r2	2	1	2	4	2
r3	5	4	5	2	1
r4	2	1	4	5	2
r5	1	2	1	2	5

Add the nine blue cells, row by row:

$\frac{5+2+4 \;+\; 2+1+2 \;+\; 5+4+5}{9} = \frac{30}{9} = 3.33 \to 3$

3	?	?
?	?	?
?	?	?

Step 2 · slide one column right

	c1	c2	c3	c4	c5
r1	5	2	4	2	1
r2	2	1	2	4	2
r3	5	4	5	2	1
r4	2	1	4	5	2
r5	1	2	1	2	5

$\frac{2+4+2 \;+\; 1+2+4 \;+\; 4+5+2}{9} = \frac{26}{9} = 2.89 \to 3$

3	3	?
?	?	?
?	?	?

Step 3 · the other cells the same way

Window sums:

30	26	23
26	28	27
25	26	27

÷ 9:

3.33	2.89	2.56
2.89	3.11	3.00
2.78	2.89	3.00

Every one rounds to 3.

Step 4 · why the output is 3×3

Across, the window can start at c1, c2, c3 → **3 positions** ($5 - 3 + 1$). Down, at r1, r2, r3 → **3 positions**. So 3×3: the border pixels get no output.

Speed tip: add each column's three cells once (r1–r3: 12, 7, 11, 8, 4), then each window = three neighbouring column sums ($12 + 7 + 11 = 30$).

What to write in the exam: Box filter $\frac{1}{9}\times$(all ones), no padding → size $5-3+1=3$, so 3×3. First value: $\frac{30}{9}=3.33\to 3$; all nine round to 3.

Q3: $\begin{bmatrix}3&3&3\\3&3&3\\3&3&3\end{bmatrix}$ **(3×3; exact values above if your instructor keeps fractions).**

Q4 · zero padding (same image, one ring of zeros, so every pixel gets an output):

Step 1 · pad, window at the first position

		c1	c2	c3	c4	c5	
	0	0	0	0	0	0	0
r1	0	5	2	4	2	1	0
r2	0	2	1	2	4	2	0
r3	0	5	4	5	2	1	0
r4	0	2	1	4	5	2	0
r5	0	1	2	1	2	5	0
	0	0	0	0	0	0	0

Grey = padding zeros. The window is centred on (r1, c1). The zeros still count as 3 of the 9 values, and you still divide by 9:

$\frac{0+0+0\ +\ 0+5+2\ +\ 0+2+1}{9}=\frac{10}{9}=1.11\to 1$

1	?	?	?	?
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?
?	?	?	?	?

Step 2 · slide one column right

		c1	c2	c3	c4	c5	
	0	0	0	0	0	0	0
r1	0	5	2	4	2	1	0
r2	0	2	1	2	4	2	0
r3	0	5	4	5	2	1	0
r4	0	2	1	4	5	2	0
r5	0	1	2	1	2	5	0
	0	0	0	0	0	0	0

Centred on (r1, c2):

$\frac{0+0+0\ +\ 5+2+4\ +\ 2+1+2}{9}=\frac{16}{9}=1.78\to 2$

1	2	?	?	?

Step 3 · the other cells the same way

Window sums:

10	16	15	15	9
19	30	26	23	12
15	26	28	27	16
15	25	26	27	17
6	11	15	19	14

÷ 9:

1.11	1.78	1.67	1.67	1.00
2.11	3.33	2.89	2.56	1.33
1.67	2.89	3.11	3.00	1.78
1.67	2.78	2.89	3.00	1.89
0.67	1.22	1.67	2.11	1.56

The middle 3×3 (sums 30, 26, …) is exactly Q3. Only the border is new.

Step 4 · why the output is 5×5

The padded grid is 7 wide, so the window can start at $7 - 3 + 1 = 5$ places across and 5 down → 5×5, the same as the image. The border values come out small because zeros were averaged in.

What to write in the exam: Zero padding $p = 1 \rightarrow 7\times7$, output 5×5. Show the padded grid and one value: $\frac{10}{9} = 1.11 \rightarrow 1$ at the top-left. Then the rounded 5×5 below.

Q4: $\begin{bmatrix} 1 & 2 & 2 & 2 & 1 \\ 2 & 3 & 3 & 3 & 1 \\ 2 & 3 & 3 & 3 & 2 \\ 2 & 3 & 3 & 3 & 2 \\ 1 & 1 & 2 & 2 & 2 \end{bmatrix}$ **(5×5). The interior 3×3 equals Q3's answer; only the border changes.**

↩ back to the card

S6.8 · 5 · Box blur by hand, without padding and with zeros — Answers

Step 1 · find the smallest window sum

In Q4's sums grid the smallest is **6**, bottom-left (r5, c1).

Step 2 · look at its window

		c1	c2	c3	c4	c5	
	0	0	0	0	0	0	0
r1	0	5	2	4	2	1	0
r2	0	2	1	2	4	2	0
r3	0	5	4	5	2	1	0
r4	0	2	1	4	5	2	0
r5	0	1	2	1	2	5	0
	0	0	0	0	0	0	0

Five of the nine cells are padding zeros (a corner window). The four real pixels are small:

$\frac{0+2+1\ +\ 0+1+2\ +\ 0+0+0}{9} = \frac{6}{9} = 0.67 \to 1$

What to write in the exam: The bottom-left corner, $6/9 = 0.67 \to 1$: its window has five padding zeros and only four small real pixels (2, 1, 1, 2).

↩ back to the card

S6.9 · 6 · The median filter: sort nine values, keep the middle one — Show the full exam answer

The rule in words. No multiplying, no dividing. For each pixel: write down the 9 numbers under the 3×3 window (padding zeros count!), sort them from small to big, and keep the one in the middle (the 5th). That middle one is the output.

Step 1 · pad, window at the first position

		c1	c2	c3	c4	c5	
	0	0	0	0	0	0	0
r1	0	5	2	4	2	1	0
r2	0	2	1	2	4	2	0
r3	0	5	4	5	2	1	0
r4	0	2	1	4	5	2	0
r5	0	1	2	1	2	5	0
	0	0	0	0	0	0	0

One ring of zeros (grey) so every pixel can be a centre. Window centred on (r1, c1).

Step 2 · list, sort, take the middle

The 9 values, row by row:

0	0	0	0	5	2	0	2	1

Sorted:

0	0	0	0	**0**	1	2	2	5

Middle (5th) = **0**.

0	?	?	?	?

Step 3 · slide one column right

Centred on (r1, c2): values 0 0 0 / 5 2 4 / 2 1 2.

0	0	0	1	**2**	2	2	4	5

Middle = **2**.

0	2	?	?	?

Step 4 · one window fully inside

		c1	c2	c3	c4	c5	
	0	0	0	0	0	0	0
r1	0	5	2	4	2	1	0
r2	0	2	1	2	4	2	0
r3	0	5	4	5	2	1	0
r4	0	2	1	4	5	2	0
r5	0	1	2	1	2	5	0
	0	0	0	0	0	0	0

Centred on (r2, c2): 5 2 4 / 2 1 2 / 5 4 5, no zeros.

1	2	2	2	**4**	4	5	5	5

Middle = **4**.

Step 5 · the size

Padded grid 7 wide → the window has $7 - 3 + 1 = 5$ positions across and 5 down → 5×5 output, one value per original pixel. Corner windows hold five zeros, so corners always come out 0.

The other cells the same way. All 25 sorted windows, for checking (middle in bold), position = (row, column):

pixel	sorted window	median
(1,1)	0 0 0 0 0 1 2 2 5	0
(1,2)	0 0 0 1 2 2 2 4 5	2
(1,3)	0 0 0 1 2 2 2 4 4	2
(1,4)	0 0 0 1 2 2 2 4 4	2
(1,5)	0 0 0 0 0 1 2 2 4	0
(2,1)	0 0 0 1 2 2 4 5 5	2
(2,2)	1 2 2 2 4 4 5 5 5	4
(2,3)	1 2 2 2 2 4 4 4 5	2
(2,4)	1 1 2 2 2 2 4 4 5	2
(2,5)	0 0 0 1 1 2 2 2 4	1
(3,1)	0 0 0 1 1 2 2 4 5	1
(3,2)	1 1 2 2 2 4 4 5 5	2
(3,3)	1 1 2 2 4 4 4 5 5	4
(3,4)	1 2 2 2 2 4 4 5 5	2
(3,5)	0 0 0 1 2 2 2 4 5	2
(4,1)	0 0 0 1 1 2 2 4 5	1
(4,2)	1 1 1 2 2 4 4 5 5	2
(4,3)	1 1 2 2 2 4 4 5 5	2
(4,4)	1 1 2 2 2 4 5 5 5	2
(4,5)	0 0 0 1 2 2 2 5 5	2
(5,1)	0 0 0 0 0 1 1 2 2	0
(5,2)	0 0 0 1 1 1 2 2 4	1
(5,3)	0 0 0 1 1 2 2 4 5	1
(5,4)	0 0 0 1 2 2 4 5 5	2

(5,5)	0 0 0 0 0 2 2 5 5	0

What to write in the exam: Zero padding $p = 1 \rightarrow 7\times7$; output 5×5. Show one window in full: (r1, c1) sorted 0 0 0 0 **0** 1 2 2 5 → 0. Then the 5×5 grid below.

Q5: $\begin{bmatrix} 0 & 2 & 2 & 2 & 0 \\ 2 & 4 & 2 & 2 & 1 \\ 1 & 2 & 4 & 2 & 2 \\ 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 1 & 2 & 0 \end{bmatrix}$ **(5×5).**

↩ back to the card

S6.10 · 6 · The median filter: sort nine values, keep the middle one — Answers

Step 1 · the 9 values

7	3	3	8	1	0	3	9	2

Step 2 · sort, take the 5th

0	1	2	3	**3**	3	7	8	9

Four values on each side of the circled one.

What to write in the exam: Sorted 0 1 2 3 **3** 3 7 8 9 → median 3.

↩ back to the card

S6.11 · 7 · Quick recall before you move on — Answers

1 · which one turns the kernel

Convolution (L6 p8). Correlation slides the kernel as printed.

2 · count window positions

x	x	x	x	x	x	x

On one 7-cell row, a 3-wide window starts at cells 1…5 → **5 positions** $(7 - 3 + 1)$, so 5×5.

0	x	x	x	x	x	x	x	0

With $p = 1$ the row is 9 long → $9 - 3 + 1 = 7$ positions, so 7×7.

3 · box vs weighted

1	1	1
1	1	1
1	1	1

1	2	1
2	**4**	2
1	2	1

The **box** blurs more: its corners count as much as its centre. The weighted one trusts the centre (4) most.

4 · salt-and-pepper

Median. Sorting pushes the outlier (e.g. 255) to the end of the list, so it is never the middle one. The mean adds it in and smears it.

What to write in the exam: 1 convolution. 2 5×5; 7×7 with $p = 1$. 3 box (equal weights). 4 median: the outlier is sorted to an end and never chosen.

↩ back to the card

LECTURE 7

Sharpening: derivatives, the Laplacian, unsharp masking, Sobel

Tier A · must ⏱ 53 min

Slides: Lec 7-Sharpening (40 p; p2–p18 sharpening, p20–p39 image features & SIFT) · Sheet 6 Q7–Q13 ↗ Szeliski Ch 3

Lecture 6 blurred images by averaging neighbours. This lecture does the opposite: it measures how fast brightness **changes** from one pixel to the next and adds that change back, so edges get crisper. Every tool here (Laplacian, unsharp mask, Sobel) is still a small mask slid over the image, so the sliding-and-multiplying you learnt in Lecture 6 does all the arithmetic.

What the exam can ask from this lecture

- First and second derivative of a 1-D row of pixels (Sheet 6 Q7, Q8), and which one is zero on a ramp.
- Derive the 3×3 Laplacian mask from $\partial^2 f/\partial x^2 + \partial^2 f/\partial y^2$ (Q10) and apply it to a small image (Q9, Q12).
- Sharpen: $g = f - \nabla^2 f$ when the mask centre is negative, $g = f + \nabla^2 f$ when it is positive (L7 p11).
- Unsharp masking and high-boost on a blurred edge (L7 p16).
- Build a 3×3 Gaussian mask from its formula (Q11); low-pass versus high-pass (Q13).
- Sobel gradient of a pixel, magnitude $|G_x| + |G_y|$ (L7 p17–p18).
- Short theory: edge / corner / blob / texture features; SIFT advantages and disadvantages (L7 p25–p39).

⏳ **Only 20 minutes?** Do cards 1, 2 and 4, in that order.

PEN & PAPER · ⏱ 9 min · Tier A

1 · Derivatives of a row of pixels (Sheet 6 Q7, Q8)

Read a row left to right: 1, 1, 1, then a jump to 3. A computer notices the jump by **subtracting neighbours**; the whole lecture is built on that.

IN PLAIN WORDS

① A **derivative** measures how much the brightness changes as you step one pixel. Pixels sit on a grid, so there is no limit to take; the slides (L6 p24, repeated in L7 p2) use differences. With $f(x)$ the brightness at position x:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x) \qquad \frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

The **first derivative** is "next minus me". The **second derivative** is "my two neighbours minus twice me", which is the change of the change. Q7 calls the first one "Laplacian first derivative"; the Laplacian is really the second-derivative idea, so read Q7 as "first derivative" and Q8 as "second derivative".

WARM-UP BY HAND

② Row $f = [2,\ 2,\ 5,\ 5,\ 3,\ 3]$, positions $x = 0 \ldots 5$.

x	0	1	2	3	4	5
f	2	2	5	5	3	3
f′ = f(x+1) − f(x)	0	3	0	−2	0	–
f″ = f(x+1)+f(x−1)−2f(x)	–	3	−3	−2	2	–

THE ONE IDEA

③ The first derivative fires once per jump; the second derivative fires twice, with opposite signs, on the two sides of the jump. Flat stretches give 0 in both. That sign flip (a **zero crossing**) is how the second derivative pinpoints an edge.

BUILD IT

④

1. $f'(1) = 5 - 2 = 3$, $f'(3) = 3 - 5 = -2$, the rest 0.
   Why: "next minus me"; the last pixel has no "next", so 6 pixels give 5 values.
2. $f''(1) = 5 + 2 - 4 = 3$, $f''(2) = 5 + 2 - 10 = -3$, $f''(3) = 3 + 5 - 10 = -2$, $f''(4) = 3 + 5 - 6 = 2$.
   Why: the two end pixels lack a neighbour, so 6 pixels give 4 values.

CHECK

⑤ Add up all the first derivatives: $0 + 3 + 0 - 2 + 0 = 1$, and indeed the row ends at 3 after starting at 2. That telescoping sum is a free check for Q7.

🧭 **Like speed and acceleration in a car.** The first derivative is the speed of the brightness climb; the second is the push you feel only when the speed changes, at the start and end of a climb. *Where it breaks:* pixels jump in whole steps, so there is no smooth limit.

EXAM PROBLEM

Your turn — Sheet 6 Q7 and Q8. Q7: "Use Laplacian first derivative in the below array." Q8: "Use laplacian second derivative in the below array."

1	1	1	3	3	5	6	6	6	2	1	1	1

✍️ Show the full exam answer: S7.1 (end of this lecture)

TRY ALONE

⑦ Second derivative of the ramp $[1, 2, 3, 4]$?

✍️ Answers: S7.2 (end of this lecture)

↗ Discrete Laplace operator (Wikipedia)

🖍 PEN & PAPER · ⏱ 6 min · Tier A

2 · From the second derivative to the 3×3 Laplacian mask (Sheet 6 Q10)

An image has rows *and* columns. What if you take card 1's second derivative across the row, again down the column, and add them?

IN PLAIN WORDS

① The **Laplacian** is that sum (L7 p2): $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$. A **mask** (or kernel) is the little grid of weights you slide over the image; at each pixel you multiply each weight by the pixel under it and add (correlation, Lecture 6). Q10 asks: which 3×3 grid of weights computes $\nabla^2 f$?

WARM-UP BY HAND

② One direction only: $f(x-1) - 2f(x) + f(x+1)$ has weights $[1\ -2\ 1]$ on (left, centre, right); the y direction gives the same weights standing up. Patch: centre 7, up 12, left 8, right 11, down 9. Across $8 + 11 - 14 = 5$, down $12 + 9 - 14 = 7$, sum $\nabla^2 f = 12$.

THE ONE IDEA

③ Stack the horizontal $[1\ -2\ 1]$ and the vertical $[1\ -2\ 1]$ on one 3×3 grid: the two centres add to -4, the four side neighbours get 1, the corners get 0.

BUILD IT

④

1. $\frac{\partial^2 f}{\partial x^2} = f(x+1,y) + f(x-1,y) - 2f(x,y)$, $\frac{\partial^2 f}{\partial y^2} = f(x,y+1) + f(x,y-1) - 2f(x,y)$
   Why: card 1's formula, holding y fixed, then x.
2. $\nabla^2 f = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)$
   Why: add them; the two $-2f(x,y)$ terms merge into $-4f(x,y)$ (L7 p3).
3. Read off the weight of each position: centre -4, the four direct neighbours 1, the four diagonals 0 (they never appear).
   Why: a mask is the list of coefficients of the formula, placed where each pixel sits.

CHECK

⑤ The weights add to 0, so a flat patch gives 0. Warm-up patch: $12 + 8 + 11 + 9 - 4 \cdot 7 = 12$, as in ②.

Cousins on L7 p9: -8 centre with diagonals, and the sign-flipped $+4$ and $+8$. The sign matters for **sharpening**, which adds the Laplacian back to the original (L7 p11):

$$g(x,y) = \begin{cases} f(x,y) - \nabla^2 f(x,y) & \text{centre of the mask negative } (-4,-8) \\ f(x,y) + \nabla^2 f(x,y) & \text{centre of the mask positive } (+4,+8) \end{cases}$$

L7 p12–p13: rows 50, 60, 90, 100 with the $+4$ mask; the "60" pixel gives $240 - 50 - 90 - 60 - 60 = -20$, so $g = 40$. The row becomes 40, 40, 110, 110: a steeper edge.

🧭 **Like the "sharpen" slider in a phone photo editor.** *Where it breaks:* it boosts grain too, since to the Laplacian a speck of TV static looks like a tiny edge.

EXAM PROBLEM

Your turn — Sheet 6 Q10. "Use Laplacian second derivative to construct 3x3 filter mask."

✍️ Show the full exam answer: S7.3 (end of this lecture)

TRY ALONE

⑦ You used the $0, -1, 0 / -1, 4, -1 / 0, -1, 0$ mask. Plus or minus when sharpening?

✍ Answers: S7.4 (end of this lecture)

↗ Discrete Laplace operator (Wikipedia) · ↗ Szeliski Ch 3

🖍 PEN & PAPER · ⏱ 7 min · Tier A

3 · Sliding the Laplacian over a small image (Sheet 6 Q9)

Q9's arithmetic is easy; the real decision is **what to do at the border**, where the mask hangs off the picture.

IN PLAIN WORDS

① Place the centre of the -4 mask on a pixel, compute up + down + left + right $-$ 4 · centre, move on. Border pixels miss neighbours; L6 p6 offers **ignore the edges** (output shrinks by 1 on each side) or **pad with zeros** (same size, fake edges). Main answer: interior; padded version as a second reading.

WARM-UP BY HAND

② Image $\begin{bmatrix} 2 & 2 & 2 & 2 \\ 5 & 5 & 5 & 5 \\ 5 & 5 & 5 & 5 \end{bmatrix}$: interior is 1×2, both pixels have up 2, down/left/right 5, centre 5, so $17 - 20 = -3$.

THE ONE IDEA

③ The output size is (rows − 2) × (cols − 2) when you ignore the edges; every number is "four neighbours minus four times the centre".

BUILD IT

④

1. Size: $3 \times 4 \to (3-2) \times (4-2) = 1 \times 2$.
   Why: a 3×3 mask needs one pixel of room on every side (Lecture 6's $n - k + 1$).
2. Pixels (1, 1) and (1, 2): $2 + 5 + 5 + 5 - 4 \cdot 5 = -3$.
   Why: the darker row above pulls the sum below $4 \cdot 5$; negative means brighter than the neighbours' average.

CHECK

⑤ Turn the 2s into 5s and you get $20 - 20 = 0$: flat gives 0.

🧭 **Like a magnifying glass sliding over a page:** it rests only where the whole lens fits; at the margins you stop early or tape blank paper around (zero padding). *Where it breaks:* the blank paper is not neutral, the Laplacian sees the jump to 0 as an edge.

EXAM PROBLEM

Your turn — Sheet 6 Q9. "Use Laplacian second derivative in the array below."

1	1	1	1	1
4	4	4	4	4
2	4	4	4	2

4	4	4	4	4

✍ Show the full exam answer: S7.5 (end of this lecture)

TRY ALONE

⑦ Zero padding: what does the top-left pixel (value 1) of Q9 give?

✍ Answers: S7.6 (end of this lecture)

↗ Kernel (image processing), Wikipedia

🖍 PEN & PAPER · ⏱ 6 min · Tier A

4 · A clean step edge, filtered and then sharpened (Sheet 6 Q12)

Q12 is the textbook edge: rows of 50 on rows of 100. Guess first: the Laplacian is non-zero **only on the two rows touching the boundary**.

IN PLAIN WORDS

① Apply the 3×3 Laplacian mask (centre −4, card 2) to every interior pixel, then, to show what it is for, sharpen with $g = f - \nabla^2 f$ (negative centre, L7 p11). An **edge** is a place where brightness changes sharply between two regions (L7 p26).

WARM-UP BY HAND

② Image with rows 10, 10, 30, 30 (4×4, every row constant). Interior = rows 1–2, columns 1–2.

row	up	down	left + right	centre	$\nabla^2 f$	$g = f - \nabla^2 f$
1 (value 10)	10	30	20	10	60 - 40 = 20	10 - 20 = -10 → clip to 0
2 (value 30)	10	30	60	30	100 - 120 = -20	30 + 20 = 50

THE ONE IDEA

③ A step gives a +/− pair on the two rows beside it and 0 everywhere else; subtracting it pushes the dark side darker and the bright side brighter. That overshoot is what your eye reads as "sharper".

BUILD IT

④

1. Dark row: $10 + 30 + 10 + 10 - 40 = 20$; bright row: $10 + 30 + 30 + 30 - 120 = -20$.
   Why: only the neighbour across the edge differs from the centre.
2. Sharpen: $10 - 20 = -10 \to 0$ (clipped), $30 + 20 = 50$.
   Why: negative centre, so subtract (L7 p11); 8-bit pixels cannot go below 0.

CHECK

⑤ Rows far from the edge have all neighbours equal to them, so they give 0.

EXAM PROBLEM

Your turn — Sheet 6 Q12. "Using the 3x3 Laplacian filter on the below sub-image."

50	50	50	50	50	50

50	50	50	50	50	50
50	50	50	50	50	50
100	100	100	100	100	100
100	100	100	100	100	100
100	100	100	100	100	100

✍ Show the full exam answer: S7.7 (end of this lecture)

TRY ALONE

⑦ Rows 20, 20, 20, 80, 80, 80: interior Laplacian values, row by row?

✍ Answers: S7.8 (end of this lecture)

↗ Discrete Laplace operator (Wikipedia)

🖍 **PEN & PAPER · ⏱ 6 min · Tier A**

5 · Unsharp masking and high-boost: sharpen by subtracting a blur

An old print-shop trick (L7 p15): to sharpen, make a *blurrier* copy; the difference is **the detail the blur destroyed**, and you add it back.

IN PLAIN WORDS

① f is the original, $\bar{f}$ (written f' on L7 p16) is a blurred copy made with the 3×3 average filter from Lecture 6. The **mask** here is not a kernel but an image: mask $= f - \bar{f}$. Then

$$g_{\text{unsharp}} = f + \text{mask}, \qquad g_{\text{high-boost}} = f + k \cdot \text{mask}, \; k > 1$$

High-boost turns the volume of the mask up (the slide uses $k = 3$).

WARM-UP BY HAND

② Row $f = [30, 30, 60, 60]$, ends copied outward: $\bar{f} = [30, 40, 50, 60]$, mask $= [0, -10, 10, 0]$, $g = f + \text{mask} = [30, 20, 70, 60]$, and with $k = 2$: $[30, 10, 80, 60]$.

THE ONE IDEA

③ $f - \bar{f}$ is zero on flat ground and a −/+ pair at an edge; adding it digs a dip before the edge and builds a bump after it.

BUILD IT

④

1. Blur: $\bar{f}(1) = (30 + 30 + 60)/3 = 40$.
   Why: identical rows turn the 3×3 average into a 3-pixel average along the row.
2. Mask $30 - 40 = -10$; $g = 30 - 10 = 20$, or $30 + 2(-10) = 10$ with $k = 2$.
   Why: pushing against the blur undoes it, and a bit more.

CHECK

⑤ The gap across the edge grew from 30 to 50 (unsharp) and 70 ($k = 2$); flat pixels did not move.

🧭 **Like an old darkroom trick:** lay an out-of-focus negative over the print and only fine detail survives (hence "unsharp mask"). *Where it breaks:* a big k amplifies noise and leaves bright halos.

EXAM PROBLEM

Your turn — L7 p16. "Apply Unsharp filter on the following blurred edge." Every row of f is $[50,\ 60,\ 90,\ 100]$ (4×4). Blur with the 3×3 standard average (all weights $1/9$), then give $g_{\text{unsharp}} = f + \text{mask}$ and $g_{\text{highboost}} = f + 3 \cdot \text{mask}$.

✍️ Show the full exam answer: S7.9 (end of this lecture)

TRY ALONE

⑦ Unsharp the row $[30, 30, 60, 60]$ with $k = 3$.

✍️ Answers: S7.10 (end of this lecture)

↗ Unsharp masking (Wikipedia)

✏️ PEN & PAPER · ⏱ 5 min · Tier A

6 · Building a 3×3 Gaussian mask from its formula (Sheet 6 Q11)

A box average gives every neighbour the same vote; a **Gaussian** mask lets near pixels count more, following the bell curve.

IN PLAIN WORDS

① $h(x,y) = \dfrac{1}{2\pi\sigma^2}\, e^{-\frac{x^2+y^2}{2\sigma^2}}$, where (x, y) is the position *relative to the mask centre* (so $x, y \in \{-1, 0, 1\}$) and σ is the **standard deviation**, the width of the bell (the sheet writes it as α). Only $x^2 + y^2$ matters, and on a 3×3 grid it takes three values: 0 (centre), 1 (the four side cells), 2 (the four corners).

WARM-UP BY HAND

② Same recipe with $\sigma = 2$: $2\sigma^2 = 8$ and $\frac{1}{2\pi\sigma^2} = \frac{1}{8\pi} = 0.0398$.

cell	$x^2 + y^2$	h	normalised ($\div 0.3042$)
centre	**0**	$0.0398 \cdot e^{0} = 0.0398$	**0.1308**
side (×4)	**1**	$0.0398 \cdot e^{-1/8} = 0.0351$	**0.1154**
corner (×4)	**2**	$0.0398 \cdot e^{-1/4} = 0.0310$	**0.1019**

THE ONE IDEA

③ Compute three numbers (centre, side, corner), place them symmetrically, then divide by their total so the nine weights add to 1. Without that last step a blur would darken the image (weights summing to 0.30 keep only 30% of the brightness).

BUILD IT

④

1. $h = 0.0398$ (centre), $0.0398 \cdot e^{-1/8} = 0.0351$ (side), $0.0398 \cdot e^{-2/8} = 0.0310$ (corner).
   Why: $x^2 + y^2$ is 0, 1, 2 for centre, side, corner.
2. Total $0.0398 + 4(0.0351) + 4(0.0310) = 0.3042$; divide each weight by it.

Why: weights summing to 1 keep flat areas at their brightness.

CHECK

⑤ $0.1308 + 4(0.1154) + 4(0.1019) = 1.000$.

🧭 **Like Instagram's soft-focus blur** versus a blocky box blur: influence fades smoothly with distance. *Where it breaks:* the true bell never reaches zero; a 3×3 cut drops its tails, hence the renormalising.

EXAM PROBLEM

Your turn — Sheet 6 Q11. "Using the gaussian function below to construct 3x3 filter mask. $h(x, y) = \frac{1}{2\pi\alpha^2} e^{-\frac{x^2+y^2}{2\alpha^2}}$, and α which is standard deviation equals 1."

✍️ Show the full exam answer: S7.11 (end of this lecture)

TRY ALONE

⑦ If σ shrinks to 0.5, does the normalised centre weight go up or down?

✍️ Answers: S7.12 (end of this lecture)

↗ Gaussian blur (Wikipedia)

📖 **READ & UNDERSTAND · ⏱ 3 min · Tier A**

7 · Low-pass versus high-pass: the two families side by side (Sheet 6 Q13)

Think of an image as slow changes (large smooth areas, the sky) plus fast changes (edges, fine texture, noise). **Frequency** here means how quickly brightness changes from pixel to pixel. A **low-pass filter** lets the slow changes through and removes the fast ones: it smooths (L6 p10). A **high-pass filter** does the reverse: it keeps the fast changes and removes the flat background, so edges and detail stand out.

The weight sum gives it away. Average and Gaussian masks have positive weights **summing to 1**: a flat area of 80 stays 80. The Laplacian and Sobel masks have weights **summing to 0**: a flat area of 80 becomes $80 \cdot 0 = 0$, only change survives. Tiny number: the row $[10, 10, 40, 40]$ under the average $[1, 1, 1]/3$ gives 20 and 30 in the middle (the jump of 30 is softened to 10 per step), while the second derivative gives $+30$ and -30 there (only the jump shows).

🧭 **Like the bass and treble knobs on a speaker.** Low-pass keeps the bass (smooth shading), high-pass keeps the treble (crisp edges, and also the hiss). *Where it breaks:* images are 2-D, so "fast change" can point in any direction, which is why Sobel needs two masks.

EXAM PROBLEM

Your turn — Sheet 6 Q13. "What is the difference between low-pass filter and high-pass filter."

✍️ Show the full exam answer: S7.13 (end of this lecture)

↗ Kernel (image processing), Wikipedia

🖍 **PEN & PAPER · ⏱ 6 min · Tier A**

8 · The Sobel gradient: how strong is the edge at this pixel?

The Laplacian ignores direction; Sobel measures change **along each axis separately**, then combines them.

IN PLAIN WORDS

① The **gradient** is the pair (change left-to-right, change top-to-bottom). Sobel computes each with a 3×3 mask (L7 p17). Using the slide's labels:

G_x (right column − left column)			G_y (bottom row − top row)		
-1	0	1	-1	-2	-1
-2	0	2	0	0	0
-1	0	1	1	2	1

Edge strength (**magnitude**) on the slide: $|\nabla f| \approx |G_x| + |G_y|$ (L7 p18), a cheap stand-in for $\sqrt{G_x^2 + G_y^2}$.

WARM-UP BY HAND

② Neighbourhood $\begin{bmatrix} 10 & 10 & 50 \\ 10 & 10 & 50 \\ 10 & 10 & 50 \end{bmatrix}$, a vertical edge: $G_x = (50-10) + 2(50-10) + (50-10) = 160$, $G_y = 80 - 80 = 0$, so $|G| \approx 160$.

THE ONE IDEA

③ Each Sobel mask is "one side minus the other side" with weights 1, 2, 1; a flat neighbourhood gives 0, an edge across the mask gives a big number. G_x reacts to vertical edges (brightness changes as you go right), G_y to horizontal ones.

BUILD IT

④

1. G_x: right column $(50, 50, 50) \cdot (1, 2, 1) = 200$, left column -40; total 160.
   Why: correlation, weight times pixel, then add.
2. $G_y = 80 - 80 = 0$; $|G| \approx 160 + 0$.
   Why: identical rows, nothing changes going down.

CHECK

⑤ All 10s gives 0 for both masks: a constant area has zero gradient (L7 p18). Not on the slides: **Prewitt** uses 1, 1, 1 instead of 1, 2, 1; **Roberts** uses 2×2 diagonal differences.

Like a spirit level laid twice, east–west and north–south; together they give the steepness. *Where it breaks:* $|G_x| + |G_y|$ overstates diagonal edges by up to about 1.4×.

EXAM PROBLEM

Your turn — L7 p18. "Find the gradient of the following shaded pixels" with the Sobel masks above and $V = |G_x| + |G_y|$. Image (4×4):

10	10	200	200
10	10	200	200

100	100	20	20
100	100	20	20

The slide asks for pixel (1,1); find it, then the other three interior pixels.

✍ Show the full exam answer: S7.14 (end of this lecture)

TRY ALONE

⑦ G_y for $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 9 & 9 & 9 \end{bmatrix}$?

✍ Answers: S7.15 (end of this lecture)

↗ Sobel operator (Wikipedia) · ↗ Prewitt · ↗ Roberts cross

📖 **READ & UNDERSTAND · ⏱ 2 min · Tier C**

9 · Image features and SIFT, in one paragraph

Edges are one of four kinds of **image features**, the distinctive parts a computer uses to categorise and match pictures (L7 p25–p29): **edges** (boundaries where intensity changes sharply), **corners** (where two or more edges meet), **blobs** (regions of similar intensity or colour, often round objects) and **texture** (a repeating pattern). **SIFT** (L7 p35–p39) finds keypoints and describes them by local gradient directions. Advantages: invariant to scale and rotation, distinctive, robust to noise and partial occlusion. Disadvantages: computationally expensive (scale-space pyramid, keypoints, descriptors) and memory-hungry. Tiny number: the corner patch $0, 0, 0/0, 9, 9/0, 9, 9$ gives Sobel $G_x = G_y = 27$, strong in both directions, while an edge is strong in one.

🧭 **Like recognising a friend by their face, not their pixel position.** *Where it breaks:* a blank wall has no keypoints to match.

↗ OpenCV SIFT tutorial (linked on L7 p35)

🔁 **RECALL DRILL · ⏱ 3 min · Tier A**

10 · Quick recall before the exam

1. Write the first and the second derivative of $f(x)$ as differences.
2. Second derivative of $[5, 5, 9, 9]$ at its two interior points?
3. You sharpen with the mask whose centre is -8. Is it $g = f + \nabla^2 f$ or $g = f - \nabla^2 f$?
4. Unsharp masking: write the mask and the high-boost formula.
5. Sum of weights of a Gaussian mask after normalising? Of a Laplacian mask?
6. Sobel gives $G_x = -30$, $G_y = 40$. Edge strength by the slides' rule?

✍ Answers: S7.16 (end of this lecture)

⚠ Traps

- Sheet 6 Q7 says "Laplacian first derivative". There is no such thing: compute the plain first derivative $f(x+1) - f(x)$ and say so in one line.

- Count your outputs: n pixels give $n-1$ first-derivative values and $n-2$ second-derivative values; a 3×3 mask with edges ignored shrinks each side by 2.
- The sharpening sign follows the mask centre: negative centre (-4, -8) means subtract, positive centre means add. Mixing them blurs instead of sharpening.
- Gaussian: (x, y) are offsets from the centre $(-1, 0, 1)$, not pixel indices 0–2; and normalise so the weights add to 1.
- Slide L7 p18 says the gradient at (1,1) is 0 "because it is in a constant area". In the 4×4 image shown, the interior pixel (1,1) is next to the 200s and 100s and has $|G_x| + |G_y| = 580$; the 0 only holds for the top-left corner with copied borders. State which reading you use. The same slide also swaps the G_x/G_y names between figure and arithmetic.
- Sharpened values can fall below 0 or rise above 255: clip them (card 4's warm-up).

Solutions and answers — Lecture 7

S7.1 · 1 · Derivatives of a row of pixels (Sheet 6 Q7, Q8) — Show the full exam answer

Step 1 · number the pixels

x	0	1	2	3	4	5	6	7	8	9	10	11	12
f	1	1	1	3	3	5	6	6	6	2	1	1	1

The top row is just the position x, counted from 0 on the left. The bottom row is the brightness f at that position. 13 pixels.

Step 2 · Q7, one first derivative in full

x	0	1	2	3	4	5	6	7	8	9	10	11	12
f	1	1	1	3	3	5	6	6	6	2	1	1	1
f′			2										

Rule in words: **next pixel minus this pixel**. For $x = 2$ you use the two blue cells: $f'(2) = f(3) - f(2) = 3 - 1 = 2$. The answer goes in the dark cell, under $x = 2$.

Step 3 · slide along: the full first-derivative row

x	0	1	2	3	4	5	6	7	8	9	10	11	12
f	1	1	1	3	3	5	6	6	6	2	1	1	1
f′	0	0	2	0	2	1	0	0	-4	-1	0	0	–

Every other cell the same way: right cell minus left cell. The last pixel ($x = 12$) has no "next", so it gets nothing: 12 values. Quick check: they add to 0, and the row starts and ends at 1.

Step 4 · Q8, one second derivative in full

x	0	1	2	3	4	5	6	7	8	9	10	11	12
f	1	1	1	3	3	5	6	6	6	2	1	1	1
f″				-2									

Rule in words: **left neighbour + right neighbour − 2 × this pixel**. Now you need three cells. For $x = 3$: $f''(3) = f(4) + f(2) - 2f(3) = 3 + 1 - 2 \cdot 3 = -2$.

Step 5 · slide along: the full second-derivative row

x	0	1	2	3	4	5	6	7	8	9	10	11	12
f	1	1	1	3	3	5	6	6	6	2	1	1	1
f″	–	0	2	-2	2	-1	-1	0	-4	3	1	0	–

The two end pixels miss a neighbour, so they get nothing: 11 values. Look at the jumps: each one gives a + and a − side by side (2, −2 at the step 1→3; −4, 3 at the drop 6→2). That sign flip marks the edge.

What to write in the exam: Q7: $f'(x) = f(x+1) - f(x)$ (the sheet's "Laplacian first derivative" means the first derivative), $f' = [0, 0, 2, 0, 2, 1, 0, 0, -4, -1, 0, 0]$. Q8: $f''(x) = f(x+1) + f(x-1) - 2f(x)$, $f'' = [0, 2, -2, 2, -1, -1, 0, -4, 3, 1, 0]$ for $x = 1 \ldots 11$. Show one subtraction in full for each.

$f' = [0,\ 0,\ 2,\ 0,\ 2,\ 1,\ 0,\ 0,\ -4,\ -1,\ 0,\ 0]$ · $f'' = [0,\ 2,\ -2,\ 2,\ -1,\ -1,\ 0,\ -4,\ 3,\ 1,\ 0]$ **(for** $x = 1 \ldots 11$**)**

↩ back to the card

S7.2 · 1 · Derivatives of a row of pixels (Sheet 6 Q7, Q8) — Answers

Step 1 · one value in full

x	0	1	2	3
f	1	2	3	4
f″		0		

Left + right − 2 × me, at $x = 1$: $f''(1) = 3 + 1 - 2 \cdot 2 = 0$.

Step 2 · the full row

x	0	1	2	3
f	1	2	3	4
f″	–	0	0	–
f′	1	1	1	–

$x = 2$: $4 + 2 - 6 = 0$. A steady ramp climbs by the same 1 each step, so the "change of the change" is 0 (L6 p26).

What to write in the exam: $f'' = [0, 0]$; on a ramp the first derivative is constant (1) and the second is 0.

↩ back to the card

S7.3 · 2 · From the second derivative to the 3×3 Laplacian mask (Sheet 6 Q10) — Show the full exam answer

Step 1 · across: card 1's rule on the middle row

0	0	0
1	-2	1
0	0	0

$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$: left gets weight 1, right gets 1, the centre (dark) gets −2. Nothing else is used, so 0.

Step 2 · down: the same rule standing up

0	1	0
0	-2	0
0	1	0

$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$: up 1, down 1, centre −2.

Step 3 · add the two grids cell by cell

0	1	0
1	-4	1
0	1	0

The Laplacian is the sum of the two: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$. Centre: $-2 + (-2) = -4$. Sides: 1. Corners: never used, 0. The nine weights add to 0.

Step 4 · test it on a window

·	12	·
8	7	11
·	9	·

Multiply each pixel by the weight in the same place and add: $12 + 8 + 11 + 9 - 4 \cdot 7 = 40 - 28 = 12$. A flat window (all equal) would give 0.

What to write in the exam: $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$, then the mask below. Add one line: with diagonals the centre is −8; flipping signs gives +4 / +8.

Laplacian mask $= \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ **(weights sum to 0)**

Worth adding for full marks: if the diagonal second derivatives are included too, the mask is $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$; multiplying by -1 gives the equivalent masks with centre $+4$ or $+8$ (L7 p9). To sharpen, use $g = f - \nabla^2 f$ for a negative centre and $g = f + \nabla^2 f$ for a positive centre (L7 p11).

↩ back to the card

S7.4 · 2 · From the second derivative to the 3×3 Laplacian mask (Sheet 6 Q10) — Answers

Step 1 · look only at the centre

0	-1	0
-1	4	-1
0	-1	0

The centre weight is +4: positive.

Step 2 · apply the sign rule

Rule (L7 p11): centre negative → subtract, $g = f - \nabla^2 f$. Centre positive → add, $g = f + \nabla^2 f$. Here: add.

What to write in the exam: Plus. The centre is positive, so $g = f + \nabla^2 f$.

↩ back to the card

S7.5 · 3 · Sliding the Laplacian over a small image (Sheet 6 Q9) — Show the full exam answer

Step 1 · the image, labelled

	c0	c1	c2	c3	c4
r0	1	1	1	1	1
r1	4	4	4	4	4
r2	2	4	4	4	2
r3	4	4	4	4	4

Rows r0–r3 counted from the top, columns c0–c4 from the left. 4 rows × 5 columns.

Step 2 · which pixels get an answer

	c0	c1	c2	c3	c4
r0	~~1~~	~~1~~	~~1~~	~~1~~	~~1~~
r1	~~4~~	4	4	4	~~4~~
r2	~~2~~	4	4	4	~~2~~
r3	~~4~~	~~4~~	~~4~~	~~4~~	~~4~~

The mask needs a neighbour on every side. Border pixels (crossed out) lack one, so we ignore them (L6 p6). Only the 2×3 amber block gets a value.

Step 3 · one pixel in full: (r2, c1)

	c0	c1	c2	c3	c4
r0	1	1	1	1	1
r1	4	4	4	4	4
r2	2	4	4	4	2
r3	4	4	4	4	4

mask		
0	1	0
1	−4	1
0	1	0

Lay the mask on the image with its centre on the dark cell. Rule: up + down + left + right − 4 × centre. Up 4, down 4, left 2, right 4, centre 4: $4 + 4 + 2 + 4 - 4 \cdot 4 = 14 - 16 = -2$.

Step 4 · the other five the same way

	c1	c2	c3
r1	−3	−3	−3
r2	−2	0	−2

Row r1: the 1 above pulls it down: $1 + 4 + 4 + 4 - 16 = -3$. (r2, c2) sits in all 4s: $16 - 16 = 0$. (r2, c3) mirrors (r2, c1): −2.

Step 5 · second reading: pad with zeros

1	2	2	2	1
−9	−3	−3	−3	−9
4	−2	0	−2	4
−10	−4	−4	−4	−10

If you put a frame of 0s around the image, every pixel gets a value (same 4×5 size). The amber middle is unchanged; the border numbers are fake, caused by the 0 frame.

What to write in the exam: Mask $\begin{bmatrix}0 & 1 & 0\\ 1 & -4 & 1\\ 0 & 1 & 0\end{bmatrix}$, $\nabla^2 f$ = up + down + left + right − 4·centre, edges ignored → 2×3 output. One pixel in full, e.g. $4+4+2+4-16=-2$, then the grid below.

$\nabla^2 f = \begin{bmatrix}-3 & -3 & -3\\ -2 & 0 & -2\end{bmatrix}$ **(2×3, edges ignored)**

If the -8 mask (with diagonals) is used instead, the interior is $\begin{bmatrix}-11 & -9 & -11\\ -2 & 0 & -2\end{bmatrix}$. With the $+4$ mask every sign flips.

↩ back to the card

S7.6 · 3 · Sliding the Laplacian over a small image (Sheet 6 Q9) — Answers

Step 1 · add the frame of zeros

0	0	0	0
0	1	1	1
0	4	4	4

Top-left corner of Q9 with the grey 0 frame around it. The dark cell is the pixel; its four neighbours are blue.

Step 2 · the sum

Up 0, left 0, right 1, down 4, centre 1: $0+0+1+4-4\cdot 1=5-4=1$.

What to write in the exam: $0+0+1+4-4=1$ (the two 0s come from the padding, so this value is partly fake).

↩ back to the card

S7.7 · 4 · A clean step edge, filtered and then sharpened (Sheet 6 Q12) — Show the full exam answer

Step 1 · the image, and where answers go

	c0	c1	c2	c3	c4	c5
r0	~~50~~	~~50~~	~~50~~	~~50~~	~~50~~	~~50~~
r1	~~50~~	50	50	50	50	~~50~~
r2	~~50~~	50	50	50	50	~~50~~
r3	~~100~~	100	100	100	100	~~100~~
r4	~~100~~	100	100	100	100	~~100~~
r5	~~100~~	~~100~~	~~100~~	~~100~~	~~100~~	~~100~~

Rows from the top, columns from the left. Border pixels (crossed out) miss a neighbour, so we ignore them: the answer is the 4×4 amber block.

Step 2 · one pixel just above the edge: (r2, c1)

	c0	c1	c2
r1	50	50	50
r2	50	50	50
r3	100	100	100

mask		
0	1	0
1	−4	1
0	1	0

Up + down + left + right − 4 × centre: $50 + 100 + 50 + 50 - 4 \cdot 50 = 250 - 200 = 50$. Only the "down" neighbour (100) is different, and it pushes the result up.

Step 3 · one pixel just below the edge: (r3, c1)

	c0	c1	c2
r2	50	50	50
r3	100	100	100
r4	100	100	100

$50 + 100 + 100 + 100 - 4 \cdot 100 = 350 - 400 = -50$. Now the "up" neighbour (50) is the odd one out, so the result is negative.

Step 4 · the full Laplacian grid

	c1	c2	c3	c4
r1	0	0	0	0
r2	50	50	50	50
r3	-50	-50	-50	-50
r4	0	0	0	0

Each row of the image is constant, so every cell in a row gives the same number. Rows r1 and r4 have all four neighbours equal to the centre: $200 - 200 = 0$.

Step 5 · sharpen: $g = f - \nabla^2 f$

	f	∇^2f	g
r1	50	0	50 - 0 = 50
r2	50	50	50 - 50 = 0
r3	100	-50	100 + 50 = 150
r4	100	0	100 - 0 = 100

The mask centre is negative (−4), so subtract (L7 p11). The dark side gets darker (0), the bright side brighter (150): the edge now jumps by 150 instead of 50.

What to write in the exam: Mask with centre −4, edges ignored → 4×4. Show one pixel: $50 + 100 + 50 + 50 - 200 = 50$. Grid below. Sharpened ($g = f - \nabla^2 f$): rows 50, 0, 150, 100.

$$\nabla^2 f = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 50 & 50 & 50 & 50 \\ -50 & -50 & -50 & -50 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$ **(4×4, edges ignored)**

With the $+4$ mask the Laplacian rows are $0, -50, 50, 0$ and you use $g = f + \nabla^2 f$, giving the same g. Full-size versions: replicate padding (the slides' style) gives 6×6 with rows $0, 0, 50, -50, 0, 0$; zero padding gives fake border values (top-left corner $50 + 50 - 200 = -100$), the artefact L6 p6 warns about.

↩ back to the card

S7.8 · 4 · A clean step edge, filtered and then sharpened (Sheet 6 Q12) — Answers

Step 1 · one row in full: row 2 (value 20, just above the edge)

	c0	c1	c2
r1	20	20	20
r2	20	20	20
r3	80	80	80

Up + down + left + right − 4 × centre: $20 + 80 + 20 + 20 - 4 \cdot 20 = 140 - 80 = 60$.

Step 2 · all four interior rows, then sharpen

row	f	$\nabla^2 f$	$g = f - \nabla^2 f$
r1	20	0	20
r2	20	60	20 - 60 = -40 → 0
r3	80	-60	80 + 60 = 140
r4	80	0	80

Row 3: $20 + 80 + 80 + 80 - 320 = -60$. Rows 1 and 4 sit among equal pixels: 0. −40 is below 0, so clip it to 0.

What to write in the exam: $\nabla^2 f$ rows $0, 60, -60, 0$; sharpened rows $20, 0, 140, 80$.

↩ back to the card

S7.9 · 5 · Unsharp masking and high-boost: sharpen by subtracting a blur — Show the full exam answer

Step 1 · one row, with the ends copied outward

x		0	1	2	3	
f	50	50	60	90	100	100

All four rows are the same, so one row is enough. The grey cells are copies of the end pixels (the slide's numbers only work this way). Since rows are equal, the 3×3 average is just the average of 3 cells along the row.

Step 2 · blur one pixel in full: x = 1

x		0	1	2	3	
f	50	50	60	90	100	100
$\bar{f}$			67			

Average of left, me, right: $\bar{f}(1) = (50 + 60 + 90)/3 = 66.7 \to 67$ (round to the nearest whole number).

Step 3 · the whole blurred row

x		0	1	2	3	
f	50	50	60	90	100	100
$\bar{f}$		53	67	83	97	

Same for the others: $(50 + 50 + 60)/3 = 53.3 \to 53$, $(60 + 90 + 100)/3 = 83.3 \to 83$, $(90 + 100 + 100)/3 = 96.7 \to 97$.

Step 4 · mask = original − blurred

x	0	1	2	3
f	50	60	90	100
$\bar{f}$	53	67	83	97
mask	−3	−7	7	3

The mask is the detail the blur threw away. At $x = 1$: $60 - 67 = -7$. Before the edge it is negative, after it positive.

Step 5 · add it back (once, then 3 times)

x	0	1	2	3
f	50	60	90	100
mask	−3	−7	7	3
f + mask	47	53	97	103
f + 3·mask	41	39	111	109

At $x = 1$: unsharp $60 + (-7) = 53$; high-boost $60 + 3 \cdot (-7) = 60 - 21 = 39$. The edge pair (60, 90), a gap of 30, becomes (53, 97), gap 44, and with $k = 3$ (39, 111), gap 72.

What to write in the exam: Borders copied, rounded: $\bar{f} = [53, 67, 83, 97]$ (show $(50 + 60 + 90)/3 = 66.7 \to 67$). Mask $= f - \bar{f} = [-3, -7, 7, 3]$. $g = f +$ mask $= [47, 53, 97, 103]$. High-boost $g = f + 3 \cdot$ mask $= [41, 39, 111, 109]$. Same for every row. The edge got steeper.

Every row: $\bar{f} = [53, 67, 83, 97]$**, mask** $= [-3, -7, 7, 3]$**,** $g_{\text{unsharp}} = [47, 53, 97, 103]$**,** $g_{\text{highboost}} = [41, 39, 111, 109]$

Comment (as on the slide): the edge pair moved from $(60, 90)$, a gap of 30, to $(53, 97)$, a gap of 44, and with high-boost to $(39, 111)$, a gap of 72. If unsharp masking is not enough, multiply the mask by $k > 1$.

↩ back to the card

S7.10 · 5 · Unsharp masking and high-boost: sharpen by subtracting a blur — Answers

Step 1 · blur one pixel: x = 1

x		0	1	2	3	
f	30	30	30	60	60	60
$\bar{f}$		30	40	50	60	

$(30 + 30 + 60)/3 = 40$; the others the same way give 30, 50, 60.

Step 2 · mask, then f + 3·mask

x	0	1	2	3
mask = f − $\bar{f}$	0	−10	10	0
f + 3·mask	30	0	90	60

At $x = 1$: $30 - 40 = -10$, then $30 + 3 \cdot (-10) = 0$. At $x = 2$: $60 + 30 = 90$.

What to write in the exam: $\bar{f} = [30, 40, 50, 60]$, mask $= [0, -10, 10, 0]$, $g = [30,\ 0,\ 90,\ 60]$.

↩ back to the card

S7.11 · 6 · Building a 3×3 Gaussian mask from its formula (Sheet 6 Q11) — Show the full exam answer

Step 1 · write each cell's offset (x, y) from the centre

(-1, -1)	(0, -1)	(1, -1)
(-1, 0)	(0, 0)	(1, 0)
(-1, 1)	(0, 1)	(1, 1)

x and y are not pixel numbers 0–2; they are steps away from the centre: −1, 0 or 1.

Step 2 · square and add: $x^2 + y^2$

2	1	2
1	0	1
2	1	2

Example corner: $(-1)^2 + (-1)^2 = 2$. Only three numbers appear: 0 (centre), 1 (the four sides), 2 (the four corners). So you only compute three values.

Step 3 · plug into the formula ($\sigma = 1$)

$\sigma = 1$, so $2\sigma^2 = 2$ and $\frac{1}{2\pi\sigma^2} = \frac{1}{2\pi} = 0.1592$. So $h = 0.1592 \cdot e^{-(x^2+y^2)/2}$.

$x^2 + y^2$	$e^{-(x^2+y^2)/2}$	h
0	$e^0 = 1$	$0.1592 \cdot 1 = 0.1592$
1	$e^{-1/2} = 0.6065$	$0.1592 \cdot 0.6065 = 0.0965$
2	$e^{-1} = 0.3679$	$0.1592 \cdot 0.3679 = 0.0585$

Step 4 · put them back on the grid (raw mask)

0.0585	0.0965	0.0585
0.0965	0.1592	0.0965
0.0585	0.0965	0.0585

Replace each 0 / 1 / 2 from Step 2 with its h.

Step 5 · normalise so the weights add to 1

0.0751	0.1238	0.0751
0.1238	0.2042	0.1238
0.0751	0.1238	0.0751

Sum $= 0.1592 + 4(0.0965) + 4(0.0585) = 0.7795$ (with the rounded values you get 0.7792; either is fine). Divide every cell by it: centre $0.1592/0.7795 = 0.2042$. Without this a blur would darken the image.

What to write in the exam: $x, y \in \{-1, 0, 1\}$, $x^2 + y^2 \in \{0, 1, 2\}$. $h = \frac{1}{2\pi} e^{-(x^2+y^2)/2}$: 0.1592, 0.0965, 0.0585. Raw grid, sum 0.7795, divide → normalised grid below.

Gaussian 3×3, $\sigma = 1$: centre 0.2042, sides 0.1238, corners 0.0751 (raw: 0.1592, 0.0965, 0.0585); weights sum to 1

↩ back to the card

S7.12 · 6 · Building a 3×3 Gaussian mask from its formula (Sheet 6 Q11) — Answers

Step 1 · same three cells, smaller σ

$\sigma = 0.5$: $2\sigma^2 = 0.5$. Side: $e^{-1/0.5} = e^{-2}$, corner: e^{-4}. These are tiny, so the sides and corners lose almost all their weight.

Step 2 · the normalised masks side by side

σ = 1			σ = 0.5		
0.0751	0.1238	0.0751	0.0113	0.0838	0.0113
0.1238	0.2042	0.1238	0.0838	0.6193	0.0838
0.0751	0.1238	0.0751	0.0113	0.0838	0.0113

The centre jumps from 0.2042 to 0.6193.

What to write in the exam: Up. A smaller σ is a narrower bell, so the centre gets most of the weight (less blur).

↩ back to the card

S7.13 · 7 · Low-pass versus high-pass: the two families side by side (Sheet 6 Q13) — Show the full exam answer

Step 1 · low-pass on a tiny edge: average 3 cells

x	0	1	2	3
f	10	10	40	40
average	–	20	30	–

$(10 + 10 + 40)/3 = 20$, then $(10 + 40 + 40)/3 = 30$. The jump of 30 is spread out into steps of 10: blurred. Weights $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$ add to 1, so a flat area keeps its value.

Step 2 · high-pass on the same row: second derivative

x	0	1	2	3
f	10	10	40	40
f″	–	30	-30	–

$40 + 10 - 2 \cdot 10 = 30$, then $40 + 10 - 2 \cdot 40 = -30$. Only the jump shows. Weights $1, -2, 1$ add to 0, so a flat area becomes 0.

Step 3 · the comparison table

	Low-pass (smoothing) filter	High-pass (sharpening) filter
Passes	low frequencies: slowly varying areas	high frequencies: edges, fine detail, noise
Removes	noise, small details, sharp transitions	constant / slowly varying background
Effect	blurs the image, reduces noise	highlights edges, sharpens (after adding back to f)
Based on	averaging (integration)	differences (derivatives)
Mask weights	positive, sum to 1	positive and negative, sum to 0
Examples	box average $\frac{1}{9}[1 \dots 1]$, weighted average, Gaussian (median is a non-linear smoother)	Laplacian, Sobel, unsharp mask $f - \bar{f}$

What to write in the exam: The table (at least passes / removes / effect / examples) plus one line: low-pass weights sum to 1 (flat stays flat), high-pass weights sum to 0 (flat becomes 0).

Low-pass keeps slow intensity changes and blurs / removes noise; high-pass keeps fast changes (edges) and removes flat regions. Low-pass weights sum to 1, high-pass weights sum to 0.

↩ back to the card

S7.14 · 8 · The Sobel gradient: how strong is the edge at this pixel? — Show the full exam answer

Step 1 · the image, labelled

	c0	c1	c2	c3
r0	10	10	200	200
r1	10	10	200	200
r2	100	100	20	20
r3	100	100	20	20

Rows from the top, columns from the left, counted from 0. The four amber cells are the interior pixels: the only ones with a full 3×3 window.

Step 2 · the window of (r1, c1) next to G_x

	c0	c1	c2
r0	10	10	200
r1	10	10	200
r2	100	100	20

G_x		
-1	0	1
-2	0	2
-1	0	1

Multiply each pixel by the weight in the same place, add all nine: $-10 + 0 + 200 - 20 + 0 + 400 - 100 + 0 + 20 = 490$. In words: right column minus left column, middle row counted twice.

Step 3 · same window, G_y

10	10	200
10	10	200
100	100	20

G_y		
-1	-2	-1
0	0	0
1	2	1

Bottom row minus top row, middle counted twice: $(100 + 200 + 20) - (10 + 20 + 200) = 320 - 230 = 90$. Edge strength $V = |G_x| + |G_y| = 490 + 90 = 580$.

Step 4 · the other three windows the same way

	G_x		G_y		V	
	c1	c2	c1	c2	c1	c2
r1	490	490	90	-450	580	940
r2	-50	-50	90	-450	140	500

Slide the window one cell right, or one down, and repeat. Take absolute values before adding: $|490| + |-450| = 940$.

Step 5 · why the slide says 0

10	10	10
10	10	10
10	10	10

The slide's "(1,1)" counts from 1, so it is the top-left corner. Copy the border outward (grey) and its window is all 10s: $G_x = -10 - 20 - 10 + 0 + 0 + 0 + 10 + 20 + 10 = 0$, $G_y = 0$, $V = 0$, "a constant area".

What to write in the exam: Say which pixel you mean. Slide's (1,1) = top-left with copied borders: window all 10s, $G_x = G_y = 0$, $V = 0$. Interior (r1, c1): $G_x = 490$, $G_y = 90$, $V = 580$; show the nine products for one mask. Other interior pixels: $V = 940, 140, 500$.

Top-left pixel with replicated borders (the slide's reading): $V = 0$. Interior pixels: $V =$ $\begin{bmatrix} 580 & 940 \\ 140 & 500 \end{bmatrix}$

Labels: on p17 the mask with rows $-1\ -2\ -1 / 0\,0\,0 / 1\,2\,1$ is called G_y, but the p18 arithmetic for "G_x" uses exactly those weights. The names are swapped between figure and arithmetic; V is the same either way.

↩ back to the card

S7.15 · 8 · The Sobel gradient: how strong is the edge at this pixel? — Answers

Step 1 · window next to G_y

0	0	0
0	0	0
9	9	9

G_y		
-1	-2	-1
0	0	0
1	2	1

Only the bottom row is non-zero: $1 \cdot 9 + 2 \cdot 9 + 1 \cdot 9 = 9 + 18 + 9 = 36$.

Step 2 · and G_x

Every column is $(0, 0, 9)$, so right column − left column $= 0$: $G_x = 0$. A horizontal edge shows only in G_y.

What to write in the exam: $G_y = 9 + 18 + 9 = 36$, $G_x = 0$.

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S7.16 · 10 · Quick recall before the exam — Answers

1. $f'(x) = f(x+1) - f(x)$ ("next minus me"); $f''(x) = f(x+1) + f(x-1) - 2f(x)$ ("left + right − 2 × me").
2.

x	0	1	2	3
f	5	5	9	9
f''	–	4	-4	–

   $x = 1$: $9 + 5 - 2 \cdot 5 = 4$; $x = 2$: $9 + 5 - 2 \cdot 9 = -4$. A +/− pair around the step.
3. $g = f - \nabla^2 f$ (negative centre, L7 p11).
4. mask $= f - \bar{f}$ (original minus blurred); $g = f + k \cdot$ mask with $k > 1$ ($k = 1$ is plain unsharp masking).
5. 1 (low-pass, keeps brightness); 0 (high-pass, flat areas vanish).
6. $|-30| + |40| = 70$ (the exact magnitude $\sqrt{900 + 1600} = 50$).

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The formula sheet — write it from memory before the exam

Cover the right column, say each line out loud, then check. Star anything you missed and redo the starred lines once more.

Lectures 2–3 · digital images, pixels, operations	
Levels and bits	$L = 2^k$ (8 bits → 256 levels, 3 bits → 8)
Image size	$M \times N \times$ channels $\times\ k$ bits; ÷ 8 = bytes; ÷ 2^{20} = MB
Transmission	bits on the line = pixels × (data bits + start + stop) per packet; time = bits ÷ baud
Neighbours	N_4: $(x \pm 1, y), (x, y \pm 1)$ · N_D: $(x \pm 1, y \pm 1)$ · $N_8 = N_4 \cup N_D$
m-adjacency (diagonal pair, both in V)	allowed iff $N_4(p) \cap N_4(q)$ has no pixel with value in V
Distances	$D_e = \sqrt{(x-s)^2 + (y-t)^2}$ · $D_4 = \lvert x-s\rvert + \lvert y-t\rvert$ · $D_8 = \max(\lvert x-s\rvert, \lvert y-t\rvert)$
Bilinear interpolation	$v(x,y) = ax + by + cxy + d$ from the 4 nearest pixels; bicubic uses 16
Linear operator	$H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$; one counter-example disproves it
ROI / noise	ROI $= f \odot$ mask (array product) · corrupted $= f + \eta$, clip to 0–255
Lectures 4–5 · intensity transformations and histograms	
Negative · log · gamma	$s = L - 1 - r$ · $s = c\log(1 + r)$ · $s = c\,r^{\gamma}$
Contrast stretching	$I_{new} = (I - I_{min})\dfrac{new_{max} - new_{min}}{I_{max} - I_{min}} + new_{min}$
Histogram	$h(r_k) = n_k$, $p(r_k) = n_k / MN$
Equalization	$s_k = (L-1)\sum_{j=0}^{k} p_r(r_j)$, rounded to the nearest level
Matching (specification)	equalize the input → s_k; $G(z_q) = (L-1)\sum_{i=0}^{q} p_z(z_i)$, rounded; map each s_k to the z_q whose $G(z_q)$ is closest
Lecture 6 · spatial filtering	
Correlation	$g(x,y) = \sum_s \sum_t w(s,t)\, f(x+s, y+t)$
Convolution	correlation with the kernel rotated 180°; to make a convolution give a correlation's output, rotate the kernel 180° first
Output size and padding	no padding: $n - k + 1$ · padding $p = (k-1)/2$ on each side keeps size n · full: $n + k - 1$
Smoothing	box: $\frac{1}{9}$ of all-ones 3×3 · weighted: $\frac{1}{16}\begin{bmatrix}1&2&1\\2&4&2\\1&2&1\end{bmatrix}$ · median: middle of the 9 sorted values
Lecture 7 · sharpening	
First derivative	$\dfrac{\partial f}{\partial x} = f(x+1) - f(x)$
Second derivative	$\dfrac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$

Laplacian	$\nabla^2 f = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y) \rightarrow$ mask $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ (with diagonals: centre -8)
Sharpening	$g = f + c\,\nabla^2 f$, with $c = -1$ when the mask's centre is negative
Gaussian, $\sigma = 1$	$h(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$: centre 0.1592, edge 0.0965, corner 0.0585; divide by their sum to normalise
Unsharp masking / high-boost	$g_{mask} = f - \bar{f}$ (blurred); $g = f + k\,g_{mask}$, $k = 1$ unsharp, $k > 1$ high-boost
Low-pass vs high-pass	low-pass (smoothing) keeps slow changes, removes detail and noise; high-pass (sharpening) keeps edges and fine detail